Regularity of the Time-to-Collision Map and Convergence of Learned Hamilton–Jacobi Solutions

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Abstract. Controllers for autonomous vehicles that depend solely on inter-vehicle spacing can behave overly aggressively by ignoring relative velocity. Time-to-collision (TTC) addresses this, but classical kinematic TTC assumes a frozen system state, making it non-predictive and unreliable as feedback. We introduce a dynamically-informed TTC, defined as the hitting time of the collision set under an optimal-velocity prediction flow with merely measurable (L^∞) leader acceleration. This makes the collision estimate consistent with the dynamics, at the cost of a feedback coefficient that is the hitting time of an auxiliary flow and is discontinuous at grazing states. The TTC map is discontinuous on the set of states whose prediction grazes the collision set; away from it the closed loop is a Carathéodory system, and across it we show that trajectories cross transversally, so that Filippov solutions exist for all physical data and are unique, and are Carathéodory solutions, through nondegenerate grazing states. Every contact of the closed loop has zero closing speed: no solution reaches the collision set with negative relative velocity, so impacts are excluded for every solution (a zero-speed touch is not excluded). The analytical core is a regularity theory for the TTC map on the domain of transversal initial data. On compact subsets it is Lipschitz, continuously differentiable in the state with an explicit gradient, and Lipschitz but in general not differentiable in time, since its time derivative jumps where the leader acceleration does (whenever the TTC is sensitive to the relative velocity, which we prove for prediction horizons below an explicit threshold); the Hamilton–Jacobi equation it satisfies holds classically at every state for almost every time. The Lipschitz constants are governed by a uniform transversality margin that degenerates as tangential collisions are approached. Since the dynamic TTC admits no closed form, online simulation introduces delay that undermines safety guarantees. Recasting the TTC map as a value function, we prove a stability estimate for a physics-informed formulation: the L^2 error of the learned value function is bounded by the square root of the total loss, with a constant that does not degenerate at the boundary of the domain, and at zero residual the approximation coincides pointwise with the value function. A two-stage architecture removes the online-simulation delay from the learned controller, and simulations on a six-vehicle platoon illustrate its behavior in the tested scenarios, including attenuation of the leader's disturbance along the platoon; no safety or string-stability guarantee is claimed for the learned controller.

Key words. Time-to-collision; car-following dynamics; hitting-time regularity; Hamilton–Jacobi–Bellman equations; physics-informed neural networks; Carathéodory and Filippov solutions; collision avoidance

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1. Introduction and Related Works.

Microscopic traffic models describe the interaction of vehicle-to-vehicle through ordinary differential equations in which each vehicle responds to its predecessor. Classical formulations [52, 47], and in particular the Optimal Velocity

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Model (OVM) [5], reproduce observed phenomena such as stop-and-go waves and spontaneous congestion [16, 32], have been analysed extensively for stability [63, 65, 42, 43], and underpin driver-assistance and autonomous-vehicle control [59, 57]. Numerous refinements improve behavioural realism [26, 29, 41, 44, 70, 69]. A limitation shared by majority of models in this family is that the control depends only on inter-vehicle spacing, and therefore carries no information about the rate of closure: the response grows exaggerated as spacing decreases even when the leader is pulling away and collision risk is low [69]. Time-to-collision (TTC) supplies exactly the missing information, and empirical studies confirm that drivers respond more strongly to TTC than to spacing alone [66, 36, 49, 61, 62].

A circularity in kinematic TTC. The classical TTC of Hayward [25] is the gap divided by the closing speed, and the refinements that followed — Potential TTC [60], Modified TTC [51, 67], two-dimensional extensions [64, 45] — relax the constant-speed assumption only through limited acceleration projections. All remain *frozen-state*: TTC is computed as though no control acts from the present moment onward [25, 9, 17, 10, 27, 2]. Embedded in a feedback law this is self contradictory — the controller is designed to prevent precisely the collision that the estimate presumes will be met passively — and several OVM-based models inherit the inconsistency by retaining an instantaneous TTC term [30, 40, 35, 1, 31, 37].

We therefore define TTC through the system's own predicted evolution. Writing Ξ for the inter-vehicle gap and Λ for the relative velocity, the *dynamic* TTC of a state $\boldsymbol{z}$ at time $\bar{t}$ is the hitting time

$$\tau(\boldsymbol{z}) \overset{\text{def}}{=} \inf\{\theta \in (0, \boldsymbol{T}_h - \bar{t}] : \ \Xi(\theta; \boldsymbol{z}) = 0, \ \Lambda(\theta; \boldsymbol{z}) < 0\}, \qquad \inf \varnothing \overset{\text{def}}{=} \boldsymbol{T}_h - \bar{t}, \tag{1.1}$$

the first time within the prediction horizon $\boldsymbol{T}_h$ at which the predicted gap reaches zero with negative predicted relative velocity (the precise definition, and why tangential touches are not counted, is definition 3.6; on the domain studied in section 4 this is simply the first zero of the predicted gap), where $\Xi(\cdot\,; \boldsymbol{z})$ evolves under OVM prediction dynamics started from $\boldsymbol{z}$, with a leader acceleration that is merely measurable and bounded, $\mathsf{a}_\ell \in L^\infty$. The OVM is the natural choice of prediction model; it is the base layer of the proposed dynamics with the TTC term removed [5, 4], so the controller anticipates the trajectory that the uncontrolled system would in fact follow. The resulting control law is genuinely predictive — but the map τ now appears in the vector field whose flow defines it, and the model is well posed only if τ is regular enough to serve as a coefficient. Three obstructions follow, and they organize the paper.

(i) Regularity of a hitting time. Hitting times are generically discontinuous; a trajectory tangent to the target set can be perturbed into one that misses it entirely. Regularity of exit- and minimum-time functions is classical [7, 11], where Lipschitz continuity is typically obtained from Petrov-type controllability conditions ensuring that some admissible control drives the state into the target at a definite rate; without such a condition the value function is continuous but need not be locally Lipschitz. Our setting is different in two respects. The prediction flow is *uncontrolled*, so the relevant nondegeneracy is not controllability but *transversality*; the predicted trajectory must cross $\{\Xi = 0\}$ with $\Lambda < 0$ rather than graze it. And the forcing is only L^∞ in time, placing the problem in the Carathéodory rather than the classical framework [23]. We prove that on the domain of transversal initial data

τ is locally Lipschitz, continuously differentiable in the state with an explicit gradient, and Lipschitz in time; the time regularity is sharp, since $\partial_s \tau$ jumps where the leader acceleration does, wherever τ is sensitive to the relative velocity (which holds for prediction horizons below $\pi/\sqrt{\alpha(1-\alpha/4)}$). The Lipschitz constants are controlled by a single uniform transversality margin that degenerates precisely as tangency is approached. Where the prediction grazes the collision set, τ is discontinuous, and the closed-loop field with it; we show that closed-loop trajectories cross this set transversally, so that well-posedness survives in the sense of Filippov and, when every grazing state met is nondegenerate, in the Carathéodory sense.

(ii) No closed form, hence delay. Eliminating Λ reduces the prediction dynamics to a forced, damped nonlinear oscillator with measurable forcing; no closed-form solution is available, and τ additionally requires inverting the flow to locate a zero of the gap. Evaluating τ therefore means simulating an ODE at every control instant. The resulting latency is not merely an inefficiency; by the time the simulation completes the state has moved, the feedback loop is broken, and the collision guarantees that the controller was designed to provide no longer apply. A real-time surrogate is thus mathematically necessary, not only practically convenient [15, 48].

(iii) Approximation with guarantees. Neural networks approximate controlled dynamics compactly and evaluate in constant time [39, 33, 12], and rigorous foundations exist for their use in safety-critical control [55, 56, 8, 34]. The physics-informed framework [54] embeds the governing equation in the loss. What is required here, however, is not merely a good fit but a *guarantee*: that a small residual implies a small error in the learned object. We obtain this by recasting τ as the value function of a minimum-time problem. The dynamic programming principle yields a first-order Hamilton–Jacobi equation that τ satisfies classically at every state for almost every time, and whose Hamiltonian is *affine* in the gradient; consequently the characteristics coincide with the OVM trajectories and do not cross. This is the structural reason why a solution that is absolutely continuous along the prediction trajectories—in particular every locally Lipschitz almost-everywhere solution—is here determined by its values on the collision set, in contrast to the general Hamilton–Jacobi setting, where gradient ambiguity admits spurious Lipschitz almost-everywhere solutions [14, 7].

Related work on learning Hamilton–Jacobi solutions. Hamilton–Jacobi reachability provides a set-based characterisation of safety through the value function of a differential game [46, 18], and learned approximations have made high-dimensional instances tractable [6]. These treat the dynamics as given and the value function as the object to be learned; the regularity of that object is inherited rather than established. Convergence theory for physics-informed approximation has developed in parallel [58, 53], largely for linear elliptic and parabolic problems. Closest to the present work is [38], which bounds solution error by residual for stationary Lyapunov and Hamilton–Jacobi–Bellman equations by integrating the residual along closed-loop trajectories — the same mechanism we use. The settings differ in ways that matter: there the value function is an infinite-horizon cost to an equilibrium, and a *pointwise relative* residual bound is required, since a merely uniform bound may integrate to infinity along trajectories; here the horizon is the finite hitting time (1.1) with a free terminal boundary, which removes that obstruction and permits an L^2 hypothesis, but requires establishing regularity of the hitting time itself. To our knowledge the TTC map has not previously been treated as a value function, and no regularity theory for it exists; the traffic literature computes TTC

kinematically throughout.

Sobolev regularity of the target is what converts approximation theory into explicit rates [68]; the Lipschitz and C^1-in-state results of Section 4 play exactly that role here. The limitation in time is inherited from the forcing: under a discontinuous leader acceleration no C^1 function can have zero strong-form residual, and the quantitative stability estimate is then the operative statement.

Contributions. The main contributions of this paper are as follows:

- **A dynamically-informed TTC and its closed-loop model.** We introduce a feedback law in which the time-to-collision is computed as the hitting time of the collision set under the system's own prediction dynamics, rather than under a frozen-state assumption. The resulting controller is genuinely predictive, and resolves the inconsistency of classical TTC feedback — at a price: the feedback coefficient τ is the hitting time of an auxiliary prediction flow, it has no closed form, and it is discontinuous at the states whose prediction grazes the collision set.
- **Well-posedness and collision avoidance.** The closed-loop vector field is locally Lipschitz away from the *grazing set*, the states whose predicted trajectory touches the collision set tangentially, and discontinuous across it. We prove existence of Filippov solutions for all physical data, uniqueness of Carathéodory solutions in the regular set, and, at nondegenerate grazing states, that every solution crosses the grazing set transversally with a definite speed, so that uniqueness and the Carathéodory solution concept survive the crossing. The mechanism is structural: the prediction changes along the closed loop only through the TTC term, whose sign is the same on both sides of the grazing set. Collision avoidance is then proved for *every* solution: at any contact the relative velocity tends to zero, with the quantitative barrier $|V| \le \sqrt{2\bar{a}X}$ near contact. The singularity $\beta V/\tau^2$ that complicates the analysis is precisely the mechanism that forbids impact.
- **Regularity of the hitting-time map.** On compact subsets of the transversal domain we prove that τ is Lipschitz with an explicit constant, continuously differentiable in the state with an explicit gradient obtained by implicit differentiation and a jointly continuous gradient, and Lipschitz in time; it is Fréchet-differentiable wherever its time derivative exists, and the Hamilton–Jacobi equation holds classically at every state for almost every time. The time regularity is sharp: $\partial_s\tau$ jumps where the leader acceleration jumps, at every state where $\partial_\Lambda\tau \neq 0$, which we prove for prediction horizons below $\pi/\sqrt{\alpha(1-\alpha/4)}$. The Lipschitz constants are governed by a uniform transversality margin $\gamma_{\boldsymbol{D}}$, which degenerates as tangential collisions are approached and accounts for the failure of global Lipschitz continuity. This regularity is not covered by existing exit-time theory, where Lipschitz continuity follows from controllability conditions unavailable for an uncontrolled prediction flow.
- **Non-degeneracy of the endpoint map.** The error analysis pushes an integral over the three-dimensional domain onto the two-dimensional collision surface through the endpoint map Ψ, which assigns to each state the point at which its prediction collides. We prove that its 2-Jacobian is bounded below, $J_2\tilde{\Psi} \ge c_* > 0$, by a Cauchy–Binet argument and a slice estimate that follows from the fixed-time flow map being bi-

Lipschitz. The constant depends only on the flow, not on the transversality margin: it does not degenerate at the boundary of the domain.

- L^2 **stability of the learned value function.** Recasting τ as the value function of a minimum-time problem, we prove that the L^2 error of a physics-informed approximation is bounded by the square root of the total loss, with a constant that is uniform up to the boundary of the transversal domain. The boundary term is controlled by the co-area formula together with the Jacobian bound above; the interior term by a change of variables along the flow, whose Jacobian is explicit ($\det Y = e^{-\alpha\kappa}$ by Liouville's formula). The estimate is one-directional: it bounds the error by the loss, not conversely. At zero residual, and under continuous forcing, the conclusion sharpens to *pointwise* identification with the value function.
- **Approximation and validation.** We propose a physics-informed architecture for approximating the full system trajectory without the latency incurred by online simulation of the prediction dynamics, and report its accuracy against reference simulations and its behavior on a six-vehicle platoon in the tested scenarios. These are numerical observations; the analytical guarantees of this paper concern the exact controller and the L^2 error of the learned value function, not the closed loop driven by the surrogate. Implementation details are given in the Supplementary Materials.

2. Mathematical Model of Dynamic-based TTC Control.

In this section, we present the general proposed control framework, which consists of the proposed AV dynamical model with dynamic-based TTC, a novel approach in computing TTC.

2.1. The Model.

As discussed in Section 1, a control law depending only on inter-vehicle spacing carries no information about the rate of closure, and incorporating the relative velocity through a time-to-collision term supplies precisely that information. We now make the resulting model precise.

We propose the nonlinear feedback control:

$$\begin{cases} \dot{x}_\ell = v_\ell \\ \dot{v}_\ell = \mathsf{a}_\ell \\ \dot{x}_n = v_n \\ \dot{v}_n = \alpha\left\{\mathcal{V}(x_{n-1}(t) - x_n(t)) - v_n(t)\right\} + \beta \frac{(v_{n-1}(t) - v_n(t))}{\left(\tau_n(t, x_{n-1}, x_n, v_{n-1}, v_n)\right)^2} \\ (x_\ell(0), x_n(0), v_\ell(0), v_n(0)) = (x_{\ell,\circ}, x_{n,\circ}, v_{\ell,\circ}, v_{n,\circ}), \end{cases} \tag{2.1}$$

for $n \in \{1, \cdots, N\}$ or in compact form, we define $\boldsymbol{\varphi}_n := (x_{n-1}, x_n, v_{n-1}, v_n)$ and the dynamical model Equation (2.1) can be written in the form of

$$\begin{cases} \dot{\Phi} = \mathsf{F}(t, \Phi) = \mathsf{F}(t, \Phi, \tau(t, \Phi)), \quad \Phi = (\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \cdots, \boldsymbol{\varphi}_N), \quad \tau = (\tau_1, \cdots, \tau_N) \\ \Phi(t_\circ) = \Phi_\circ. \end{cases} \tag{2.2}$$

The index $n \in \{1, \cdots, N\}$, indicates the vehicle, and the position and velocity of n-th vehicle are denoted by x_n and v_n, respectively. The preceding vehicle of vehicle $n \in \{1, \cdots, N\}$ is indexed by $n - 1$. The position, velocity, and acceleration of the leading vehicle are denoted

by x_ℓ, v_ℓ, and a_ℓ, respectively. For notational consistency, the exogenous leader, which is the preceding vehicle of vehicle 1, is identified as vehicle 0, making $x_0 = x_\ell$ and $v_0 = v_\ell$. Here, τ_n is the time-to-collision (TTC) control strategy which will be defined later in this section—see Def. 3.6. The proposed platoon may include all interacting AVs or AVs interacting with a human-driven vehicle (HV) leader (See the schematic in Figure 1).

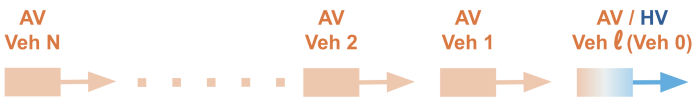


Figure 1: Platoon Schematic

The optimal velocity function $\mathcal{V} : \mathbb{R} \to \mathbb{R}$ is a smooth, bounded, strictly increasing Lipschitz function with $\mathcal{V}(0) = 0$ and $\mathcal{V} \geq 0$ on $[0, \infty)$ (its values at negative gaps are used only by the prediction flow, which is continued through the collision set). In this paper, we consider

$$\mathcal{V}(x_{n-1} - x_n) \overset{\text{def}}{=} \tanh(x_{n-1} - x_n - 2) + \tanh(2), \tag{2.3}$$

see [48, 5] for properties of this function, and [44] for model's sensitivity analysis with respect to the coefficients α and β.

In addition, we assume a reasonable behavior for the leading vehicle from a practical point of view; that is, the leader's acceleration is a prescribed measurable function of time,

$$\mathsf{a}_\ell \in L^\infty(0, \boldsymbol{T}_h), \qquad \text{with} \qquad 0 \leq v_{\ell,\circ} + \int_0^t \mathsf{a}_\ell(s)\, ds \leq v_{\max} < \infty \quad \text{for all } t \in [0, \boldsymbol{T}_h], \tag{2.4}$$

where $v_{\max}$ is the maximum permitted speed. In particular the leader does not react to the follower. See Section SM4 for some details on calculations of the leading acceleration.

Since the leader enters the analysis only through a prescribed bounded acceleration profile a_ℓ and a speed in $[0, v_{\max}]$, the analytical parts consider $N = 1$ (a leader–follower pair). Extending the analytical statements to a platoon in which each predecessor is itself a controlled follower is not immediate: a controlled predecessor is only proved to be absolutely continuous, and the prediction of pair n would depend on the predecessor's future motion, hence on upstream states; we do not pursue this here (cf. [44] for the uncontrolled case), and the platoon experiments of Section 6 are numerical. Therefore, from hereon in this paper, we focus our analysis on a general leader-follower control dynamics in the compact form

$$\begin{cases} \dot{\boldsymbol{\varphi}} = \mathsf{F}(t, \boldsymbol{\varphi}), & t \in [0, \boldsymbol{T}_h] \\ \boldsymbol{\varphi}(t_\circ) = \boldsymbol{\varphi}_\circ. \end{cases} \tag{2.5}$$

for $\boldsymbol{\varphi} = (x_\ell, x, v_\ell, v)$.

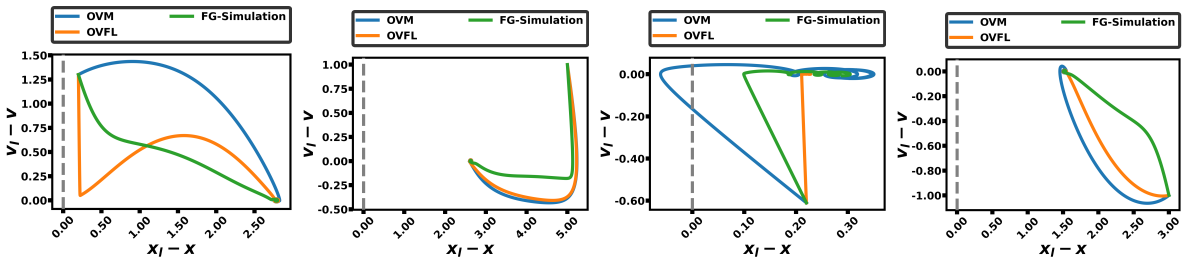
2.2. The Dynamic-Based TTC and its Approximation. In this paper, the dynamic-based TTC mapping τ (definition 3.6) is computed from the OVM-dynamics. More precisely, given any state $(\bar{t}, \bar{\boldsymbol{\varphi}})$ of system Equation (2.5), the OVM-dynamical model Equation (2.6) predicts

the time to collision of system Equation (2.2) starting from such current state of the system. The choice of OVM for predicting the time to collision can be justified by the fact that the OVM-dynamics is a simple characterization of dynamic evolution of acceleration/deceleration as a function of distance between consecutive vehicles. In addition, the OVM is essentially part of the dynamical model (2.1), excluding the β-control term that potentially is used in AV's dynamics to prevent the collision. This dynamical model is proven to be a proper representation of critical properties of the system. It should also be noted that the choice of OVM dynamics is not restrictive in any sense and can simply be replaced by any other dynamical model as long as it satisfied a certain properties—e.g. Lipschitz solution.

To continue with the motivation of the proposed dynamical model, it would be helpful to further compare its behavior with the existing microscopic traffic models—OVM, OVFL—considering different initial conditions. Let us recall

$$\text{OVM} : \dot{v}(t) = \alpha \{\mathcal{V}(x_\ell(t) - x(t)) - v(t)\} \tag{2.6}$$

$$\text{OVFL} : \dot{v}(t) = \alpha \{\mathcal{V}(x_\ell(t) - x(t)) - v(t)\} + \beta \frac{v_\ell(t) - v(t)}{(x_\ell(t) - x(t))^2}. \tag{2.7}$$



(a) (b) (c) (d)

Figure 2: Comparison between the trajectories of OVFL, OVM, and the proposed dynamics with dynamic-based TTC (Simulation approaches are detailed in Def. SM2.1). Parameters: α =2, β = 3, x = 0, Simulation Time = 50, $\mathsf{a}_\ell = 0.01\sin(0.5t)$ Initial Conditions: (a) $x_\ell = 0.2$, $v = 0.3$, $v_\ell = 1.6$; (b) $x_\ell = 5$, v =0.5, $v_\ell = 1.5$; (c) $x_\ell = 0.22$, v =0.62, $v_\ell = 0.01$ (d) $x_\ell = 3$, v =1.9, $v_\ell = 0.1$

Figure 2 compares the properties of these models as discussed in Section 1.

3. Well-posedness and collision avoidance of the closed loop. In this section we study the two-vehicle closed loop (2.5). The TTC τ that enters the feedback is the hitting time of the collision set under the OVM prediction flow started from the current state; it is defined in definition 3.6 below (this is the definition anticipated in (1.1)). The difficulty is twofold. The feedback coefficient $1/\tau^2$ is defined through a flow that the feedback itself perturbs, and τ is discontinuous on the set of states whose prediction *grazes* the collision set. We show that away from this grazing set the closed loop is a Carathéodory system with unique solutions; that across it solutions exist in the sense of Filippov and, at nondegenerate grazing states, cross transversally so that uniqueness survives; and that no solution ever reaches the collision set with positive closing speed.

Notation for this section.. We work in the difference variables $X = x_\ell - x$ (gap) and $V = v_\ell - v$ (relative velocity), $\xi = (X, V)$, so that the follower speed is $v = v_\ell(t) - V$. The predicted gap and relative velocity under the OVM flow are written (Ξ, Λ), as in (1.1). Throughout, $z = (\bar{t}, \bar{X}, \bar{V})$ denotes a state (a point of space-time), and $|\cdot|$ the Euclidean norm.

3.1. Setting.

Assumption 3.1 (Standing assumptions).

(*i*) $\alpha > 0$ *and* $\beta > 0$.

(*ii*) $\mathsf{a}_\ell \in L^\infty(0, \boldsymbol{T}_h)$, *and the leader velocity* $v_\ell(t) \overset{\textit{def}}{=} v_{\ell,\circ} + \int_0^t \mathsf{a}_\ell(r)\, dr$ *satisfies* $0 \le v_\ell(t) \le v_{\max}$ *for all* $t \in [0, \boldsymbol{T}_h]$. *The leader position is* $x_\ell(t) \overset{\textit{def}}{=} x_{\ell,\circ} + \int_0^t v_\ell(r)\, dr$. *We fix a representative of* a_ℓ *with* $|\mathsf{a}_\ell(t)| \le \|\mathsf{a}_\ell\|_{L^\infty}$ *for* every t. *This makes the pointwise-in-*t *bounds below hold at every* t, *not just a.e.*

(*iii*) $v_{\max} \ge \overline{\mathcal{V}} \overset{\textit{def}}{=} 1 + \tanh 2$.

Remark 3.2. *Item (iii) says that the maximal permitted speed is at least the supremum of the optimal velocity function; it is exactly what makes* $[0, v_{\max}]$ *invariant for the follower speed (lemmas* 3.5 *and* 3.18*).*

Facts on $\mathcal{V}$*..* $\mathcal{V}(r) = \tanh(r-2) + \tanh 2$ is smooth and strictly increasing, with $0 < \mathcal{V}' \le 1$, $\mathcal{V}(0) = 0$, and

$$\tanh 2 - 1 < \mathcal{V}(r) < \overline{\mathcal{V}} \quad (r \in \mathbb{R}), \qquad \mathcal{V}(r) > 0 \quad (r > 0). \tag{3.1}$$

In particular $|\mathcal{V}| < \overline{\mathcal{V}}$ and $-\mathcal{V} < 1 - \tanh 2 < \overline{\mathcal{V}}$ on $\mathbb{R}$.

The deceleration constant.. We set

$$\bar{a} \overset{\textbf{def}}{=} \|\mathsf{a}_\ell\|_{L^\infty} + \alpha(\overline{\mathcal{V}} + v_{\max} + 1). \tag{3.2}$$

(The "+1" gives room to work on an open neighbourhood of the physical states; see (3.7).)

Reduction to the difference variables.. Since the leader trajectory (x_ℓ, v_ℓ) is prescribed, the map $(x, v) \mapsto (X, V) \overset{\textbf{def}}{=} (x_\ell(t) - x,\ v_\ell(t) - v)$ is, for each t, an affine bijection of $\mathbb{R}^2$, and system (2.5) is equivalent to

$$\dot{X} = V, \qquad \dot{V} = \mathsf{f}(t, X, V) \overset{\textbf{def}}{=} \mathsf{g}(t, X, V) - \beta \frac{V}{\tau(t, X, V)^2}, \tag{3.3}$$

where

$$\mathsf{g}(t, X, V) \overset{\textbf{def}}{=} \mathsf{a}_\ell(t) - \alpha\big[\mathcal{V}(X) + V - v_\ell(t)\big] = \mathsf{a}_\ell(t) - \alpha\big[\mathcal{V}(X) - v\big], \qquad v = v_\ell(t) - V. \tag{3.4}$$

Indeed $\dot{v} = \alpha(\mathcal{V}(X) - v) + \beta(v_\ell - v)/\tau^2$ and $\dot{V} = \mathsf{a}_\ell - \dot{v}$. We write $\xi = (X, V)$ and $\mathsf{F}(t, \xi) \overset{\textbf{def}}{=} (V, \mathsf{f}(t, \xi))$. The follower speed is always $v = v_\ell(t) - V$.

Sets.. We use

$$\Omega \overset{\textbf{def}}{=} \{(t, X, V) :\ t \in [0, \boldsymbol{T}_h),\ X > 0,\ V \in \mathbb{R}\}, \tag{3.5}$$

$$\mathcal{P} \overset{\textbf{def}}{=} \{(t, X, V) \in \Omega :\ 0 \le v_\ell(t) - V \le v_{\max}\} \qquad \text{(physical states)}, \tag{3.6}$$

$$\mathcal{P}^+ \overset{\textbf{def}}{=} \{(t, X, V) \in \Omega :\ -1 < v_\ell(t) - V < v_{\max} + 1\} \qquad \text{(open slab, } \mathcal{P} \subset \mathcal{P}^+\text{)}. \tag{3.7}$$

Ω and $\mathcal{P}^+$ are relatively open in $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$ (v_ℓ is continuous).

Extension of the leader data.. For the prediction flow only, we extend a_ℓ by 0 on $(\boldsymbol{T}_h, \boldsymbol{T}_h+1]$, so that $v_\ell(s) = v_\ell(\boldsymbol{T}_h)$ there. This is a technical device used in lemma 3.11 to treat predicted contacts exactly at $s = \boldsymbol{T}_h$; it does not affect the closed loop.

3.2. The prediction flow. For a datum $z = (\bar{t}, \bar{X}, \bar{V}) \in [0, \boldsymbol{T}_h) \times \mathbb{R}^2$, the predicted (OVM) relative motion $(\Xi, \Lambda)(s; z)$, $s \in [0, \boldsymbol{T}_h + 1]$, solves

$$\partial_s \Xi = \Lambda, \qquad \partial_s \Lambda = \mathsf{g}(s, \Xi, \Lambda) = \mathsf{a}_\ell(s) - \alpha\big[\mathcal{V}(\Xi) + \Lambda - v_\ell(s)\big], \qquad (\Xi, \Lambda)(\bar{t}; z) = (\bar{X}, \bar{V}). \tag{3.8}$$

It uses the same leader data as the closed loop and omits the TTC term. We write $\mathsf{G}(s, Z) \overset{\text{def}}{=} (\Lambda, \mathsf{g}(s, Z))$ for $Z = (\Xi, \Lambda)$, and

$$\zeta(s; z) \overset{\text{def}}{=} v_\ell(s) - \Lambda(s; z) \qquad \text{(predicted follower speed)}, \qquad \bar{\zeta} \overset{\text{def}}{=} v_\ell(\bar{t}) - \bar{V}.$$

Lemma 3.3 (Flow).

(a) *For every z, (3.8) has a unique absolutely continuous solution on $[0, \boldsymbol{T}_h+1]$. Moreover $\Xi(\cdot; z) \in C^1$ with $\partial_s \Xi = \Lambda$.*

(b) *For every bounded $B \subset [0, \boldsymbol{T}_h) \times \mathbb{R}^2$ there are constants $m_B, L_B < \infty$ such that, for all $z, z_1, z_2 \in B$ and $s, s_1, s_2 \in [0, \boldsymbol{T}_h + 1]$,*

$$|(\Xi, \Lambda)(s_1; z) - (\Xi, \Lambda)(s_2; z)| \le m_B |s_1 - s_2|, \tag{3.9}$$

$$|(\Xi, \Lambda)(s; z_1) - (\Xi, \Lambda)(s; z_2)| \le L_B |z_1 - z_2|. \tag{3.10}$$

In particular $(s, z) \mapsto (\Xi, \Lambda)(s; z)$ is continuous.

Proof. (a) G is measurable in s, and for all s and Z_1, Z_2,

$$|\mathsf{G}(s, Z_1) - \mathsf{G}(s, Z_2)| \le |\Lambda_1 - \Lambda_2| + \alpha|\mathcal{V}(\Xi_1) - \mathcal{V}(\Xi_2)| + \alpha|\Lambda_1 - \Lambda_2| \le L_{\mathsf{G}} |Z_1 - Z_2|, \qquad L_{\mathsf{G}} \overset{\text{def}}{=} 1 + 2\alpha,$$

using $\mathcal{V}' \le 1$. By (3.1) and $0 \le v_\ell \le v_{\max}$, $|\mathsf{g}(s, Z)| \le \bar{a} + \alpha|\Lambda|$, so $|\mathsf{G}(s, Z)| \le (1 + \alpha)|Z| + \bar{a}$. Carathéodory's theorem with a globally Lipschitz right-hand side gives a unique global solution, forward and backward from $\bar{t}$ [24, Ch. I, §5]. Since Λ is continuous and $\Xi(s) = \bar{X} + \int_{\bar{t}}^s \Lambda$, Ξ is C^1.

(b) Let $\varrho_B \overset{\text{def}}{=} \sup_{z \in B} |(\bar{X}, \bar{V})|$. Grönwall's inequality (forward and backward) gives $|Z(s; z)| \le R_B \overset{\text{def}}{=} (\varrho_B + \bar{a}(\boldsymbol{T}_h + 1)) e^{(1+\alpha)(\boldsymbol{T}_h+1)}$, hence (3.9) with $m_B \overset{\text{def}}{=} (1 + \alpha) R_B + \bar{a}$.

For (3.10), let $z_i = (\bar{t}_i, \bar{Z}_i)$ and assume $\bar{t}_1 \le \bar{t}_2$. By (3.9), $|Z(\bar{t}_2; z_1) - \bar{Z}_1| \le m_B |\bar{t}_2 - \bar{t}_1|$. The functions $Z(\cdot; z_1)$ and $Z(\cdot; z_2)$ solve the same ODE with values $Z(\bar{t}_2; z_1)$ and $\bar{Z}_2$ at time $\bar{t}_2$, so Grönwall on $[0, \boldsymbol{T}_h + 1]$ gives

$$|Z(s; z_1) - Z(s; z_2)| \le e^{L_{\mathsf{G}}(\boldsymbol{T}_h+1)} |Z(\bar{t}_2; z_1) - \bar{Z}_2| \le e^{L_{\mathsf{G}}(\boldsymbol{T}_h+1)} (|\bar{Z}_1 - \bar{Z}_2| + m_B |\bar{t}_1 - \bar{t}_2|),$$

which is (3.10) with $L_B \overset{\text{def}}{=} e^{L_{\mathsf{G}}(\boldsymbol{T}_h+1)}(1 + m_B)$. ∎

Remark 3.4. *The initial-time estimate is obtained by a "startup gap" plus initial-data stability. It does* not *use any time-shift invariance, which the system lacks because a_ℓ and v_ℓ depend on time.*

Lemma 3.5 (One-sided bounds on the prediction). *Let $z = (\bar{t}, \bar{X}, \bar{V})$ and $\hat{v} \overset{\text{def}}{=} \max\{\bar{\zeta}, v_{\max}\}$. For $s \in [\bar{t}, \boldsymbol{T}_h + 1]$:*

(a) *$\zeta(s; z) \le \hat{v}$; consequently $\Lambda(s; z) \ge -\hat{v}$.*

(b) *If $\bar{\zeta} < v_{\max} + 1$, then $\partial_s \Lambda(s; z) \le \bar{a}$ for a.e. s.*

Proof. From (3.8) and $\dot{v}_\ell = \mathsf{a}_\ell$ a.e., $\partial_s \zeta = \mathsf{a}_\ell - \partial_s \Lambda = \alpha(\mathcal{V}(\Xi) - \zeta)$ a.e. The right-hand side is continuous in s, so $\zeta \in C^1$.

(a) Let $u \overset{\text{def}}{=} (\zeta - \hat{v})^+$. Then u is absolutely continuous with $u' = \mathbf{1}_{\{\zeta > \hat{v}\}} \partial_s \zeta$ a.e. On $\{\zeta > \hat{v}\}$ we have $\mathcal{V}(\Xi) < \overline{\mathcal{V}} \le v_{\max} \le \hat{v} < \zeta$, so $\partial_s \zeta < 0$. Hence $u' \le 0$ a.e., and $u(\bar{t}) = 0$ gives $u \equiv 0$. Then $\Lambda = v_\ell - \zeta \ge -\zeta \ge -\hat{v}$, using $v_\ell \ge 0$.

(b) $\partial_s \Lambda = \mathsf{a}_\ell - \alpha \mathcal{V}(\Xi) + \alpha \zeta \le \|\mathsf{a}_\ell\|_\infty + \alpha(1 - \tanh 2) + \alpha \hat{v}$ by (3.1) and (a). If $\bar{\zeta} < v_{\max} + 1$ then $\hat{v} < v_{\max} + 1$, and $1 - \tanh 2 < \overline{\mathcal{V}}$ gives the bound $\bar{a}$. ∎

3.3. The TTC map.

Definition 3.6 (TTC, first predicted contact, regular and grazing sets). *For $z = (\bar{t}, \bar{X}, \bar{V}) \in \Omega$:*

$$\tilde{\mathcal{T}}(z) \overset{\text{def}}{=} \inf\{s \in [\bar{t}, \boldsymbol{T}_h] : \ \Xi(s; z) = 0, \ \Lambda(s; z) < 0\}, \quad \inf \varnothing \overset{\text{def}}{=} \boldsymbol{T}_h, \quad \tau(z) \overset{\text{def}}{=} \tilde{\mathcal{T}}(z) - \bar{t}, \tag{3.11}$$

$$\sigma(z) \overset{\text{def}}{=} \inf\{s \in [\bar{t}, \boldsymbol{T}_h] : \ \Xi(s; z) = 0\}, \quad \inf \varnothing \overset{\text{def}}{=} +\infty. \tag{3.12}$$

The regular set *is* $\mathcal{R} \overset{\text{def}}{=} \{z \in \Omega : \ \sigma(z) = +\infty \ \text{ or } \ \Lambda(\sigma(z); z) < 0\}$, *and the* grazing set *is* $\Gamma \overset{\text{def}}{=} \Omega \setminus \mathcal{R}$.

Remark 3.7. *Since $\Xi(\bar{t}; z) = \bar{X} > 0$ and $\Xi > 0$ on $[\bar{t}, \sigma(z))$, and we have $\Lambda(\sigma(z); z) = \partial_s \Xi(\sigma(z); z) \le 0$ whenever $\sigma(z) < \infty$. Hence*

$$\Gamma = \{z \in \Omega : \ \sigma(z) < \infty, \ \Lambda(\sigma(z); z) = 0\} :$$

these are the states whose first *predicted contact has zero relative velocity. This includes tangential touches, where the gap reopens. It also includes zero-speed* crossings, *where the predicted gap becomes negative with $\Lambda = 0$ at the crossing point. The latter are not counted by $\tilde{\mathcal{T}}$, which only registers contacts with $\Lambda < 0$.*

Remark 3.8 (First contact versus first transversal contact). *The set $\{\tilde{\mathcal{T}} < \boldsymbol{T}_h, \ \Lambda(\tilde{\mathcal{T}}) < 0\}$ of states whose first* transversal *predicted contact occurs before the horizon is* not *contained in $\mathcal{R}$: a prediction may touch 0 tangentially at $\sigma(z)$, reopen, and cross transversally later. Such states lie in Γ, where τ is discontinuous (remark 3.12). Explicit examples with smooth, admissible leader data are easy to construct (e.g. $\alpha = 0.5$, $\boldsymbol{T}_h = 3$, a predicted gap $0.01(s-1)^2(2-s)$ realised by a suitable a_ℓ: first contact at $s = 1$ with $\Lambda = 0$, transversal crossing at $s = 2$). For this reason the domain $\mathbb{D}$ on which the regularity theory of section 4 is developed requires the* first *predicted contact to be transversal (definition 4.1); on $\mathbb{D}$ the two notions coincide.*

Lemma 3.9 (Bounds). *For $z \in \Omega$, with $\hat{v} = \max\{\bar{\zeta}, v_{\max}\}$,*

$$\min\Big\{\frac{\bar{X}}{\hat{v}}, \ \boldsymbol{T}_h - \bar{t}\Big\} \ \le \ \tau(z) \ \le \ \boldsymbol{T}_h - \bar{t}.$$

In particular $\tau > 0$ on Ω, and $\tau(z) \ge \min\{\bar{X}/v_{\max}, \boldsymbol{T}_h - \bar{t}\}$ for $z \in \mathcal{P}$.

Proof. The upper bound holds because $\tilde{\mathcal{T}} \le \boldsymbol{T}_h$. By lemma 3.5(a), $\Xi(s;z) \ge \bar{X} - \hat{v}(s-\bar{t}) > 0$ for $s < \bar{t} + \bar{X}/\hat{v}$. Hence either the set in (3.11) is empty, so that $\tau = \boldsymbol{T}_h - \bar{t}$, or all its elements are $\ge \bar{t} + \bar{X}/\hat{v}$. ∎

Lemma 3.10 (Upper semicontinuity; formula on $\mathcal{R}$). *$\tilde{\mathcal{T}}$ is upper semicontinuous on Ω; in particular $\tilde{\mathcal{T}}$ and τ are Borel. For $z \in \mathcal{R}$ we have $\tilde{\mathcal{T}}(z) = \min\{\sigma(z), \boldsymbol{T}_h\}$.*

Proof. Fix $z_0 = (\bar{t}_0, \bar{X}_0, \bar{V}_0) \in \Omega$ and $\varepsilon > 0$. If $\tilde{\mathcal{T}}(z_0) = \boldsymbol{T}_h$, then $\tilde{\mathcal{T}}(z) \le \boldsymbol{T}_h = \tilde{\mathcal{T}}(z_0)$ for every z, and there is nothing to prove.

Otherwise there is s' with $\Xi(s'; z_0) = 0$, $\Lambda(s'; z_0) < 0$ and $s' < \tilde{\mathcal{T}}(z_0) + \varepsilon$. Note $s' > \bar{t}_0$, because $\Xi(\bar{t}_0; z_0) = \bar{X}_0 > 0$. By continuity choose $h \in (0, \min\{\varepsilon, 1\})$, so that $s' + h \le \boldsymbol{T}_h + 1$, with $s' - h > \bar{t}_0$, $\Lambda(\cdot; z_0) < 0$ on $[s'-h, s'+h]$, and $\Xi(s'-h; z_0) > 0 > \Xi(s'+h; z_0)$. By lemma 3.3(b) there is $\rho > 0$ such that for $|z - z_0| < \rho$: $\bar{t} < s' - h$, $\Lambda(\cdot; z) < 0$ on $[s'-h, s'+h]$, and $\Xi(s'-h; z) > 0 > \Xi(s'+h; z)$. The intermediate value theorem gives $s_z \in (s'-h, s'+h)$ with $\Xi(s_z; z) = 0$ and $\Lambda(s_z; z) < 0$.

- If $s_z \le \boldsymbol{T}_h$, then $\tilde{\mathcal{T}}(z) \le s_z < s' + h < \tilde{\mathcal{T}}(z_0) + 2\varepsilon$.
- If $s_z > \boldsymbol{T}_h$, then $\tilde{\mathcal{T}}(z) \le \boldsymbol{T}_h < s_z < \tilde{\mathcal{T}}(z_0) + 2\varepsilon$.

Hence $\limsup_{z \to z_0} \tilde{\mathcal{T}}(z) \le \tilde{\mathcal{T}}(z_0)$. An upper semicontinuous function has open sublevel sets $\{\tilde{\mathcal{T}} < c\}$, so it is Borel.

For $z \in \mathcal{R}$: if $\sigma(z) = \infty$, Ξ has no zero in $[\bar{t}, \boldsymbol{T}_h]$ and $\tilde{\mathcal{T}} = \boldsymbol{T}_h$. If $\sigma(z) < \infty$, then $\sigma(z)$ belongs to the set in (3.11) (because $\Lambda(\sigma) < 0$) and is its smallest element, so $\tilde{\mathcal{T}} = \sigma \le \boldsymbol{T}_h$. ∎

Lemma 3.11 ($\mathcal{R}$ is open and τ is locally Lipschitz on $\mathcal{R}$). *$\mathcal{R}$ is relatively open in $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$. Every $z_0 \in \mathcal{R}$ has a neighbourhood on which τ is Lipschitz. More precisely, let B be the closed unit ball around z_0 and L_B the constant of lemma 3.3. All radii ρ below are taken ≤ 1, so that (3.10) applies. Then:*

(a) *if $\sigma(z_0) = \infty$, then $\tau(z) = \boldsymbol{T}_h - \bar{t}$ near z_0;*
(b) *if $\sigma(z_0) < \infty$ and $\Lambda(\sigma(z_0); z_0) = -2\gamma < 0$, then $\mathrm{Lip}(\tau) \le L_B/\gamma + 1$ near z_0.*

Proof. Case (a). $\Xi(\cdot; z_0)$ is continuous on $[0, \boldsymbol{T}_h + 1]$, positive at $\bar{t}_0$, and has no zero on $[\bar{t}_0, \boldsymbol{T}_h]$. Hence it is positive on $[\bar{t}_0, \boldsymbol{T}_h]$ and, by continuity, on $[\max\{\bar{t}_0 - h', 0\}, \boldsymbol{T}_h]$ for some $h' > 0$. Let $2c > 0$ be its minimum there, and let $\rho \overset{\text{def}}{=} \min\{h', c/L_B\}$. For $|z - z_0| < \rho$ we have $\bar{t} > \bar{t}_0 - h'$, and (3.10) gives $\Xi(s; z) \ge 2c - L_B\rho > 0$ on $[\bar{t}, \boldsymbol{T}_h]$. Thus $\sigma(z) = \infty$, $z \in \mathcal{R}$, and $\tau(z) = \boldsymbol{T}_h - \bar{t}$, which is 1-Lipschitz.

Case (b). Let $\sigma_0 \overset{\text{def}}{=} \sigma(z_0) \in (\bar{t}_0, \boldsymbol{T}_h]$. Choose $h \in (0, \min\{1, (\sigma_0 - \bar{t}_0)/2\})$ such that $\Lambda(s; z_0) \le -\frac{3}{2}\gamma$ for $|s - \sigma_0| \le h$. This is possible because $\Lambda(\cdot; z_0)$ is continuous, and $\sigma_0 + h \le \boldsymbol{T}_h + 1$. Then $\Xi(\sigma_0 + h; z_0) \le -\frac{3}{2}\gamma h$. Since σ_0 is the first zero, $\Xi(\cdot; z_0) > 0$ on $[\bar{t}_0, \sigma_0 - h]$, hence on $[\bar{t}_0 - h', \sigma_0 - h]$ for some $h' \in (0, h)$ (intersected with $[0, \infty)$). Let $2c > 0$ be the minimum there. Pick $\rho \in (0, h')$ with $L_B\rho < \min\{c,\ \gamma/2,\ \frac{3}{4}\gamma h\}$. For $|z - z_0| < \rho$:

(i) $\bar{t} < \bar{t}_0 + h' < \sigma_0 - h$ and $\Xi(s; z) > c$ on $[\bar{t}, \sigma_0 - h]$;
(ii) $\Lambda(s; z) \le -\gamma$ on $[\sigma_0 - h, \sigma_0 + h]$;
(iii) $\Xi(\sigma_0 + h; z) < 0$.

By (i)–(iii) and strict monotonicity on $[\sigma_0 - h, \sigma_0 + h]$, the extended prediction $\Xi(\cdot; z)$ has exactly one zero $\hat{\sigma}(z)$ in $[\bar{t}, \sigma_0 + h]$. It lies in $(\sigma_0 - h, \sigma_0 + h)$ and is transversal. If $\hat{\sigma}(z) \le \boldsymbol{T}_h$ then $\sigma(z) = \hat{\sigma}(z)$; otherwise $\sigma(z) = \infty$. In both cases $z \in \mathcal{R}$, and by lemma 3.10 $\tilde{\mathcal{T}}(z) = \min\{\hat{\sigma}(z), \boldsymbol{T}_h\}$.

For z_1, z_2 in this ball, write $\hat{\sigma}_i = \hat{\sigma}(z_i)$. Then $\Xi(\hat{\sigma}_1; z_1) = 0 = \Xi(\hat{\sigma}_2; z_2)$, and $\partial_s \Xi = \Lambda \le -\gamma$ on the interval containing $\hat{\sigma}_1, \hat{\sigma}_2$. By the mean value theorem and (3.10),

$$\gamma|\hat{\sigma}_1 - \hat{\sigma}_2| \le |\Xi(\hat{\sigma}_1; z_1) - \Xi(\hat{\sigma}_2; z_1)| = |\Xi(\hat{\sigma}_2; z_2) - \Xi(\hat{\sigma}_2; z_1)| \le L_B|z_1 - z_2|.$$

Since $r \mapsto \min\{r, \boldsymbol{T}_h\}$ is 1-Lipschitz and $\tau = \tilde{\mathcal{T}} - \bar{t}$, we get $|\tau(z_1) - \tau(z_2)| \le (L_B/\gamma + 1)|z_1 - z_2|$.

Cases (a) and (b) exhaust $\mathcal{R}$, and both produce a neighbourhood contained in $\mathcal{R}$. So $\mathcal{R}$ is open. ∎

Remark 3.12 (Discontinuity on Γ). *τ is in general discontinuous on Γ. Here is the typical picture. Let $z_0 \in \Gamma$, and suppose the predicted gap touches 0 at σ_0 with $\sigma_0 + h \le \boldsymbol{T}_h$ and reopens, so that $\Xi(\cdot; z_0) > 0$ on $(\sigma_0, \sigma_0 + h]$. Assume the touch is nondegenerate (e.g. $\operatorname{ess\,inf}(\mathsf{a}_\ell + \alpha v_\ell) > 0$ near σ_0). Then $\tau(z_0) \ge \sigma_0 + h - \bar{t}_0$.*

The state gradient $D_{(\bar{X},\bar{V})}\Xi(\sigma_0; z_0) = (\phi_1(\sigma_0), \delta_0(\sigma_0))$, the first row of the fundamental matrix of the variational equation of (3.8) along the prediction (its Wronskian is $e^{-\alpha(\sigma_0 - \bar{t}_0)} \neq 0$), is nonzero; no derivative with respect to the initial time $\bar{t}$ is used, and none exists in general at a jump of a_ℓ (remark 4.11). So there is a state direction $d = (0, d_X, d_V)$ with $D_{(\bar{X},\bar{V})}\Xi(\sigma_0; z_0) \cdot (d_X, d_V) < 0$. Along $z_0 + \varepsilon d$ the predicted gap dips below 0 near σ_0 and crosses transversally, so $\tau \approx \sigma_0 - \bar{t}_0$. Note that the direction matters: the linearised gap is a damped oscillator, and $\partial\Xi(\sigma_0)/\partial\bar{X}$ can be negative; there are examples (small α, long prediction horizon) where lowering $\bar{X}$ removes *the contact and raising it creates one.*

Hence, when $\bar{V}_0 \neq 0$, f jumps across Γ, and no Lipschitz property of the closed-loop field can hold across Γ. It holds on compact subsets of $\mathcal{R}$ (lemma 3.16). If instead $\sigma_0 = \boldsymbol{T}_h$, both one-sided values tend to $\boldsymbol{T}_h - \bar{t}_0$, and τ is continuous there.

Lemma 3.13 (TTC in the approach region). *Let $z = (\bar{t}, \bar{X}, \bar{V}) \in \mathcal{P}^+$ with*

$$\bar{V} < 0 \qquad \text{and} \qquad \bar{X} < \frac{\bar{V}^2}{2\bar{a}}.$$

Then $z \in \mathcal{R}$ and

$$\tau(z) \le \frac{2\bar{X}}{|\bar{V}|}. \tag{3.13}$$

Proof. Since $z \in \mathcal{P}^+$, $\bar{\zeta} < v_{\max} + 1$, so lemma 3.5(b) gives $\partial_s \Lambda \le \bar{a}$ a.e. Integrating twice, for $s \ge \bar{t}$,

$$\Lambda(s; z) \le \bar{V} + \bar{a}(s - \bar{t}), \qquad \Xi(s; z) \le \bar{X} + \bar{V}(s - \bar{t}) + \tfrac{\bar{a}}{2}(s - \bar{t})^2.$$

Let $s^* \overset{\text{def}}{=} 2\bar{X}/|\bar{V}|$ and $s_1 \overset{\text{def}}{=} \min\{\bar{t} + s^*, \boldsymbol{T}_h\}$. Since $\bar{a}s^* = 2\bar{a}\bar{X}/|\bar{V}| < |\bar{V}|$, we have $\Lambda(s; z) < 0$ for $s \in [\bar{t}, s_1]$.

Case $\bar{t} + s^* \ge \boldsymbol{T}_h$. Lem. 3.9 gives $\tau \le \boldsymbol{T}_h - \bar{t} \le s^*$. Moreover $\Lambda < 0$ on $[\bar{t}, \boldsymbol{T}_h]$, so any first zero in $[\bar{t}, \boldsymbol{T}_h]$ is transversal and $z \in \mathcal{R}$.

Case $\bar{t} + s^* < \boldsymbol{T}_h$. Then $\bar{t} + s^*$ lies in the domain of the prediction, and

$$\Xi(\bar{t} + s^*; z) \le \bar{X}\Big(\frac{2\bar{a}\bar{X}}{\bar{V}^2} - 1\Big) < 0.$$

So $\Xi(\cdot; z)$ changes sign on $[\bar{t}, \bar{t} + s^*]$, and $\sigma(z) \in (\bar{t}, \bar{t} + s^*)$ with $\Lambda(\sigma(z); z) < 0$. Thus $z \in \mathcal{R}$, and by lemma 3.10 $\tau = \sigma(z) - \bar{t} < s^*$. ∎

3.4. Solution concepts for the closed loop.

Definition 3.14 (Solutions). *Let $J \subset [0, \boldsymbol{T}_h)$ be an interval with left endpoint $t_\circ \in J$, and let $\xi = (X, V) : J \to \mathbb{R}^2$ be absolutely continuous on compact subintervals, with $(t, \xi(t)) \in \Omega$ for $t \in J$ and $\xi(t_\circ) = \xi_\circ$.*

(a) *ξ is a* Carathéodory solution *of (3.3) if $\dot\xi(t) = \mathsf{F}(t, \xi(t))$ for a.e. $t \in J$; equivalently, $\xi(t) = \xi_\circ + \int_{t_\circ}^t \mathsf{F}(r, \xi(r))\, dr$ for all $t \in J$.*

(b) *ξ is a* Filippov solution *if $\dot\xi(t) \in \mathfrak{F}[\mathsf{F}](t, \xi(t))$ for a.e. $t \in J$, where*

$$\mathfrak{F}[\mathsf{F}](t, \xi) \overset{def}{=} \bigcap_{\delta>0} \bigcap_{|N|=0} \overline{\mathrm{co}}\, \mathsf{F}(t, B(\xi, \delta) \setminus N),$$

$\overline{\mathrm{co}}$ denotes the closed convex hull and $|N|$ the Lebesgue measure in $\mathbb{R}^2$ [22, Ch. 2, §4]. Only δ small enough that $B(\xi, \delta) \subset \{X > 0\}$ are used, so that F is defined on the ball.

Lemma 3.15 (Two facts on Filippov sets). *Fix t and $\xi = (X, V)$.*

(i) *If $\mathsf{f}(t, \cdot)$ is continuous at ξ, then $\mathfrak{F}[\mathsf{F}](t, \xi) = \{\mathsf{F}(t, \xi)\}$.*

(ii) *Let $W \ni \xi$ be open and $h : W \to \mathbb{R}$ continuous, and suppose $\mathsf{f}(t, \eta) \geq h(\eta)$ for a.e. $\eta \in W$. Then every $w = (w_1, w_2) \in \mathfrak{F}[\mathsf{F}](t, \xi)$ satisfies $w_1 = V$ and $w_2 \geq h(\xi)$. The same holds with "$\leq$" in both places.*

Proof. (ii) Take $\delta > 0$ with $B(\xi, \delta) \subset W$, and let $\omega(\delta) \overset{\mathbf{def}}{=} \sup_{|\eta-\xi|<\delta} |h(\eta) - h(\xi)|$. Let N be the null set off which $\mathsf{f}(t, \cdot) \geq h$. The values $\mathsf{F}(t, \eta) = (\eta_2, \mathsf{f}(t, \eta))$, for $\eta \in B(\xi, \delta) \setminus N$, lie in the closed convex set $\{w : |w_1 - V| \leq \delta,\ w_2 \geq h(\xi) - \omega(\delta)\}$. So does their closed convex hull, and hence $\mathfrak{F}[\mathsf{F}](t, \xi)$. Letting $\delta \downarrow 0$ gives the claim.

(i) In the same way $\mathfrak{F}[\mathsf{F}](t, \xi)$ is contained in every closed ball around $\mathsf{F}(t, \xi)$ of radius

$$\sup_{|\eta-\xi|<\delta} |\mathsf{F}(t, \eta) - \mathsf{F}(t, \xi)| \xrightarrow[\delta\downarrow 0]{} 0.$$

It also contains $\mathsf{F}(t, \xi)$. By continuity at ξ, this point lies in the closure of $\mathsf{F}(t, B(\xi, \delta) \setminus N)$ for every $\delta > 0$ and every null set N, because $B(\xi, \delta) \setminus N$ accumulates at ξ. Hence $\mathfrak{F}[\mathsf{F}](t, \xi) = \{\mathsf{F}(t, \xi)\}$. ∎

3.5. Local well-posedness in the regular set.

Lemma 3.16 (Carathéodory conditions on $\mathcal{R}$). *Let $K \subset \mathcal{R}$ be compact. Then, there are $m_K, L_K < \infty$ such that*

$$|\mathsf{F}(t, \xi)| \leq m_K \quad \text{for a.e. } t, \qquad |\mathsf{F}(t, \xi_1) - \mathsf{F}(t, \xi_2)| \leq L_K |\xi_1 - \xi_2|,$$

whenever $(t, \xi), (t, \xi_i) \in K$. Moreover, for each ξ, $t \mapsto \mathsf{F}(t, \xi)$ is measurable on $\{t : (t, \xi) \in \mathcal{R}\}$.

Proof. *τ on K.* By lemma 3.11, τ is locally Lipschitz, hence continuous, on $\mathcal{R}$. By lemma 3.9 it is positive, so $\tau_K \overset{\mathbf{def}}{=} \min_K \tau > 0$.

τ is also Lipschitz on K. If not, there would be $p_n, q_n \in K$ with $|\tau(p_n) - \tau(q_n)| > n|p_n - q_n|$. Since $0 < \tau \leq \boldsymbol{T}_h$, this forces $|p_n - q_n| < \boldsymbol{T}_h/n$. Along a subsequence $p_n, q_n \to p \in K$, which contradicts the Lipschitz bound of τ on a neighbourhood of p. Let L_K^τ be the constant, and $R_K \overset{\mathbf{def}}{=} \max_K |V|$.

Bounds on F. By (3.1), a.e. on K, $|\mathsf{g}| \le \|\mathsf{a}_\ell\|_\infty + \alpha(\overline{\mathcal{V}} + R_K + v_{\max})$, and $|\beta V/\tau^2| \le \beta R_K/\tau_K^2$. For two points of K with the same t, $\mathsf{a}_\ell(t)$ and $v_\ell(t)$ cancel, so

$$|\mathsf{f}(t,\xi_1) - \mathsf{f}(t,\xi_2)| \le \alpha|X_1 - X_2| + \alpha|V_1 - V_2| + \beta\Big(\frac{|V_1 - V_2|}{\tau_K^2} + R_K\,\frac{|\tau_2^2 - \tau_1^2|}{\tau_K^4}\Big),$$

with $|\tau_2^2 - \tau_1^2| \le 2\boldsymbol{T}_h L_K^\tau |\xi_1 - \xi_2|$. This gives L_K.

Measurability. For fixed ξ, the set $\{t : (t,\xi) \in \mathcal{R}\}$ is relatively open and $\tau(\cdot,\xi)$ is continuous on it. Also a_ℓ is measurable and v_ℓ continuous. ∎

Theorem 3.17 (Local well-posedness in $\mathcal{R}$). *Let $(t_\circ,\xi_\circ) \in \mathcal{R}$. There is a unique maximal Carathéodory solution $\xi : [t_\circ, T_\mathcal{R}) \to \mathbb{R}^2$ with graph in $\mathcal{R}$. Its graph eventually leaves every compact subset of $\mathcal{R}$: for every compact $K \subset \mathcal{R}$ there is $t_K < T_\mathcal{R}$ with $(t,\xi(t)) \notin K$ for $t \in (t_K, T_\mathcal{R})$. Moreover, every Filippov solution $\tilde\xi : J \to \mathbb{R}^2$ starting at $(t_\circ,\xi_\circ)$ coincides with ξ on the common interval $J \cap [t_\circ, T_\mathcal{R})$.*

Proof. $\mathcal{R}$ is only relatively open in $[0,\boldsymbol{T}_h)\times\mathbb{R}^2$, because it may contain points with $t = 0$. To apply the classical theorems on an open set, extend a_ℓ by 0 and v_ℓ by $v_\ell(0)$ to $t \in (-1,0)$. The prediction flow and τ are then defined for $\bar t \in (-1,\boldsymbol{T}_h)$, all the lemmas above hold verbatim, and solutions starting at $t_\circ \ge 0$ are unaffected. The same remark applies to theorem 3.23.

By lemma 3.16, F satisfies the Carathéodory conditions with local Lipschitz bounds on the open set $\mathcal{R}$. The classical existence, uniqueness and extension theorem [24, Ch. I, §5] gives the first two statements.

For the last statement, let $\tilde\xi : J \to \mathbb{R}^2$ be a Filippov solution starting at $(t_\circ,\xi_\circ)$. Let $S \overset{\text{def}}{=} \sup\{s \in J : (t,\tilde\xi(t)) \in \mathcal{R} \text{ for } t \in [t_\circ, s]\}$, which is $> t_\circ$ because $\mathcal{R}$ is open. On $[t_\circ, S)$, lemma 3.11 and lemma 3.15(i) give $\mathfrak{F}[\mathsf{F}] = \{\mathsf{F}\}$ along $\tilde\xi$, so $\tilde\xi$ is a Carathéodory solution in $\mathcal{R}$. By uniqueness, $\tilde\xi = \xi$ on $[t_\circ, \min\{S, T_\mathcal{R}\})$.

Suppose $S < T_\mathcal{R}$ and $S \in J$. By continuity $\tilde\xi(S) = \xi(S)$, and $(S,\xi(S)) \in \mathcal{R}$. Openness of $\mathcal{R}$ and continuity of $\tilde\xi$ then keep the graph of $\tilde\xi$ in $\mathcal{R}$ slightly beyond S, contradicting the definition of S. Hence either $S \ge T_\mathcal{R}$, or $S = \sup J \notin J$. In both cases $\tilde\xi = \xi$ on $J \cap [t_\circ, T_\mathcal{R})$. ∎

3.6. Invariance of the velocity box.

Lemma 3.18 (Maximum principle). *Let ξ be a Carathéodory or Filippov solution on J with $(t_\circ,\xi_\circ) \in \mathcal{P}$. Then $(t,\xi(t)) \in \mathcal{P}$ for all $t \in J$; that is, $0 \le v(t) \le v_{\max}$. Consequently $|V| \le v_{\max}$, X is $v_{\max}$-Lipschitz, and X has a finite left limit at the right endpoint of J.*

Proof. The function $v = v_\ell - V$ is absolutely continuous, with $\dot v = \mathsf{a}_\ell - \dot V$ a.e.

Upper bound. Fix t and let $W_+ \overset{\text{def}}{=} \{\eta = (X,V) : X > 0,\ v_\ell(t) - V > v_{\max}\}$, which is open. For $\eta \in W_+$, with $v = v_\ell(t) - V$, we have $\mathsf{a}_\ell(t) - \mathsf{f}(t,\eta) = \alpha(\mathcal{V}(X) - v) + \beta(v_\ell(t) - v)/\tau^2$. The second term is ≤ 0, because $v_\ell(t) \le v_{\max} < v$. Hence

$$\mathsf{f}(t,\eta) \;\ge\; \mathsf{a}_\ell(t) - \alpha(\overline{\mathcal{V}} - (v_\ell(t) - V)) \overset{\text{def}}{=} h_+(\eta),$$

which is continuous in η. For a Carathéodory solution this holds pointwise. For a Filippov solution, lemma 3.15(ii) gives $\dot V \ge h_+(\xi(t))$ at a.e. t with $\xi(t) \in W_+$. In both cases $\dot v \le \alpha(\overline{\mathcal{V}} - v) < 0$ at a.e. t with $v(t) > v_{\max}$, using $\overline{\mathcal{V}} \le v_{\max}$. With $u \overset{\text{def}}{=} (v - v_{\max})^+$ we have $u' = \mathbf{1}_{\{v > v_{\max}\}}\dot v \le 0$ a.e. and $u(t_\circ) = 0$, so $u \equiv 0$.

Lower bound. On $W_- \overset{\text{def}}{=} \{X > 0,\ v_\ell(t) - V < 0\}$, we have $\mathsf{a}_\ell(t) - \mathsf{f} = \alpha(\mathcal{V}(X) - v) + \beta(v_\ell(t) - v)/\tau^2 \geq \alpha(\mathcal{V}(X) - v)$, because $v_\ell(t) \geq 0 > v$. Hence $\mathsf{f} \leq \mathsf{a}_\ell(t) - \alpha(\mathcal{V}(X) - v)$, a continuous bound. As above, $\dot{v} \geq \alpha(\mathcal{V}(X) - v) > 0$ at a.e. t with $v(t) < 0$, by (3.1) and $X > 0$. Then $(-v)^+$ is nonincreasing and vanishes at $t_\circ$.

The remaining statements follow from $V = v_\ell - v \in [v_\ell - v_{\max}, v_\ell] \subset [-v_{\max}, v_{\max}]$ and $\dot{X} = V$. ∎

3.7. Impact-freeness. Throughout this subsection $\xi = (X, V)$ is a Carathéodory or Filippov solution on $[t_\circ, t^*)$ with $t^* \leq \boldsymbol{T}_h$ and $(t_\circ, \xi_\circ) \in \mathcal{P}$. By lemma 3.18 its graph stays in $\mathcal{P} \subset \mathcal{P}^+$, and $X(t^*-)$ exists. We use the open *approach region*

$$\mathcal{A} \overset{\text{def}}{=} \left\{(t, X, V) \in \mathcal{P}^+ :\ V < 0,\ X < \frac{V^2}{2\bar{a}}\right\} \subset \mathcal{R}, \tag{3.14}$$

where the inclusion is lemma 3.13.

Lemma 3.19 (One-sided acceleration bounds). *At a.e. $t \in [t_\circ, t^*)$:*

(a) *if $V(t) < 0$, then $\dot{V}(t) \geq -\bar{a}$;*
(b) *if $V(t) > 0$, then $\dot{V}(t) \leq \bar{a}$;*
(c) *if $(t, \xi(t)) \in \mathcal{A}$, then $\dot{V}(t) \geq -\bar{a} + \beta \dfrac{|V(t)|^3}{4X(t)^2}$.*

Consequently $t \mapsto V(t) + \bar{a}t$ is nondecreasing on every interval where $V < 0$, and $t \mapsto V(t) - \bar{a}t$ is nonincreasing on every interval where $V > 0$.

Proof. On $\mathcal{P}^+$ we have $v = v_\ell(t) - V \in (-1, v_{\max} + 1)$, so by (3.1)

$$-\bar{a} \ \leq\ \mathsf{g}(t, X, V) = \mathsf{a}_\ell(t) - \alpha\mathcal{V}(X) + \alpha v \ \leq\ \bar{a} \qquad \text{at every point of } \mathcal{P}^+. \tag{3.15}$$

(a) On the open set $\{\eta : (t, \eta) \in \mathcal{P}^+,\ \eta_2 < 0\}$ the TTC term is $-\beta\eta_2/\tau^2 > 0$, so $\mathsf{f}(t, \eta) \geq \mathsf{g}(t, \eta) \geq -\bar{a}$ at every point. For a Carathéodory solution this gives $\dot{V} \geq -\bar{a}$ wherever $V < 0$. For a Filippov solution, lemma 3.15(ii) with the constant function $h \equiv -\bar{a}$ gives the same. (b) is symmetric: $\mathsf{f} \leq \mathsf{g} \leq \bar{a}$ where $\eta_2 > 0$. (c) On $\mathcal{A}$, lemma 3.13 gives $1/\tau^2 \geq V^2/(4X^2)$, hence

$$\mathsf{f}(t, X, V) \ \geq\ -\bar{a} + \beta \frac{|V|^3}{4X^2} \ \overset{\text{def}}{=}\ k(X, V) \qquad \text{at every point of } \mathcal{A}; \tag{3.16}$$

k is continuous on $\{V < 0, X > 0\}$, and lemma 3.15(ii) applies on the open set $\{\eta : (t, \eta) \in \mathcal{A}\}$. The monotonicity statements follow because V is absolutely continuous. ∎

Theorem 3.20 (Every contact has zero closing speed). *If $X(t^*-) = 0$, then $V(t) \to 0$ as $t \uparrow t^*$.*

Proof. Step 1: $\liminf_{t \uparrow t^*} V(t) \geq 0$. Exactly one of the following holds.

Case A: $V < 0$ on some $(t^* - \delta, t^*)$. By lemma 3.19, $V + \bar{a}t$ is nondecreasing there. Since $|V| \leq v_{\max}$, the left limit $V^* \overset{\text{def}}{=} V(t^*-)$ exists. Suppose $V^* < 0$. Then there is $t_1 < t^*$ with $V \leq V^*/2$ on $[t_1, t^*)$ and, since $X \to 0$, with $X < V^{*2}/(8\bar{a}) \leq V^2/(2\bar{a})$ there. So $(t, \xi(t)) \in \mathcal{A}$ on $[t_1, t^*)$, and lemma 3.19(c) gives, a.e.,

$$\dot{V} \ \geq\ -\bar{a} + \beta \frac{|V^*|^3}{32\, X^2}.$$

Because $X(t^*-)=0$ and X is $v_{\max}$-Lipschitz (lemma 3.18), $X(s)\le v_{\max}(t^*-s)$, so

$$V(t)-V(t_1)\ \ge\ -\bar a(t-t_1)+\frac{\beta|V^*|^3}{32\,{v_{\max}}^2}\Big(\frac{1}{t^*-t}-\frac{1}{t^*-t_1}\Big)\ \longrightarrow\ +\infty\qquad (t\uparrow t^*),$$

contradicting $V\le v_{\max}$. Hence $V^*=0$.

Case B: there are $t_n\uparrow t^*$ with $V(t_n)\ge 0$. Fix $\varepsilon>0$ and n with $\bar a(t^*-t_n)<\varepsilon$. Let $t\in(t_n,t^*)$ with $V(t)<0$, and set $s_2\overset{\text{def}}{=}\sup\{s\in[t_n,t]:V(s)\ge 0\}$. By continuity $V(s_2)=0$ and $V<0$ on $(s_2,t]$. Lem. 3.19 gives $V(t)+\bar a t\ge V(s_2)+\bar a s_2$, i.e.

$$V(t)\ \ge\ -\bar a(t-s_2)\ \ge\ -\bar a(t^*-t_n)\ >\ -\varepsilon.$$

So $V>-\varepsilon$ on (t_n,t^*), and $\liminf V\ge 0$.

Step 2: $\limsup_{t\uparrow t^*}V(t)\le 0$. Let $t<t^*$ with $X(t)<X(t_\circ)$ and $c\overset{\text{def}}{=}V(t)>0$. V cannot be positive on all of $[t_\circ,t]$, since that would give $X(t)>X(t_\circ)$. So $t'\overset{\text{def}}{=}\sup\{s\le t:V(s)\le 0\}$ satisfies $V(t')=0$ and $V>0$ on $(t',t]$. By lemma 3.19(b), $V(s)\ge c-\bar a(t-s)$ on $[t',t]$; at $s=t'$ this gives $t-t'\ge c/\bar a$. Hence

$$X(t)\ \ge\ X(t')+\int_{t-c/\bar a}^{t}(c-\bar a(t-s))\,ds\ \ge\ \frac{c^2}{2\bar a},\qquad \text{i.e.}\qquad V(t)\le\sqrt{2\bar a X(t)}. \tag{3.17}$$

Since $X(t)\to 0$, $\limsup V\le 0$. Together with Step 1, $V\to 0$. ∎

The following quantitative version is not needed for the safety claim, but it describes the approach to contact precisely.

Proposition 3.21 (Parabolic barrier near contact). *If $X(t^*-)=0$, there is $t_4<t^*$ such that*

$$|V(t)|\ \le\ \sqrt{2\bar a\,X(t)}\qquad \textit{for all } t\in[t_4,t^*). \tag{3.18}$$

Moreover, the trajectory cannot enter $\mathcal{A}$ through a point with $X\le\beta^2\bar a/8$: if $(t_1,\xi(t_1))\notin\mathcal{A}$ and $X(t_1)\le\beta^2\bar a/8$, there is no $t>t_1$ with $(s,\xi(s))\in\mathcal{A}$ for all $s\in(t_1,t]$. In particular, on any interval throughout which $X<\beta^2\bar a/8$, a trajectory that is outside $\mathcal{A}$ at some time stays outside $\mathcal{A}$ for the rest of that interval; this is the form (3.24) used below.

The proof uses the functions

$$\Phi(x)\overset{\text{def}}{=}\frac{\beta}{4}\frac{1}{x}-\frac{1}{\sqrt{2\bar a}}\frac{1}{\sqrt{x}},\qquad \psi(x)\overset{\text{def}}{=}\Phi(x)-\frac{1}{\sqrt{2\bar a x}}=\frac{\beta}{4}\frac{1}{x}-\frac{2}{\sqrt{2\bar a}}\frac{1}{\sqrt{x}},\qquad x>0. \tag{3.19}$$

Proof. Step 1: a differential inequality in $\mathcal{A}$. Let $I\subset[t_\circ,t^*)$ be an interval on which $(t,\xi(t))\in\mathcal{A}$. By lemma 3.19(c), $\dot V\ge k(X,V)$ a.e. on I. On I, $V<0$, so $w\overset{\text{def}}{=}1/|V|=-1/V$ is absolutely continuous on compact subintervals, with

$$\dot w=\frac{\dot V}{V^2}\ \ge\ -\frac{\bar a}{V^2}+\frac{\beta}{4}\frac{|V|}{X^2}\qquad \text{a.e. on } I.$$

On $\mathcal{A}$, $V^2 > 2\bar{a}X$, so $\bar{a}/V^2 < 1/(2X)$. Since $\dot{X} = V = -|V|$, we have $|V|/X^2 = \frac{d}{dt}(1/X)$. Therefore

$$\dot{w} \;\geq\; \frac{\beta}{4}\frac{d}{dt}\Big(\frac{1}{X}\Big) - \frac{1}{2X} \qquad \text{a.e. on } I. \tag{3.20}$$

Step 2: integrated form. Let $[t_1, t_2] \subset [t_\circ, t^*)$ with $(t, \xi(t)) \in \mathcal{A}$ for $t \in (t_1, t_2]$, and $V(t_1) < 0$. We claim

$$w(t_2) - w(t_1) \;\geq\; \Phi(X(t_2)) - \Phi(X(t_1)). \tag{3.21}$$

On $[t_1, t_2]$, V is continuous and nonzero, so w is absolutely continuous there, and $X \geq X(t_2) > 0$. Also $|V| > \sqrt{2\bar{a}X}$ on $(t_1, t_2]$. By the chain rule for $x \mapsto x^{-1/2}$ composed with X,

$$\int_{t_1}^{t_2} \frac{dt}{2X(t)} = \int_{t_1}^{t_2} \frac{|\dot{X}(t)|}{2X(t)|V(t)|}\,dt \leq \int_{t_1}^{t_2} \frac{-\dot{X}(t)}{2\sqrt{2\bar{a}}\,X(t)^{3/2}}\,dt = \frac{1}{\sqrt{2\bar{a}}}\Big(X(t_2)^{-1/2} - X(t_1)^{-1/2}\Big).$$

Integrating (3.20) over $[t_1, t_2]$ and using this bound gives (3.21).

Step 3: properties of ψ. We have $\psi'(x) = -\frac{\beta}{4}x^{-2} + \frac{1}{\sqrt{2\bar{a}}}x^{-3/2}$, which is < 0 iff $x < x_1 \overset{\text{def}}{=} \beta^2\bar{a}/8$. So ψ is strictly decreasing on $(0, x_1]$. Moreover

$$\psi(x) \;\longrightarrow\; +\infty \qquad (x \downarrow 0). \tag{3.22}$$

Since $w < 1/\sqrt{2\bar{a}X}$ in $\mathcal{A}$, (3.21) implies, under the hypotheses of Step 2,

$$\psi(X(t_2)) \;<\; \Phi(X(t_1)) - w(t_1). \tag{3.23}$$

Step 4: the solution cannot stay in $\mathcal{A}$ up to t^*. Suppose $(t, \xi(t)) \in \mathcal{A}$ for all $t \in [t_1, t^*)$. Applying (3.23) on $[t_1, t]$ for every $t \in (t_1, t^*)$ gives $\psi(X(t)) < \Phi(X(t_1)) - w(t_1)$. The right-hand side is a fixed number, while the left-hand side tends to $+\infty$ as $t \uparrow t^*$, by (3.22) and $X(t) \to 0$. This is a contradiction.

Step 5: once close to contact, no re-entry into $\mathcal{A}$. Choose $t_3 < t^*$ such that $X(t) < \min\{x_1, X(t_\circ)\}$ for all $t \in [t_3, t^*)$. By Step 4 there is $t_4 \in [t_3, t^*)$ with $(t_4, \xi(t_4)) \notin \mathcal{A}$. We claim

$$(t, \xi(t)) \notin \mathcal{A} \qquad \text{for all } t \in [t_4, t^*). \tag{3.24}$$

Suppose not, and pick $t \in (t_4, t^*)$ with $(t, \xi(t)) \in \mathcal{A}$. Let $t_1 \overset{\text{def}}{=} \sup\{s \in [t_4, t] : (s, \xi(s)) \notin \mathcal{A}\}$. This supremum is attained, because the set of such s is the preimage of a closed set under a continuous map. Then $t_1 < t$ and $(s, \xi(s)) \in \mathcal{A}$ for $s \in (t_1, t]$.

By continuity, $V(t_1) \leq 0$ and $X(t_1) \leq V(t_1)^2/(2\bar{a})$. If $V(t_1) = 0$ this forces $X(t_1) \leq 0$, which is impossible. So $V(t_1) < 0$. Since $(t_1, \xi(t_1)) \in \mathcal{P} \setminus \mathcal{A} \subset \mathcal{P}^+ \setminus \mathcal{A}$, we get $X(t_1) = V(t_1)^2/(2\bar{a})$, that is, $w(t_1) = 1/\sqrt{2\bar{a}X(t_1)}$, and the right-hand side of (3.23) equals $\psi(X(t_1))$. So (3.23) on $[t_1, t]$ reads $\psi(X(t)) < \psi(X(t_1))$.

But X is strictly decreasing on $(t_1, t]$, because $V < 0$ there, so $X(t) < X(t_1) < x_1$. Step 3 then gives $\psi(X(t)) > \psi(X(t_1))$, a contradiction. This proves (3.24). The argument used only $X(t_1) \leq x_1$ at the entry time t_1 (so that ψ is strictly decreasing on $(0, X(t_1)]$) and $V < 0$

inside $\mathcal{A}$; it therefore proves the entry statement of the proposition for an arbitrary entry point with $X(t_1) \le x_1$, not only on $[t_3, t^*)$. It does *not* show that a trajectory which leaves $\mathcal{A}$ with $X < x_1$, then opens the gap beyond x_1, cannot re-enter $\mathcal{A}$ later through a point with $X > x_1$.

Step 6: the barrier. Let $t \in [t_4, t^*)$. If $V(t) < 0$, then by (3.24) and $(t, \xi(t)) \in \mathcal{P} \subset \mathcal{P}^+$, $X(t) \ge V(t)^2/(2\bar{a})$, i.e. $V(t) \ge -\sqrt{2\bar{a}X(t)}$. If $V(t) > 0$, (3.17) applies because $X(t) < X(t_\circ)$. This is (3.18). ∎

Remark 3.22 (Interpretation). *The parabola $X = V^2/(2\bar{a})$ is the stopping-distance curve at maximal deceleration $\bar{a}$. Once $X \le \beta^2\bar{a}/8$, the vector field points out of $\mathcal{A}$ across it: there* (3.16) *gives $\dot{V} \ge -\bar{a} + \beta(2\bar{a})^{3/2}/(4\sqrt{X}) \ge \bar{a}$. On any time interval spent in $\mathcal{A}$, a Filippov solution is a Carathéodory solution, by $\mathcal{A} \subset \mathcal{R}$ and lemma 3.15(i). We do not claim uniqueness near contact from this alone: by* (3.24) *the solution is outside $\mathcal{A}$ just before contact, and there it may meet Γ. Uniqueness is addressed in subsection 3.9.*

3.8. Global existence and behaviour at the horizon.

Theorem 3.23 (Filippov existence and continuation). *Let $(t_\circ, \xi_\circ) \in \mathcal{P}$. A* non-extendable *Filippov solution is one that is not the restriction of a Filippov solution defined on a strictly larger interval. Non-extendable Filippov solutions with initial value $\xi_\circ$ at $t_\circ$ exist, and every such solution ξ, on its maximal interval $[t_\circ, T^*)$ with $T^* = T^*_\xi \in (t_\circ, \boldsymbol{T}_h]$, has the following properties.*

(a) *Either $T^* = \boldsymbol{T}_h$, or $X(T^*-) = 0$. In the latter case, $V(t) \to 0$ as $t \uparrow T^*$ (theorem 3.20), and the barrier* (3.18) *holds near T^* (proposition 3.21).*

(b) *$(t, \xi(t)) \in \mathcal{P}$ for all t.*

(c) *If $(t_\circ, \xi_\circ) \in \mathcal{R}$, then every Filippov solution coincides with the unique Carathéodory solution of theorem 3.17 on their common interval of existence. In particular $T^* \ge T_{\mathcal{R}}$, and every non-extendable Filippov solution starting at $(t_\circ, \xi_\circ)$ extends the Carathéodory one.*

Proof. Measurability and local boundedness of F *on* Ω. τ is Borel on Ω (lemma 3.10), a_ℓ is Lebesgue measurable and v_ℓ is continuous, so F is Lebesgue measurable on Ω. Let $K \subset \Omega$ be compact. Then on K we have $X \ge X_K > 0$, $t \le \boldsymbol{T}_h - \delta_K$ and $|V| \le R_K$. Hence $\hat{v} \le v_{\max} + R_K$, and lemma 3.9 gives $\tau \ge \tau_K \overset{\text{def}}{=} \min\{X_K/(v_{\max} + R_K), \delta_K\} > 0$. Therefore $|\mathsf{F}| \le R_K + \|\mathsf{a}_\ell\|_\infty + \alpha(\overline{\mathcal{V}} + R_K + v_{\max}) + \beta R_K/\tau_K^2$ a.e. on K.

Local existence. We use Filippov's existence theorem for his definition (a), which convexifies in the state variable only. It requires F to be Lebesgue measurable on an open set of (t, ξ) and locally essentially bounded, and gives a Filippov solution through every point [21], [22, Ch. 2, §7] (the version for a right-hand side that is merely measurable in t; the set-valued formulation under "basic conditions" requires joint upper semicontinuity, which fails here since a_ℓ is only L^∞). The open-set requirement is met by the extension to $t \in (-1, 0)$ described in the proof of theorem 3.17.

Continuation. By Zorn's lemma, applied to Filippov solutions through $(t_\circ, \xi_\circ)$ ordered by extension, every Filippov solution is the restriction of a non-extendable one; in particular non-extendable solutions exist. Now let ξ be *any* non-extendable solution, on $[t_\circ, T^*)$. Suppose its graph does not eventually leave some compact $K \subset \Omega$. Then there are $t_k \uparrow T^*$ with $(t_k, \xi(t_k)) \in K$, and we may assume $T^* < \boldsymbol{T}_h$.

Let $K' \supset K$ be a compact neighbourhood with $|\mathsf{F}| \le m$ a.e. on K', and let $d > 0$ be the distance from K to the complement of K'. Then $|\dot\xi| \le m$ as long as the graph stays in K'. So for k large, with $T^* - t_k < d/(1+m)$, the graph remains in K' on $[t_k, T^*)$, and $\xi(T^*-) \overset{\text{def}}{=} \lim_{t\uparrow T^*} \xi(t)$ exists by the Cauchy criterion. Moreover $(T^*, \xi(T^*-)) \in K' \subset \Omega$. Restarting the local existence theorem at this point extends the solution, contradicting non-extendability. Hence the graph eventually leaves every compact subset of Ω.

(b) is lemma 3.18.

(a). Suppose $T^* < \boldsymbol{T}_h$ and $X(T^*-) > 0$; the limit exists by lemma 3.18. Then $X_{\min} \overset{\text{def}}{=} \inf_{[t_\circ, T^*)} X > 0$, and the graph lies in the compact set $[t_\circ, T^*] \times [X_{\min}, X_\circ + v_{\max}\boldsymbol{T}_h] \times [-v_{\max}, v_{\max}] \subset \Omega$, a contradiction. The statements on V are theorem 3.20 and proposition 3.21.

(c) is the last assertion of theorem 3.17. For $T^* \ge T_{\mathcal{R}}$: if $T^* < T_{\mathcal{R}}$, the Carathéodory solution, which is also a Filippov solution, would extend ξ beyond T^*. ∎

Proposition 3.24 (Behaviour at the horizon). *In the situation of theorem 3.23, if $T^* = \boldsymbol{T}_h$, then $V(t) \to 0$ as $t \uparrow \boldsymbol{T}_h$. Hence ξ extends continuously to $[t_\circ, \boldsymbol{T}_h]$ with $V(\boldsymbol{T}_h) = 0$.*

Proof. By lemma 3.9, $\tau \le \boldsymbol{T}_h - t$. On $\mathcal{P}^+$ we have $|\mathsf{g}| \le \|\mathsf{a}_\ell\|_\infty + \alpha(\overline{\mathcal{V}} + v_{\max} + 1) = \bar a$, by (3.1). Hence, for every point of $\mathcal{P}^+$ with $V \ne 0$,

$$\operatorname{sgn}(V)\,\mathsf{f}(t,X,V) = \operatorname{sgn}(V)\,\mathsf{g} - \beta\frac{|V|}{\tau^2} \;\le\; \bar a - \beta\,\frac{|V|}{(\boldsymbol{T}_h - t)^2},$$

with a right-hand side continuous in (X,V) on $\{V>0\}$ and on $\{V<0\}$. By lemma 3.15(ii) (or directly, for Carathéodory solutions), and since $\frac{d}{dt}|V| = \operatorname{sgn}(V)\dot V$ a.e. on $\{V \ne 0\}$, we get

$$\frac{d}{dt}|V| \;\le\; \bar a - \beta\,\frac{|V|}{(\boldsymbol{T}_h - t)^2} \qquad \text{a.e. on } \{V \ne 0\}. \tag{3.25}$$

Fix $\eta > 0$ and $t_\eta \overset{\text{def}}{=} \boldsymbol{T}_h - \sqrt{\beta\eta/(2\bar a)}$. Let $u \overset{\text{def}}{=} (|V| - \eta)^+$. On $(t_\eta, \boldsymbol{T}_h) \cap \{|V| > \eta\}$, (3.25) gives $\frac{d}{dt}|V| \le \bar a - 2\bar a < 0$, so u is nonincreasing on $(t_\eta, \boldsymbol{T}_h)$.

If $u \ge c > 0$ near $\boldsymbol{T}_h$, then (3.25) gives $\frac{d}{dt}|V| \le \bar a - \beta(\eta + c)/(\boldsymbol{T}_h - t)^2$ there. This is not integrable up to $\boldsymbol{T}_h$, so $|V|$ would become negative, which is absurd. Hence $u \to 0$, that is, $\limsup|V| \le \eta$. Since $\eta > 0$ is arbitrary, $V \to 0$. Finally X has a limit because it is Lipschitz. ∎

Remark 3.25 (Uniqueness near the horizon). *If $X(\boldsymbol{T}_h-) > 0$, then for t close to $\boldsymbol{T}_h$ we have $\boldsymbol{T}_h - t < X(t)/v_{\max}$. By lemma 3.5(a), $\Xi(s) \ge X(t) - v_{\max}(s-t) > 0$ for $s \in [t, \boldsymbol{T}_h]$, so $\sigma = \infty$, the state lies in $\mathcal{R}$, and $\tau = \boldsymbol{T}_h - t$ exactly. So on a final interval the solution is the unique Carathéodory solution, and τ is smooth along it.*

3.9. Crossing the grazing set. Thm. 3.23 gives existence through Γ in Filippov's sense, and uniqueness only while the trajectory stays in $\mathcal{R}$. This subsection shows that at a *non-degenerate* grazing state the trajectory crosses Γ transversally, spends no time on it, and remains unique. The mechanism is structural: the prediction changes along the closed loop only through the TTC term, and the sign of that term is the same on both sides of Γ.

Three facts on the prediction flow.

Lemma 3.26 (Cocycle property). *For $z = (\bar{t}, \bar{Z})$ and $s_1 \in [0, \boldsymbol{T}_h + 1]$, $z^{s_1} \overset{\text{def}}{=} (s_1, Z(s_1; z))$. Then $Z(s; z^{s_1}) = Z(s; z)$ for all $s \in [0, \boldsymbol{T}_h + 1]$.*

Proof. Both sides solve (3.8) and take the value $Z(s_1; z)$ at $s = s_1$; the solution is unique by lemma 3.3(a), whose proof covers initial times in $[0, \boldsymbol{T}_h + 1]$ verbatim. No time-shift invariance is used. ∎

Lemma 3.27 (Difference of two predictions). *Let $z_a, z_b \in [0, \boldsymbol{T}_h) \times \mathbb{R}^2$ and $D(s) \overset{\text{def}}{=} \Xi(s; z_a) - \Xi(s; z_b)$. Then $D \in C^1[0, \boldsymbol{T}_h + 1]$, $D' = \Lambda(\cdot; z_a) - \Lambda(\cdot; z_b)$ is absolutely continuous, and* (3.26)

$$D'' + \alpha D' + \alpha\, \tilde{c}(s)\, D = 0 \quad a.e., \qquad \tilde{c}(s) \overset{\text{def}}{=} \int_0^1 \mathcal{V}'(\theta\, \Xi(s; z_a) + (1 - \theta)\, \Xi(s; z_b))\, d\theta \in (0, 1].$$

The leader data a_ℓ, v_ℓ do not appear in (3.26).

Proof. From (3.8), $D'' = \partial_s \Lambda(\cdot; z_a) - \partial_s \Lambda(\cdot; z_b) = -\alpha[\mathcal{V}(\Xi_a) - \mathcal{V}(\Xi_b)] - \alpha[\Lambda_a - \Lambda_b]$ a.e.; the terms $\mathsf{a}_\ell(s)$ and $\alpha v_\ell(s)$ cancel. The mean value theorem gives $\mathcal{V}(\Xi_a) - \mathcal{V}(\Xi_b) = \tilde{c}\, D$ with $\tilde{c} \in (0, 1]$ because $0 < \mathcal{V}' \le 1$. Continuity of $\tilde{c}$ follows from that of $\Xi(\cdot; z_a)$, $\Xi(\cdot; z_b)$ and $\mathcal{V}'$. ∎

Lemma 3.28 (Fundamental solutions). *Let $c : [0, \boldsymbol{T}_h + 1] \to (0, 1]$ be continuous and $t_0 \in [0, \boldsymbol{T}_h + 1]$. Let ϕ_1, ϕ_2 solve $\phi'' + \alpha\phi' + \alpha c\phi = 0$ with $(\phi_1, \phi_1')(t_0) = (1, 0)$ and $(\phi_2, \phi_2')(t_0) = (0, 1)$.*

(a) $|\phi_i(s)| + |\phi_i'(s)| \le e^{(1+2\alpha)|s - t_0|}$ for $i = 1, 2$.

(b) If $\alpha \ge 4$, then $\phi_2(s) \ge (s - t_0)\, e^{-\alpha(s - t_0)/2} > 0$ for all $s > t_0$.

(c) If $\alpha < 4$, set $\omega_\alpha \overset{\text{def}}{=} \sqrt{\alpha(1 - \alpha/4)}$. Then $\phi_2(s) \ge e^{-\alpha(s - t_0)/2} \sin(\omega_\alpha(s - t_0))/\omega_\alpha > 0$ for $s \in (t_0, t_0 + \pi/\omega_\alpha)$.

Proof. (a) is Grönwall for the first-order system (ϕ, ϕ') in the ℓ^1-norm: the column sums of the matrix $\begin{pmatrix} 0 & 1 \\ -\alpha c & -\alpha \end{pmatrix}$ are $\le \max\{\alpha c, 1 + \alpha\} \le 1 + 2\alpha$.

(b), (c). Put $u(s) \overset{\text{def}}{=} \phi_2(s) e^{\alpha(s - t_0)/2}$. A direct computation gives $u'' = (\frac{\alpha^2}{4} - \alpha c(s))u$, with $u(t_0) = 0$, $u'(t_0) = 1$.

If $\alpha \ge 4$, then $\frac{\alpha^2}{4} - \alpha c \ge \frac{\alpha^2}{4} - \alpha \ge 0$. Let $s_1 > t_0$ be the first zero of u after t_0, if any. On (t_0, s_1), $u > 0$, so $u'' \ge 0$, so $u' \ge 1$, so $u(s) \ge s - t_0$. Then $u(s_1) \ge s_1 - t_0 > 0$, a contradiction. Hence $u(s) \ge s - t_0$ for all $s > t_0$, which is (b).

If $\alpha < 4$, then $\frac{\alpha^2}{4} - \alpha c \ge -\omega_\alpha^2$. Let $w(s) \overset{\text{def}}{=} \sin(\omega_\alpha(s - t_0))/\omega_\alpha$, so $w'' = -\omega_\alpha^2 w$, $w(t_0) = 0$, $w'(t_0) = 1$, and $w > 0$ on $(t_0, t_0 + \pi/\omega_\alpha)$. While $u > 0$,

$$(u'w - uw')' = u''w - uw'' = (\tfrac{\alpha^2}{4} - \alpha c + \omega_\alpha^2)uw \;\ge\; 0,$$

and $u'w - uw' = 0$ at t_0. Hence $(u/w)' = (u'w - uw')/w^2 \ge 0$ where $u > 0$ and $w > 0$. Since $u/w \to 1$ as $s \downarrow t_0$, we get $u \ge w$ as long as $u > 0$ and $s < t_0 + \pi/\omega_\alpha$. If u had a first zero $s_1 < t_0 + \pi/\omega_\alpha$, then $u(s_1) \ge w(s_1) > 0$, a contradiction. This is (c). ∎

Nondegenerate grazing states.

Definition 3.29 (Nondegenerate grazing state). *Let $z_0 = (\bar{t}_0, \bar{X}_0, \bar{V}_0) \in \Gamma \cap \mathcal{P}^+$ with $\sigma_0 \overset{\text{def}}{=} \sigma(z_0)$, so that $\bar{t}_0 < \sigma_0$, $\Xi(\sigma_0; z_0) = 0 = \Lambda(\sigma_0; z_0)$. Let δ_0 be the solution of the variational*

equation

$$\delta'' + \alpha\delta' + \alpha\,\mathcal{V}'(\Xi(s;z_0))\,\delta = 0, \qquad \delta(\bar{t}_0) = 0, \quad \delta'(\bar{t}_0) = 1, \tag{3.27}$$

which is the derivative $\partial_{\bar{V}}\Xi(\cdot;z_0)$ *of the predicted gap with respect to the initial relative velocity. (Indeed, by lemma 3.27 below,* $D_\varepsilon \overset{\text{def}}{=} \Xi(\cdot;\bar{t}_0,\bar{X}_0,\bar{V}_0+\varepsilon)-\Xi(\cdot;z_0)$ *solves (3.26) with* $D_\varepsilon(\bar{t}_0)=0$, $D'_\varepsilon(\bar{t}_0)=\varepsilon$, *and* $\tilde{c}_\varepsilon \to \mathcal{V}'(\Xi(\cdot;z_0))$ *uniformly by (3.10); so* $D_\varepsilon/\varepsilon \to \delta_0$ *by Grönwall.) We call* z_0 nondegenerate *if*

(*ND1*) $\sigma_0 < \boldsymbol{T}_h$, *and there are* $h_0, c_0 > 0$ *with* $\mathsf{a}_\ell(s) + \alpha v_\ell(s) \geq 2c_0$ *for a.e.* $s \in [\sigma_0 - h_0, \sigma_0 + h_0]$;
(*ND2*) $\delta_0(\sigma_0) \neq 0$;
(*ND3*) $z_0^+ \overset{\text{def}}{=} (\sigma_0 + h, Z(\sigma_0 + h; z_0)) \in \mathcal{R}$ *for all sufficiently small* $h > 0$;
(*ND4*) $\bar{V}_0 \neq 0$.

We write $\varsigma \overset{\text{def}}{=} -\operatorname{sgn}(\bar{V}_0)\cdot\operatorname{sgn}(\delta_0(\sigma_0)) \in \{\pm 1\}$.

Remark 3.30 (Meaning of the conditions).

(*a*) *At the graze,* $\partial_s\Lambda = \mathsf{g}(s,0,0) = \mathsf{a}_\ell(s) + \alpha v_\ell(s)$, *since* $\mathcal{V}(0) = 0$. *So (ND1) implies that the predicted relative velocity passes through zero with positive slope: the predicted gap has a strict, quadratic minimum at* σ_0. *It excludes zero-speed down-crossings (where the gap would go negative) and flat contacts. Physically: at the predicted graze time, the leader is not braking harder than* αv_ℓ. *For a stopped leader (*$v_\ell \equiv 0$, $\mathsf{a}_\ell \equiv 0$ *near* σ_0*) the point* $(0,0)$ *is an equilibrium of the prediction, so by backward uniqueness no first-contact graze can occur there at all; (ND1) only restricts leaders that brake at the predicted touch.*

(*b*) *(ND2) says that the predicted gap at the graze time responds to the current relative velocity. Lem. 3.28 gives* $\delta_0(\sigma_0) > 0$ *whenever* $\alpha \geq 4$, *or* $\alpha < 4$ *and* $\sigma_0 - \bar{t}_0 < \pi/\omega_\alpha$. *For small* α *and long prediction horizons* $\delta_0(\sigma_0)$ *can be negative (an example with* $\alpha = 0.2$ *and a graze about* $8\,s$ *ahead is easy to produce). Then the crossing goes the other way; it is still a crossing.*

(*c*) *(ND3) excludes a second graze on the remainder of the prediction. By lemma 3.26, the prediction from* z_0^+ *is the continuation of the prediction from* z_0, *so (ND3) says: after the graze, the predicted gap either stays positive up to* $\boldsymbol{T}_h$ *or crosses zero transversally. Under (ND1),* $\Xi(\cdot;z_0) > 0$ *on* $(\sigma_0, \sigma_0 + h_1]$ *for some* $h_1 > 0$, *so by lemma 3.26 the first zero of* $\Xi(\cdot;z_0)$ *on* $[\sigma_0 + h, \boldsymbol{T}_h]$, *and* Λ *there, are the same for all* $h \in (0, h_1]$. *Hence the condition holds for all* $h \in (0, h_1]$ *as soon as it holds for one.*

(*d*) *(ND4): if* $\bar{V}_0 = 0$, *the TTC term vanishes at* z_0 *and the field is continuous there; the crossing direction is then not determined at first order.*

(*e*) *The excluded cases* $\sigma_0 = \boldsymbol{T}_h$, $\delta_0(\sigma_0) = 0$, *a double graze and* $\bar{V}_0 = 0$ *are each one further scalar condition on top of* $z_0 \in \Gamma$. *(ND1) is of a different nature: it is an essential lower bound on the forcing on a whole neighbourhood of* σ_0, *and it can fail for admissible* L^∞ *forcing even when the predicted gap has a strict quadratic minimum at* σ_0 *(for instance a predicted gap* $\varepsilon(s-\sigma_0)^2\big(2+\sin(3\log|s-\sigma_0|)\big)$, *which is* $C^{1,1}$ *with a strict minimum, is realised by admissible leader data whose* $\mathsf{a}_\ell + \alpha v_\ell$ *has negative essential values in every neighbourhood of* σ_0*). We make no claim that nondegeneracy holds generically in any precise sense; theorem 3.38(B) is conditional on it.*

Local structure near a nondegenerate grazing state. Fix a nondegenerate z_0. Let B be the closed unit ball around z_0, intersected with $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$, and L_B the constant of lemma 3.3(b).

Let $\phi_{\min} > 0$ and $h_1 \in (0, h_0]$ be such that $\operatorname{sgn}(\delta_0(\sigma_0))\,\delta_0 \geq 2\phi_{\min}$ on $[\sigma_0 - h_1, \sigma_0 + h_1]$; they exist by (ND2) and continuity of δ_0, and depend only on z_0. Put $L_{\mathsf{G}} = 1 + 2\alpha$ and let C_Λ be a bound for $|\partial_s \Lambda(s; z)|$ over $z \in B$, $s \in [0, \boldsymbol{T}_h + 1]$ (it exists by lemma 3.3(b)).

Lemma 3.31 (Window and sides). *There are $h \in (0, h_1]$ and $\rho \in (0, 1]$, with $\sigma_0 + h < \boldsymbol{T}_h$ and $\bar{t}_0 + \rho < \sigma_0 - h$, such that, writing $I \overset{\text{def}}{=} [\sigma_0 - h, \sigma_0 + h]$ and $U \overset{\text{def}}{=} \{z \in \mathcal{P}^+ : |z - z_0| < \rho\}$, the following hold for every $z = (\bar{t}, \bar{X}, \bar{V}) \in U$. (Here h is chosen first, in (a); all later steps only shrink ρ.)*

(a) *$\partial_s \Lambda(s; z) \geq c_0$ for a.e. $s \in I$; $\Lambda(\sigma_0 - h; z) \leq -c_0 h/2$ and $\Lambda(\sigma_0 + h; z) \geq c_0 h/2$. Hence $\Lambda(\cdot; z)$ has exactly one zero $s^*(z)$ in I, it lies in the open interval, and $s^*(z)$ is the strict minimiser of $\Xi(\cdot; z)$ on I. Moreover $\Xi(\sigma_0 + h; z) \geq c_0 h^2/4 > 0$.*

(b) *$\Xi(s; z) \geq c > 0$ for $s \in [\bar{t}, \sigma_0 - h]$, for a constant c independent of z.*

(c) *The* minimum-gap function *$m(z) \overset{\text{def}}{=} \Xi(s^*(z); z) = \min_I \Xi(\cdot; z)$ is L_B-Lipschitz on U, and $|s^*(z_1) - s^*(z_2)| \leq (L_B/c_0)|z_1 - z_2|$.*

(d) *With $U^- \overset{\text{def}}{=} \{m < 0\}$, $U^0 \overset{\text{def}}{=} \{m = 0\}$, $U^+ \overset{\text{def}}{=} \{m > 0\}$ (all intersected with U):*
- *$z \in U^-$ implies $z \in \mathcal{R}$, $\sigma(z) \in (\sigma_0 - h, s^*(z))$, $\Lambda(\sigma(z); z) < 0$ and $\tau(z) = \sigma(z) - \bar{t}$;*
- *$z \in U^0$ implies $z \in \Gamma$ with $\sigma(z) = s^*(z)$; in particular $\Gamma \cap U = U^0$;*
- *$z \in U^+$ implies $\Xi(\cdot; z) > 0$ on $[\bar{t}, \sigma_0 + h]$, and $z \in \mathcal{R}$.*

(e) *There is a Lipschitz function $\tau_+ : U \to (0, \boldsymbol{T}_h]$ with $\tau_+ = \tau$ on $U^0 \cup U^+$.*

(f) *$\tau \geq \tau_U > 0$ on U; $|\mathsf{F}(t, \xi)| \leq M_U$ at* every *point $(t, \xi) \in U$ (recall that the representative of a_ℓ is bounded everywhere), and $|\mathsf{G}(s, Z(s; z))| \leq M_U$ for $z \in U$, $s \in [0, \boldsymbol{T}_h + 1]$.*

(g) *$|\bar{V}| \geq \bar{v} \overset{\text{def}}{=} |\bar{V}_0|/2$ on U, and $\bar{V}$ has the sign of $\bar{V}_0$.*

Proof. (a) Along z_0, $Z(\sigma_0; z_0) = (0, 0)$, and $Z(\cdot; z_0)$ is continuous, so there is $h \leq h_0$ with $|Z(s; z_0)| \leq c_0/(2L_{\mathsf{G}})$ on I. For $|z - z_0| < \rho$ and $L_B \rho \leq c_0/(2L_{\mathsf{G}})$, (3.10) gives $|Z(s; z)| \leq c_0/L_{\mathsf{G}}$ on I. Since $|\mathsf{g}(s, Z) - \mathsf{g}(s, 0)| \leq L_{\mathsf{G}}|Z|$ and $\mathsf{g}(s, 0) = \mathsf{a}_\ell(s) + \alpha v_\ell(s) \geq 2c_0$ a.e. on $I \subset [\sigma_0 - h_0, \sigma_0 + h_0]$ by (ND1), we get $\partial_s \Lambda(s; z) = \mathsf{g}(s, Z(s; z)) \geq c_0$ a.e. on I. Along z_0 this gives $\Lambda(\sigma_0 \mp h; z_0) \lessgtr \mp c_0 h$; shrinking ρ so that $L_B \rho \leq c_0 h/2$ transfers this to all $z \in U$ with the constant $c_0 h/2$. The claims on s^* follow from strict monotonicity of $\Lambda(\cdot; z)$ on I and $\partial_s \Xi = \Lambda$. Finally $\Xi(\sigma_0 + h; z_0) = \int_{\sigma_0}^{\sigma_0 + h} \Lambda(s; z_0)\, ds \geq \int_0^h c_0 r\, dr = c_0 h^2/2$, and shrinking ρ so that $L_B \rho \leq c_0 h^2/4$ gives the last claim on U.

Throughout, ρ is also taken small enough that $U \subset \mathcal{P}^+$; this is possible because $\mathcal{P}^+$ is relatively open and $z_0 \in \mathcal{P}^+$.

(b) $\Xi(\cdot; z_0)$ is positive on $[\bar{t}_0, \sigma_0 - h]$ because σ_0 is its first zero, hence positive on $[\bar{t}_0 - h', \sigma_0 - h]$ for some $h' > 0$ (if $\bar{t}_0 > 0$), with minimum $2c > 0$ there. Shrink $\rho \leq \min\{h', c/L_B\}$.

(c) $|m(z_1) - m(z_2)| \leq \sup_{s \in I} |\Xi(s; z_1) - \Xi(s; z_2)| \leq L_B |z_1 - z_2|$, as the minimum of a family of uniformly Lipschitz functions. For s^*: $0 = \Lambda(s^*(z_1); z_1) = \Lambda(s^*(z_2); z_2)$, so by (a) and (3.10), $c_0 |s^*(z_1) - s^*(z_2)| \leq |\Lambda(s^*(z_1); z_2) - \Lambda(s^*(z_2); z_2)| = |\Lambda(s^*(z_1); z_2) - \Lambda(s^*(z_1); z_1)| \leq L_B |z_1 - z_2|$.

(d) By (a), $\Xi(\cdot; z)$ is strictly decreasing on $[\sigma_0 - h, s^*(z)]$ from $\Xi(\sigma_0 - h; z) \geq c > 0$ to $m(z)$, and strictly increasing after $s^*(z)$ on I. By (b) there is no zero on $[\bar{t}, \sigma_0 - h]$. If $m(z) < 0$, the first zero $\sigma(z)$ lies in $(\sigma_0 - h, s^*(z))$ where $\Lambda < 0$; so $z \in \mathcal{R}$ and, by lemma 3.10, $\tau = \sigma(z) - \bar{t}$. If $m(z) = 0$, the first zero is $s^*(z)$ with $\Lambda(s^*(z); z) = 0$, so $z \in \Gamma$. If $m(z) > 0$, $\Xi(\cdot; z) > 0$ on $[\bar{t}, \sigma_0 + h]$, so by lemma 3.26 $\sigma(z) = \sigma(z^+)$ with $z^+ = (\sigma_0 + h, Z(\sigma_0 + h; z))$, and $\Lambda(\sigma(z); z) = \Lambda(\sigma(z^+); z^+)$; since $z^+ \in \mathcal{R}$ (see (e)), $z \in \mathcal{R}$.

(e) By (ND3), $z_0^+ \in \mathcal{R}$. Lem. 3.11 gives a neighbourhood N of z_0^+ on which τ is Lipschitz, with constant L_N. The map $z \mapsto z^+$ is L_B-Lipschitz by (3.10) (its time coordinate is constant), so after shrinking ρ, $z^+ \in N$ for all $z \in U$; also $\Xi(\sigma_0 + h; z) > 0$ on U by (a), so $z^+ \in \Omega$. Define

$$\tau_+(z) \overset{\text{def}}{=} \tau(z^+) + (\sigma_0 + h - \bar{t}), \qquad z \in U.$$

It is Lipschitz with constant $L_N L_B + 1$. For $z \in U^0 \cup U^+$, the set in (3.11) contains no $s \le \sigma_0 + h$: for $z \in U^+$ there is no zero there, and for $z \in U^0$ the only zero is $s^*(z)$, where $\Lambda = 0$. Hence by lemma 3.26 $\tilde{\mathcal{T}}(z) = \tilde{\mathcal{T}}(z^+)$, i.e. $\tau(z) = \tilde{\mathcal{T}}(z^+) - \bar{t} = \tau(z^+) + (\sigma_0 + h) - \bar{t} = \tau_+(z)$.

(f) On U, $\bar{X} \ge \bar{X}_0 - \rho > 0$ and $\bar{t} \le \bar{t}_0 + \rho < \boldsymbol{T}_h$, so lemma 3.9 gives $\tau \ge \tau_U > 0$. The bounds on F and G follow as in lemma 3.16 and lemma 3.3(b).

(g) Shrink $\rho \le |\bar{V}_0|/2$. ∎

Lemma 3.32 (Uniform sign of the sensitivity). *In the setting of lemma 3.31, after shrinking ρ (with h and $\phi_{\min}$ fixed as there), for all $z_a, z_b \in U$, all $t \in [\bar{t}_0 - \rho, \bar{t}_0 + \rho]$ and all $\sigma \in I$, the fundamental solution ϕ_2 of (3.26) (with $\tilde{c} = \tilde{c}_{ab}$, initial time t) satisfies*

$$\operatorname{sgn}(\delta_0(\sigma_0))\,\phi_2(\sigma) \;\ge\; \phi_{\min}, \qquad \text{and} \qquad |\phi_1(\sigma)| \le C_1 \overset{\text{def}}{=} e^{(1+2\alpha)(\boldsymbol{T}_h+1)}.$$

Proof. The bound on ϕ_1 is lemma 3.28(a). By the choice of $h \le h_1$ in lemma 3.31, $\operatorname{sgn}(\delta_0(\sigma_0))\,\delta_0 \ge 2\phi_{\min}$ on I. Here δ_0 denotes the solution of the linear equation (3.27) on all of $[0, \boldsymbol{T}_h + 1]$ (it is defined backward from $\bar{t}_0$ as well, since the equation is linear with bounded coefficients).

Let $e \overset{\text{def}}{=} \phi_2 - \delta_0$ on $[t, \boldsymbol{T}_h + 1]$. Then

$$e'' + \alpha e' + \alpha \tilde{c}_{ab}\, e = -\alpha(\tilde{c}_{ab} - \mathcal{V}'(\Xi(\cdot; z_0)))\delta_0, \qquad e(t) = -\delta_0(t), \quad e'(t) = 1 - \delta_0'(t).$$

By (3.10), $|\theta\Xi(s; z_a) + (1-\theta)\Xi(s; z_b) - \Xi(s; z_0)| \le L_B\rho$, so $|\tilde{c}_{ab} - \mathcal{V}'(\Xi(\cdot; z_0))| \le \|\mathcal{V}''\|_\infty L_B \rho$ uniformly in s. Since $\delta_0(\bar{t}_0) = 0$, $\delta_0'(\bar{t}_0) = 1$ and $\delta_0 \in C^1$, the initial data of e are bounded by $\omega(\rho) \to 0$ as $\rho \to 0$. Grönwall for the first-order system (e, e') on $[t, \boldsymbol{T}_h + 1]$ gives $\sup_I |e| \le C(\omega(\rho) + \rho)$ with C depending only on $\alpha, \boldsymbol{T}_h$, $\|\mathcal{V}''\|_\infty$, L_B and $\sup|\delta_0|$. Shrink ρ so that this is $\le \phi_{\min}$. ∎

The crossing lemma.

Lemma 3.33 (Crossing). *Let z_0 be nondegenerate and let $h, \rho, I, U, m, \varsigma$ be as above. There are $h^* > 0$ and $c_3 > 0$, depending only on z_0 and the constants of lemmas 3.31 and 3.32, with the following property. Let ξ be a Filippov solution on an interval J with $z(t) \overset{\text{def}}{=} (t, \xi(t)) \in U$ for all $t \in J$. Then for all $t_1 < t$ in J with $t - t_1 \le h^*$,*

$$\varsigma\,[m(z(t)) - m(z(t_1))] \;\ge\; c_3\,(t - t_1). \tag{3.28}$$

Consequently $\varsigma\, m \circ z$ is strictly increasing on J with slope at least c_3, and $z(t) \in \Gamma$ for at most one $t \in J$.

Proof. We treat $\bar{V}_0 < 0$; the case $\bar{V}_0 > 0$ is identical with all inequalities reversed. Fix $t_1 < t$ in J, write $\Delta \overset{\text{def}}{=} t - t_1$, and let

$$D(s) \overset{\text{def}}{=} \Xi(s; z(t)) - \Xi(s; z(t_1)), \qquad s \in [0, \boldsymbol{T}_h + 1].$$

By lemma 3.27, D solves (3.26) with $\tilde{c} = \tilde{c}_{z(t),z(t_1)}$.

Step 1: data of D at $s = t$. By definition $Z(t; z(t)) = \xi(t)$, so $D(t) = X(t) - \Xi(t; z(t_1))$ and $D'(t) = V(t) - \Lambda(t; z(t_1))$. Since ξ and $Z(\cdot; z(t_1))$ both start from $\xi(t_1)$ at time t_1 and have speeds bounded by M_U (lemma 3.31(f); for the Filippov solution, $\dot{\xi} \in \mathfrak{F}[\mathsf{F}] \subset \overline{B}(0, M_U)$ a.e.),

(3.29)

$$|\xi(r) - Z(r; z(t_1))| \le 2M_U(r - t_1) \quad (r \in [t_1, t]), \qquad |D(t)| \le \int_{t_1}^{t} |V - \Lambda(\cdot; z(t_1))|\, dr \le M_U \Delta^2.$$

For $D'(t)$, write $V(t) = V(t_1) + \int_{t_1}^{t} \dot{V}\, dr$ and $\Lambda(t; z(t_1)) = V(t_1) + \int_{t_1}^{t} \mathsf{g}(r, Z(r; z(t_1)))\, dr$. On U, $V < 0$ (lemma 3.31(g)), and $\tau \le \boldsymbol{T}_h$ (lemma 3.9), so at every point of U the TTC term satisfies $-\beta V/\tau^2 \ge \beta|V|/\boldsymbol{T}_h^2 \ge \beta\bar{v}/\boldsymbol{T}_h^2$. The function $\eta \mapsto \mathsf{g}(r, \eta) + \beta|\eta_2|/\boldsymbol{T}_h^2$ is continuous and bounds $\mathsf{f}(r, \cdot)$ from below at every point of $U_r \overset{\text{def}}{=} \{\eta : (r, \eta) \in U\}$, so by lemma 3.15(ii) (or directly for Carathéodory solutions)

$$\dot{V}(r) \;\ge\; \mathsf{g}(r, \xi(r)) + \beta\, \frac{\bar{v}}{\boldsymbol{T}_h^2} \qquad \text{for a.e. } r \in [t_1, t].$$

Therefore, using $|\mathsf{g}(r, \eta_1) - \mathsf{g}(r, \eta_2)| \le L_{\mathsf{G}}|\eta_1 - \eta_2|$ and (3.29),

$$D'(t) \;\ge\; \int_{t_1}^{t} \big[\mathsf{g}(r, \xi(r)) - \mathsf{g}(r, Z(r; z(t_1)))\big] dr + \frac{\beta\bar{v}}{\boldsymbol{T}_h^2}\, \Delta \;\ge\; \frac{\beta\bar{v}}{\boldsymbol{T}_h^2}\, \Delta - L_{\mathsf{G}} M_U \Delta^2. \tag{3.30}$$

Step 2: D on the window. Set $c_3 \overset{\text{def}}{=} \beta\bar{v}\phi_{\min}/(2\boldsymbol{T}_h^2)$ and $h^* \overset{\text{def}}{=} c_3/((L_{\mathsf{G}}\phi_{\min} + C_1)M_U)$, and assume $\Delta \le h^*$. Then $L_{\mathsf{G}} M_U \Delta \le c_3/\phi_{\min} = \beta\bar{v}/(2\boldsymbol{T}_h^2)$, so (3.30) gives $D'(t) \ge \beta\bar{v}\Delta/(2\boldsymbol{T}_h^2) > 0$.

Since $t \le \bar{t}_0 + \rho < \sigma_0 - h$, every $\sigma \in I$ satisfies $\sigma > t$, and $D(\sigma) = D(t)\phi_1(\sigma) + D'(t)\phi_2(\sigma)$, where ϕ_1, ϕ_2 are the fundamental solutions of (3.26) with initial time t (lemma 3.28). By lemma 3.32, $\varsigma\phi_2(\sigma) \ge \phi_{\min}$ and $|\phi_1(\sigma)| \le C_1$, where $\varsigma = \operatorname{sgn}(\delta_0(\sigma_0))$ because $\bar{V}_0 < 0$. Using $D'(t) > 0$, (3.29) and (3.30), for all $\sigma \in I$,

$$\begin{aligned} \varsigma D(\sigma) \;&\ge\; D'(t)\, \phi_{\min} - |D(t)|\, C_1 \;\ge\; \Big(\frac{\beta\bar{v}}{\boldsymbol{T}_h^2}\Delta - L_{\mathsf{G}} M_U \Delta^2\Big)\phi_{\min} - C_1 M_U \Delta^2 \\ &=\; 2c_3\Delta - (L_{\mathsf{G}}\phi_{\min} + C_1) M_U \Delta^2 \;\ge\; c_3\, \Delta. \end{aligned}$$

Step 3: from D to m. Suppose $\varsigma = +1$. Then $\Xi(\sigma; z(t)) \ge \Xi(\sigma; z(t_1)) + c_3\Delta$ for all $\sigma \in I$. Taking the minimum over I on both sides, $m(z(t)) \ge m(z(t_1)) + c_3\Delta$. If $\varsigma = -1$, the same argument gives $m(z(t)) \le m(z(t_1)) - c_3\Delta$. This is (3.28).

Step 4: consequences. Chaining (3.28) over a partition of $[t_1, t]$ into pieces of length $\le h^*$ shows that (3.28) holds for all $t_1 < t$ in J. By lemma 3.31(d), $z(t) \in \Gamma$ iff $m(z(t)) = 0$, and a strictly monotone function vanishes at most once. ∎

Remark 3.34 (What the proof uses). *Only three things enter: the cocycle property, which makes the prediction from $z(t)$ and from $z(t_1)$ differ only through what the closed loop did on $[t_1, t]$; the fact that this difference is $-\beta \int V/\tau^2$ to leading order, whose sign is that of $-V$ on* both *sides of Γ (only its size jumps); and the sign of the sensitivity $\delta_0(\sigma_0)$. The value of τ on Γ plays no role, and no Lipschitz property of τ across Γ is used.*

Lemma 3.35 (Carathéodory solutions are Filippov solutions near a nondegenerate graze). *Let z_0 be nondegenerate and U as in lemmas 3.31 and 3.32. Then $\mathsf{F}(t,\xi) \in \mathfrak{F}[\mathsf{F}](t,\xi)$ for every $(t,\xi) \in U$. Consequently every Carathéodory solution of (3.3) whose graph lies in U is a Filippov solution.*

Proof. On $U^- \cup U^+ \subset \mathcal{R}$ the claim is lemma 3.15(i). Let $(t,\xi) = (t,X,V) \in U^0$. Fix $\varepsilon \neq 0$ small and put $z_\varepsilon \overset{\text{def}}{=} (t,X,V+\varepsilon) \in U$. By lemma 3.27 with $z_a = z_\varepsilon$, $z_b = (t,\xi)$, the difference $D = \Xi(\cdot;z_\varepsilon) - \Xi(\cdot;t,\xi)$ solves (3.26) with $D(t) = 0$, $D'(t) = \varepsilon$, so $D(\sigma) = \varepsilon\phi_2(\sigma)$ with ϕ_2 as in lemma 3.32 (initial time $t \in [\bar{t}_0 - \rho, \bar{t}_0 + \rho]$), so that $\varsigma_0\phi_2 \geq \phi_{\min}$ on I with $\varsigma_0 \overset{\text{def}}{=} \operatorname{sgn}(\delta_0(\sigma_0))$. Take ε with $\varsigma_0\varepsilon > 0$, i.e. $\varepsilon = \varsigma_0|\varepsilon|$. Then $D = |\varepsilon|\,\varsigma_0\phi_2 \geq |\varepsilon|\phi_{\min} > 0$ on I (for either sign of ς_0), and taking the minimum over I as in Step 3 of lemma 3.33, $m(z_\varepsilon) - m(t,\xi) \geq |\varepsilon|\phi_{\min} > 0$. Since $m(t,\xi) = 0$, this gives $m(z_\varepsilon) > 0$ whenever $\varsigma_0\varepsilon > 0$: the point (t,ξ) is a limit of points of $U_t^+ \overset{\text{def}}{=} \{\eta : (t,\eta) \in U^+\}$, which is open.

On $U_t^0 \cup U_t^+$ we have $\tau(t,\cdot) = \tau_+(t,\cdot)$ (lemma 3.31(e)), which is continuous, so $\mathsf{F}(t,\cdot)$ restricted to $U_t^0 \cup U_t^+$ is continuous. Given $\delta > 0$ and a null set $N \subset \mathbb{R}^2$, pick $\eta_n \to \xi$ with $\eta_n \in U_t^+$; since U_t^+ is open we may move each η_n off N without leaving $B(\xi,\delta) \cap U_t^+$ and keeping $\eta_n \to \xi$. Then $\mathsf{F}(t,\eta_n) \to \mathsf{F}(t,\xi)$, so $\mathsf{F}(t,\xi) \in \overline{\mathsf{F}(t, B(\xi,\delta) \setminus N)} \subset \overline{\operatorname{co}}\,\mathsf{F}(t, B(\xi,\delta) \setminus N)$. Intersecting over δ and N gives $\mathsf{F}(t,\xi) \in \mathfrak{F}[\mathsf{F}](t,\xi)$.

For the last statement: a Carathéodory solution satisfies $\dot{\xi} = \mathsf{F}(t,\xi)$ a.e., and at those times $\mathsf{F}(t,\xi) \in \mathfrak{F}[\mathsf{F}](t,\xi)$. ∎

Uniqueness through a nondegenerate crossing.

Proposition 3.36 (Uniqueness through the crossing). *Let $z_1 = (t_1,\xi_1) \in \Gamma$ be nondegenerate, and let ξ^a, ξ^b be Filippov solutions on $[t_1, t_1+\eta)$ with $\xi^a(t_1) = \xi^b(t_1) = \xi_1$. Then there is $\eta' \in (0,\eta]$ such that $\xi^a = \xi^b$ on $[t_1, t_1+\eta')$, the graph of ξ^a lies in $\mathcal{R}$ for $t \in (t_1, t_1+\eta')$, and ξ^a is a Carathéodory solution of (3.3) on $[t_1, t_1+\eta')$.*

Proof. Apply lemmas 3.31 to 3.33 with $z_0 = z_1$, and shrink η' so that both graphs stay in U on $[t_1, t_1+\eta')$ and $\eta' \leq h^*$. By lemma 3.33, for $t \in (t_1, t_1+\eta')$ and $i \in \{a,b\}$,

$$\varsigma\, m(z^i(t)) \;\geq\; c_3(t-t_1) \;>\; 0. \tag{3.31}$$

So both graphs lie in U^+ (if $\varsigma = +1$) or in U^- (if $\varsigma = -1$) for $t > t_1$; in either case in $\mathcal{R}$ (lemma 3.31(d)). On $\mathcal{R}$, $\mathsf{F}(t,\cdot)$ is continuous, so $\mathfrak{F}[\mathsf{F}] = \{\mathsf{F}\}$ (lemma 3.15(i)) and both ξ^i are Carathéodory solutions of (3.3) on $(t_1, t_1+\eta')$, hence on $[t_1, t_1+\eta')$ (a single point does not affect the a.e. identity). It remains to show $\xi^a = \xi^b$.

Case $\varsigma = +1$ (entering the miss side). For $t \in [t_1, t_1+\eta')$, $z^i(t) \in U^0 \cup U^+$, so $\tau(z^i(t)) = \tau_+(z^i(t))$ with τ_+ Lipschitz on U (lemma 3.31(e)). Hence both ξ^i solve $\dot{\xi} = \mathsf{F}_+(t,\xi) \overset{\text{def}}{=} (V,\ \mathsf{g}(t,\xi) - \beta V/\tau_+(t,\xi)^2)$ on $[t_1, t_1+\eta')$. Since $\tau_+(z) \geq \sigma_0 + h - \bar{t} > 2h > 0$ on U (as $\bar{t} < \sigma_0 - h$) and τ_+ is Lipschitz, F_+ is Lipschitz in ξ on U (as in lemma 3.16), and Grönwall gives $\xi^a = \xi^b$.

Case $\varsigma = -1$ (entering the crossing side). Here τ is not Lipschitz up to Γ (remark 3.12), and we use a Lipschitz bound with an integrable singularity. Fix $t \in (t_1, t_1+\eta')$ and write $\mu_i \overset{\text{def}}{=} -m(z^i(t)) \geq c_3(t-t_1)$ and $\sigma_i \overset{\text{def}}{=} \sigma(z^i(t))$, $s_i^* \overset{\text{def}}{=} s^*(z^i(t))$, so that $\tau(z^i(t)) = \sigma_i - t$ by lemma 3.31(d).

(i) The first zero is at distance $\gtrsim \sqrt{\mu_i}$ *from* s_i^*. On $[\sigma_i, s_i^*]$, $\Lambda(\cdot; z^i(t)) \le 0$ and, since $\partial_s \Lambda \le C_\Lambda$ and $\Lambda(s_i^*; z^i(t)) = 0$, $|\Lambda(s; z^i(t))| \le C_\Lambda(s_i^* - s)$. Therefore

$$\mu_i = \Xi(\sigma_i; z^i(t)) - \Xi(s_i^*; z^i(t)) = \int_{\sigma_i}^{s_i^*} |\Lambda(s; z^i(t))|\, ds \le \frac{C_\Lambda}{2}(s_i^* - \sigma_i)^2,$$

that is, $s_i^* - \sigma_i \ge \sqrt{2\mu_i/C_\Lambda} \stackrel{\text{def}}{=} 2\delta_i$.

(ii) Lipschitz bound on τ *between the two states.* Let $\delta \stackrel{\text{def}}{=} \min\{\delta_a, \delta_b\} \ge \frac{1}{2}\sqrt{2c_3(t - t_1)/C_\Lambda}$ and $d \stackrel{\text{def}}{=} |\xi^a(t) - \xi^b(t)| = |z^a(t) - z^b(t)|$. By lemma 3.31(c), $|s_a^* - s_b^*| \le (L_B/c_0)d$. Suppose $(L_B/c_0)d \le \delta$. Then $\sigma_b \le s_b^* - 2\delta \le s_a^* + \delta - 2\delta = s_a^* - \delta$, and $\sigma_a \le s_a^* - 2\delta \le s_a^* - \delta$. Both σ_a, σ_b lie in $[\sigma_0 - h,\ s_a^* - \delta]$, where $\Lambda(\cdot; z^a(t)) \le -c_0\delta$ by lemma 3.31(a) (strict increase with slope $\ge c_0$ up to its zero s_a^*). Since $\Xi(\sigma_a; z^a(t)) = 0 = \Xi(\sigma_b; z^b(t))$,

$$\begin{aligned} c_0\delta\,|\sigma_a - \sigma_b| \ &\le\ \Big|\int_{\sigma_b}^{\sigma_a} \Lambda(s; z^a(t))\, ds\Big| = |\Xi(\sigma_a; z^a(t)) - \Xi(\sigma_b; z^a(t))| \\ &= |\Xi(\sigma_b; z^b(t)) - \Xi(\sigma_b; z^a(t))| \ \le\ L_B\, d, \end{aligned}$$

so $|\tau(z^a(t)) - \tau(z^b(t))| = |\sigma_a - \sigma_b| \le L_B d/(c_0\delta)$.

(iii) The smallness condition holds for t *close to* t_1. Both solutions have speed $\le M_U$ and coincide at t_1, so $d \le 2M_U(t - t_1)$. The condition $(L_B/c_0)d \le \delta$ is implied by $2L_B M_U(t - t_1)/c_0 \le \frac{1}{2}\sqrt{2c_3(t - t_1)/C_\Lambda}$, i.e. by $\sqrt{t - t_1} \le c_0\sqrt{2c_3/C_\Lambda}/(4L_B M_U)$. Shrink η' accordingly. Then, for all $t \in (t_1, t_1 + \eta')$,

$$|\tau(z^a(t)) - \tau(z^b(t))| \ \le\ \frac{C_\tau}{\sqrt{t - t_1}}\,|\xi^a(t) - \xi^b(t)|, \qquad C_\tau \stackrel{\text{def}}{=} \frac{2L_B}{c_0}\sqrt{\frac{C_\Lambda}{2c_3}}. \tag{3.32}$$

(iv) Grönwall with an integrable coefficient. With $\tau \ge \tau_U$ and $|V| \le v_{\max} + 1$ on $U \subset \mathcal{P}^+$, the same computation as in lemma 3.16 gives, for a.e. $t \in (t_1, t_1 + \eta')$,

$$|\mathsf{F}(t, \xi^a(t)) - \mathsf{F}(t, \xi^b(t))| \ \le\ L(t)\,|\xi^a(t) - \xi^b(t)|, \quad L(t) \stackrel{\text{def}}{=} 1 + 2\alpha + \frac{\beta}{\tau_U^2} + \frac{2\beta(v_{\max} + 1)\boldsymbol{T}_h}{\tau_U^4}\,\frac{C_\tau}{\sqrt{t - t_1}}.$$

$L \in L^1(t_1, t_1 + \eta')$. Let $u \stackrel{\text{def}}{=} |\xi^a - \xi^b|$; then $u(t_1) = 0$ and $u(t) \le \int_{t_1}^t L(s)u(s)\, ds$. Iterating, $u(t) \le \sup u \cdot \big(\int_{t_1}^t L\big)^n/n!$ for every n, so $u \equiv 0$. ∎

Remark 3.37. *The exponent* $\frac{1}{2}$ *in* (3.32) *is the Hölder exponent, at* U^0*, of the crossing-side branch* $\tau^-(z) \stackrel{\text{def}}{=} \sigma(z) - \bar{t}$*, which equals* τ *on* U^- *and extends continuously to* U^0 *by* $s^*(z) - \bar{t}$ *(lemma* 3.31*(d)); the value* τ_+ *on* U^0 *is discontinuous there (remark* 3.12*). The first zero of a gap* $\approx a(s - \sigma_0)^2 - \mu$ *moves like* $\sqrt{\mu}$*. What rescues uniqueness is that the crossing is* transversal *with a definite speed,* $\mu \ge c_3(t - t_1)$*, so the singular Lipschitz coefficient is* $O((t - t_1)^{-1/2})$ *and integrable. This is the one place where the quantitative form* (3.28) *of the crossing lemma is needed.*

3.10. Main theorem.

Theorem 3.38 (Well-posedness and safety of the closed loop). *Let $(t_\circ, \xi_\circ) \in \mathcal{P}$.*

(A) ***(Unconditional.)*** *Non-extendable Filippov solutions of (3.3) with initial value $\xi_\circ$ at $t_\circ$ exist.* Every *non-extendable Filippov solution ξ, on its maximal interval $[t_\circ, T^*)$, $T^* = T^*_\xi \in (t_\circ, \boldsymbol{T}_h]$, satisfies:*

(i) $(t, \xi(t)) \in \mathcal{P}$ for all t; in particular $0 \le v \le v_{\max}$ and $|V| \le v_{\max}$;

(ii) either $T^* = \boldsymbol{T}_h$, or $X(T^*-) = 0$; in the latter case $V(t) \to 0$ as $t \uparrow T^*$, and $|V| \le \sqrt{2\bar{a}X}$ on a left neighbourhood of T^*;

(iii) if $T^* = \boldsymbol{T}_h$, then ξ extends continuously to $[t_\circ, \boldsymbol{T}_h]$ with $V(\boldsymbol{T}_h) = 0$.

In particular no impact (contact with negative relative velocity) ever occurs, for any Filippov solution whatsoever (theorem 3.20 is proved for an arbitrary solution).

(B) ***(Uniqueness under nondegenerate grazing.)*** *Let ξ be a non-extendable Filippov solution as in (A), and suppose that every grazing state on its graph, i.e. every $(t, \xi(t)) \in \Gamma$, is nondegenerate in the sense of definition 3.29. Then:*

(i) the set of times t with $(t, \xi(t)) \in \Gamma$ is discrete in $[t_\circ, T^*)$;

(ii) ξ is a Carathéodory solution of (3.3) on $[t_\circ, T^*)$;

(iii) ξ is the unique Filippov, hence the unique Carathéodory, solution with initial value $\xi_\circ$ at $t_\circ$, on $[t_\circ, T^*)$.

Proof. (A) is theorem 3.23 together with theorem 3.20 and propositions 3.21 and 3.24.

(B)(i). Write $z(t) \overset{\text{def}}{=} (t, \xi(t))$. Let t_1 be a grazing time. By hypothesis $z(t_1)$ is nondegenerate. Lems. 3.31 and 3.33 applied at $z_0 = z(t_1)$ give a neighbourhood U and, by continuity of ξ, an interval $(t_1 - \eta, t_1 + \eta)$ on which the graph stays in U. On that interval $\varsigma m \circ z$ is strictly increasing and vanishes at t_1, so $z(t) \notin \Gamma$ for $t \neq t_1$ there. Thus grazing times are isolated. They cannot accumulate at a point $t^\dagger < T^*$ either: if $z(t^\dagger) \in \mathcal{R}$, then, $\mathcal{R}$ being open, $z(t) \in \mathcal{R}$ near $t^\dagger$; if $z(t^\dagger) \in \Gamma$, it is nondegenerate and isolated by what was just shown.

(B)(ii). Off the grazing times the graph lies in $\mathcal{R}$, where $\mathfrak{F}[\mathsf{F}] = \{\mathsf{F}\}$, so $\dot{\xi} = \mathsf{F}(t, \xi)$ a.e.

(B)(iii). *Filippov competitors.* Let $\tilde{\xi}$ be a Filippov solution on $J \ni t_\circ$ with $\tilde{\xi}(t_\circ) = \xi_\circ$, and let $S \overset{\text{def}}{=} \sup\{s \in J \cap [t_\circ, T^*) : \tilde{\xi} = \xi \text{ on } [t_\circ, s]\}$. Suppose $S < \min\{T^*, \sup J\}$. By continuity $\tilde{\xi}(S) = \xi(S)$. If $(S, \xi(S)) \in \mathcal{R}$, theorem 3.17 (applied from time S) gives $\tilde{\xi} = \xi$ on a right neighbourhood of S. If $(S, \xi(S)) \in \Gamma$, it is nondegenerate by hypothesis, and proposition 3.36 gives the same. Either way the definition of S is contradicted. Hence $\tilde{\xi} = \xi$ on $J \cap [t_\circ, T^*)$. Finally $\sup J \le T^*$: otherwise $\tilde{\xi}$ would extend ξ beyond T^*, contradicting maximality.

Carathéodory competitors. Let ξ_C be a Carathéodory solution on J with $\xi_C(t_\circ) = \xi_\circ$, and define S as above with ξ_C in place of $\tilde{\xi}$. Suppose $S < \min\{T^*, \sup J\}$. If $(S, \xi(S)) \in \mathcal{R}$, then on a right neighbourhood of S the graph of ξ_C lies in $\mathcal{R}$, where $\mathfrak{F}[\mathsf{F}] = \{\mathsf{F}\}$, so ξ_C is a Filippov solution there, and theorem 3.17 applies. If $(S, \xi(S)) \in \Gamma$, it is nondegenerate; on a right neighbourhood of S the graph of ξ_C lies in the set U of lemma 3.31 for $z_0 = (S, \xi(S))$, so by lemma 3.35 ξ_C is a Filippov solution there, and proposition 3.36 applies. Either way S is contradicted, so $\xi_C = \xi$ on $J \cap [t_\circ, T^*)$. ∎

Remark 3.39 (Why both layers). *Layer (A) is the physically relevant statement and holds without any nondegeneracy assumption: whatever the leader does, no admissible motion of the closed loop ends in an impact. Layer (B) says that the generalised solution concept is only a*

safety net: whenever every grazing state met is nondegenerate, the solution is a Carathéodory solution and is unique. It is in general not C^1 at a crossing time t_1, even when a_ℓ is continuous there: τ itself jumps along the trajectory (from the miss-side value τ_+ to the crossing-side value $s^* - t_1$), so at a genuine TTC jump with $V(t_1) \neq 0$ ((ND4)) the one-sided accelerations, when they exist, differ by $-\beta V(t_1)(\tau_{\text{after}}^{-2} - \tau_{\text{before}}^{-2}) \neq 0$. Under L^∞ forcing one-sided accelerations need not exist at all, and if a_ℓ has one-sided limits at t_1 their difference $\mathsf{a}_\ell(t_1+) - \mathsf{a}_\ell(t_1-)$ adds to this expression and may cancel it; so non-C^1 behaviour is not automatic from (ND4) alone, only for forcing continuous at t_1. A Carathéodory solution that crosses Γ transversally is automatically a Filippov solution, so (B) is a statement about the same object as (A), not about a different solution concept.

Corollary 3.40 (No collision). *Let ξ be a non-extendable Filippov solution as in theorem 3.38(A).*

(a) *There is no impact. If $X(T^*-) = 0$, then $V(t) \to 0$ as $t \uparrow T^*$. Since $X > 0$ on $[t_\circ, T^*)$, this is the only time at which a contact could occur. In particular, no collision, in the sense of a contact with negative relative velocity, occurs.*

(b) *If $X(T^*-) > 0$, then $T^* = \boldsymbol{T}_h$, and the solution, defined on $[t_\circ, \boldsymbol{T}_h)$, extends continuously to $[t_\circ, \boldsymbol{T}_h]$ with $V(\boldsymbol{T}_h) = 0$ (proposition 3.24).*
Thus the only obstruction to existence on all of $[t_\circ, \boldsymbol{T}_h]$ is a contact with zero closing speed; see remark 3.41.

Remark 3.41 (What is not proved).

(a) Contact with zero closing speed. *theorem 3.20 excludes every contact with negative relative velocity, which is the notion of collision used in this paper. It does not exclude a contact with $X \to 0$ and $V \to 0$ (a touch without impact). Extensive simulations, including bang-bang leader accelerations switching every few milliseconds, show only exponential approaches to the leader, never a finite-time contact, and a scaling argument suggests the same, but we have no proof.*

(b) Degenerate grazing states. *theorem 3.38(B) gives uniqueness under the nondegeneracy conditions of definition 3.29. The excluded cases (remark 3.30(e)) are each one further condition on top of $z_0 \in \Gamma$; Filippov solutions still exist through them, so safety is unaffected.*

(c) The horizon convention. *With $\inf \varnothing = \boldsymbol{T}_h$ in (3.11), the gain $\beta/\tau^2 \geq \beta/(\boldsymbol{T}_h - t)^2$ is unbounded as $t \uparrow \boldsymbol{T}_h$, which is what forces $V(\boldsymbol{T}_h) = 0$ in proposition 3.24. Two alternatives avoid this: a sliding prediction horizon H (replace $\boldsymbol{T}_h$ by $\bar{t} + H$ in (3.11), with the leader data extended to $[\boldsymbol{T}_h, \boldsymbol{T}_h + H]$), or $\inf \varnothing = +\infty$ with $1/\tau^2 \overset{\text{def}}{=} 0$ and the infimum taken over $[\bar{t}, \infty)$. lemma 3.13 and theorem 3.20 hold unchanged under either; with the second, the case split in the proof of lemma 3.13 disappears. (The hybrid convention $\inf \varnothing = +\infty$ with the window $[\bar{t}, \boldsymbol{T}_h]$ is* not *admissible: it creates a new discontinuity of $1/\tau^2$ inside $\mathcal{R}$ along which a Filippov solution can slide into an impact at $\boldsymbol{T}_h$.)*

4. Regularity of the TTC map. We now study the TTC map τ of definition 3.6 as a function of the initial state, on the domain of states whose first predicted contact is transversal. Everything in this section concerns the prediction flow alone; the closed loop does not enter. We write $z = (s, \boldsymbol{y})$ with $\boldsymbol{y} = (\Xi, \Lambda) \in \mathbb{R}^2$ for a state, $Z(\kappa; s, \boldsymbol{y}) = (\Xi, \Lambda)(\kappa; s, \boldsymbol{y})$ for the predicted state at *elapsed* time κ, and $Z^{\text{abs}}(t; z)$ when the absolute time t is meant

(so $Z(\kappa; s, \boldsymbol{y}) = Z^{\mathrm{abs}}(s+\kappa; z)$; in section 3 only the absolute-time notation was used). The prediction flow is global: by lemma 3.3(a) it is defined on $[0, \boldsymbol{T}_h + 1]$ for all data, through and beyond the collision set, and we use this freely.

4.1. The domain $\mathbb{D}$ and the optimality principle.

Definition 4.1 (The domain $\mathbb{D}$).

$$\mathbb{D} \overset{\boldsymbol{def}}{=} \{z = (\bar{t}, \bar{X}, \bar{V}) \in \Omega : \ \sigma(z) < \boldsymbol{T}_h, \ \Lambda(\sigma(z); z) < 0\}, \tag{4.1}$$

the states whose first *predicted contact occurs before the horizon and is transversal. By lemma 3.10, $\tilde{\mathcal{T}} = \sigma$ and $\tau = \sigma - \bar{t}$ on $\mathbb{D}$: the TTC (3.11) entering the control law (3.3) and the first zero of the predicted gap agree on $\mathbb{D}$, which is the form anticipated in (1.1); moreover $\mathbb{D} \subset \mathcal{R}$.*

Lemma 4.2 ($\mathbb{D}$ is open; invariance along the prediction; optimality principle).

(*a*) *$\mathbb{D}$ is relatively open in $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$.*

(*b*) *(Invariance.) If $z = (\bar{t}, \bar{y}) \in \mathbb{D}$ and $0 \le \kappa < \tau(z)$, then $z^\kappa \overset{\boldsymbol{def}}{=} (\bar{t} + \kappa, Z(\kappa; z)) \in \mathbb{D}$ (writing $Z(\kappa; z)$ for the predicted state at elapsed time κ), and*

$$\tau(z^\kappa) = \tau(z) - \kappa. \tag{4.2}$$

Proof. (a) Let $z_0 \in \mathbb{D}$, $\sigma_0 = \sigma(z_0) < \boldsymbol{T}_h$, $\Lambda(\sigma_0; z_0) = -2\gamma < 0$. In the proof of lemma 3.11(b), choose h additionally with $\sigma_0 + h < \boldsymbol{T}_h$. For z in the ball constructed there, the first zero $\hat{\sigma}(z)$ of the predicted gap lies in $(\sigma_0 - h, \sigma_0 + h)$, hence $< \boldsymbol{T}_h$, and $\Lambda(\hat{\sigma}(z); z) \le -\gamma < 0$. So $\sigma(z) = \hat{\sigma}(z) < \boldsymbol{T}_h$ is transversal, i.e. $z \in \mathbb{D}$.

(b) By the cocycle property (lemma 3.26) the prediction from z^κ is the continuation of the prediction from z. Since $\kappa < \tau(z)$, the gap is positive at z^κ and its first zero is still $\sigma(z)$, with the same transversal velocity. Hence $z^\kappa \in \mathbb{D}$ and $\tau(z^\kappa) = \sigma(z) - (\bar{t} + \kappa) = \tau(z) - \kappa$. ∎

Remark 4.3 (Optimality principle). *(4.2) is the dynamic programming identity for the value function $\boldsymbol{V} \overset{\boldsymbol{def}}{=} \tau$: along every predicted characteristic the value decreases at unit rate until the collision set is reached.*

4.2. Uniform constants on a compact set. Throughout this section $\boldsymbol{D} \Subset \mathbb{D}$ is compact. Let $B \supset \boldsymbol{D}$ be a bounded set (for definiteness, the closed 1-neighbourhood of $\boldsymbol{D}$ intersected with $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$) and let m_B, L_B be the constants of lemma 3.3(b): for $z, z_1, z_2 \in B$ and all absolute times $s, s_1, s_2 \in [0, \boldsymbol{T}_h + 1]$,

$$|Z(s_1; z) - Z(s_2; z)| \le m_B |s_1 - s_2|, \qquad |Z(s; z_1) - Z(s; z_2)| \le L_B |z_1 - z_2|. \tag{4.3}$$

Lemma 4.4 (Comparison at a common elapsed time). *For $z_i = (\bar{t}_i, \bar{y}_i) \in B$ and every elapsed time $\theta \ge 0$ with $\bar{t}_i + \theta \in [0, \boldsymbol{T}_h + 1]$ for $i = 1, 2$,*

$$|Z(\theta; z_1) - Z(\theta; z_2)| \ \le \ (L_B + m_B)\, |z_1 - z_2|, \tag{4.4}$$

where $Z(\theta; z_i)$ denotes the predicted state of z_i at elapsed time θ, i.e. at absolute time $\bar{t}_i + \theta$.

Proof. Insert the state of trajectory 1 at the absolute time $\bar{t}_2 + \theta$:

$$|Z^{\text{abs}}(\bar{t}_1 + \theta; z_1) - Z^{\text{abs}}(\bar{t}_2 + \theta; z_2)| \le |Z^{\text{abs}}(\bar{t}_1 + \theta; z_1) - Z^{\text{abs}}(\bar{t}_2 + \theta; z_1)| + |Z^{\text{abs}}(\bar{t}_2 + \theta; z_1) - Z^{\text{abs}}(\bar{t}_2 + \theta; z_2)|,$$

and apply (4.3) to each term: $m_B|\bar{t}_1 - \bar{t}_2| + L_B|z_1 - z_2|$. ∎

Lemma 4.5 (Continuity and positivity of τ; uniform transversality).

(a) *τ is locally Lipschitz, hence continuous, on $\mathbb{D}$, and $\tau_{\min}(\boldsymbol{D}) \overset{\text{def}}{=} \min_{\boldsymbol{D}} \tau > 0$, $\tau_{\max}(\boldsymbol{D}) \overset{\text{def}}{=} \max_{\boldsymbol{D}} \tau < \boldsymbol{T}_h$.*

(b) *The* terminal approach rate $g(z) \overset{\text{def}}{=} -\Lambda(\tau(z); z)$ *is continuous and positive on $\mathbb{D}$, with*

$$|g(z_1) - g(z_2)| \le m_B|\tau(z_1) - \tau(z_2)| + (L_B + m_B)|z_1 - z_2| \qquad (z_1, z_2 \in B \cap \mathbb{D},\ |\bar{t}_1 - \bar{t}_2| \le 1). \tag{4.5}$$

Hence the transversality margin

$$\gamma_{\boldsymbol{D}} \overset{\text{def}}{=} \min_{z \in \boldsymbol{D}} g(z) = \min_{z \in \boldsymbol{D}} \big(-\Lambda(\tau(z); z)\big) \tag{4.6}$$

is attained and positive: $\Lambda(\tau(z); z) \le -\gamma_{\boldsymbol{D}} < 0$ for all $z \in \boldsymbol{D}$.

Proof. (a) $\mathbb{D} \subset \mathcal{R}$ and lemma 3.11. Positivity of τ is lemma 3.9; $\tau < \boldsymbol{T}_h - \bar{t} \le \boldsymbol{T}_h$ on $\mathbb{D}$ since $\sigma < \boldsymbol{T}_h$; the extrema are attained by continuity and compactness.

(b) Positivity is the definition of $\mathbb{D}$. For (4.5), write $g(z_1) - g(z_2) = -[\Lambda(\tau_1; z_1) - \Lambda(\tau_2; z_1)] - [\Lambda(\tau_2; z_1) - \Lambda(\tau_2; z_2)]$ with $\tau_i = \tau(z_i)$, and bound the first bracket by $m_B|\tau_1 - \tau_2|$ (elapsed-time Lipschitz, same trajectory) and the second by lemma 4.4. Continuity of g follows from that of τ, and the minimum over the compact $\boldsymbol{D}$ is attained. ∎

Lemma 4.6 (Terminal window). *Set*

$$h_0 = h_0(\boldsymbol{D}) \overset{\text{def}}{=} \min\Big\{\frac{\gamma_{\boldsymbol{D}}}{2m_B},\ \tau_{\min}(\boldsymbol{D})\Big\} > 0. \tag{4.7}$$

Then for every $z \in \boldsymbol{D}$:

$$\Lambda(\theta; z) \le -\frac{\gamma_{\boldsymbol{D}}}{2} \qquad \text{for all } \theta \in [\tau(z) - h_0, \tau(z)], \tag{4.8}$$

$$\Xi(\tau(z) - h; z) \ge \frac{\gamma_{\boldsymbol{D}}}{2}\, h \qquad \text{for all } h \in [0, h_0]. \tag{4.9}$$

Proof. $\Lambda(\cdot; z)$ is m_B-Lipschitz in elapsed time and equals $-g(z) \le -\gamma_{\boldsymbol{D}}$ at $\tau(z)$; so on the window it is $\le -\gamma_{\boldsymbol{D}} + m_B h_0 \le -\gamma_{\boldsymbol{D}}/2$. The second term in (4.7) keeps the window inside $[0, \tau(z)]$. Integrating $\partial_\theta \Xi = \Lambda$ from $\tau(z) - h$ to $\tau(z)$, where $\Xi = 0$, gives (4.9). ∎

4.3. Lipschitz regularity of τ with explicit constant.

Proposition 4.7 (Finite-difference Lipschitz bound). *There is $\varrho_0 = \varrho_0(\boldsymbol{D}) > 0$ such that for all $z_1, z_2 \in \boldsymbol{D}$ with $|z_1 - z_2| < \varrho_0$,*

$$|\tau(z_1) - \tau(z_2)| \ \le\ L_\tau(\boldsymbol{D})\, |z_1 - z_2|, \qquad L_\tau(\boldsymbol{D}) \overset{\text{def}}{=} \frac{2\,(L_B + m_B)}{\gamma_{\boldsymbol{D}}}. \tag{4.10}$$

Proof. By lemma 4.5(a), τ is uniformly continuous on $\boldsymbol{D}$; choose ϱ_0 so that $|z_1 - z_2| < \varrho_0$ implies $|\tau(z_1) - \tau(z_2)| \le h_0$. Let $\nu_i = \tau(z_i)$, and without loss of generality $\nu_1 \le \nu_2$, $h \stackrel{\text{def}}{=} \nu_2 - \nu_1 \in [0, h_0]$.

Step 1. By (4.9) applied to z_2 at h, $\Xi(\nu_1; z_2) = \Xi(\nu_2 - h; z_2) \ge \frac{\gamma_{\boldsymbol{D}}}{2} h$.

Step 2. By lemma 4.4 at the common elapsed time ν_1 (both $\bar{t}_i + \nu_1 \le \boldsymbol{T}_h$), and $\Xi(\nu_1; z_1) = 0$,

$$\Xi(\nu_1; z_2) = |\Xi(\nu_1; z_2) - \Xi(\nu_1; z_1)| \le (L_B + m_B)|z_1 - z_2|.$$

Step 3. Combining, $\frac{\gamma_{\boldsymbol{D}}}{2} h \le (L_B + m_B)|z_1 - z_2|$, which is (4.10). ∎

Theorem 4.8 (τ is Lipschitz on compacts). *For every compact $\boldsymbol{D} \Subset \mathbb{D}$, τ is Lipschitz on $\boldsymbol{D}$ with constant*

$$\widehat{L}_\tau(\boldsymbol{D}) \stackrel{\text{def}}{=} \max\Big\{L_\tau(\boldsymbol{D}),\ \frac{2\tau_{\max}(\boldsymbol{D})}{\varrho_0(\boldsymbol{D})}\Big\}. \tag{4.11}$$

In particular $\tau \in \mathcal{W}^{1,\infty}$ on a neighbourhood of $\boldsymbol{D}$, and τ is differentiable at a.e. point of $\boldsymbol{D}$.

Proof. If $|z_1 - z_2| < \varrho_0$, use proposition 4.7. If $|z_1 - z_2| \ge \varrho_0$, then $|\tau(z_1) - \tau(z_2)| \le 2\tau_{\max}(\boldsymbol{D}) \le (2\tau_{\max}/\varrho_0)|z_1 - z_2|$. For the $\mathcal{W}^{1,\infty}$ statement, apply the same to a compact neighbourhood of $\boldsymbol{D}$ in $\mathbb{D}$ (which exists since $\mathbb{D}$ is open); Rademacher's theorem gives a.e. differentiability. ∎

4.4. C^1 regularity in the state. We now show that $\boldsymbol{V} = \tau$ is continuously differentiable in the state variables, with a jointly continuous gradient, and clarify in which sense it is differentiable in time.

Global prediction flow. Recall from lemma 3.3 that the prediction field $\mathsf{G}(s, Z) = (\Lambda,\ \mathsf{a}_\ell(s) - \alpha[\mathcal{V}(\Xi) + \Lambda - v_\ell(s)])$ is globally Lipschitz in Z, so the prediction from any $z = (s, \boldsymbol{y})$ exists on all of $[0, \boldsymbol{T}_h + 1]$, through and beyond the collision set. In Parts II–IV we use this global flow freely and write $Z(\kappa; s, \boldsymbol{y}) = (\Xi, \Lambda)(\kappa; s, \boldsymbol{y})$ for the state at *elapsed* time $\kappa \in [0, \boldsymbol{T}_h + 1 - s]$. Using the global flow is what allows differentiation at $\kappa = \tau(z)$ in the proof of lemma 4.10: the map $\boldsymbol{y}' \mapsto \Xi(\tau(z); s, \boldsymbol{y}')$ must be defined on a full neighbourhood of $\boldsymbol{y}$, which includes states whose own collision occurs before $\tau(z)$.

The state derivative of the field is

$$\partial_{\boldsymbol{y}} \mathsf{G}(s, Z) = \begin{pmatrix} 0 & 1 \\ -\alpha \mathcal{V}'(\Xi) & -\alpha \end{pmatrix}, \qquad \|\partial_{\boldsymbol{y}} \mathsf{G}\| \le G_1 \stackrel{\text{def}}{=} \sqrt{1 + 2\alpha^2}, \tag{4.12}$$

continuous in the state, bounded, Lipschitz in the state (since $\mathcal{V}''$ is bounded), and *independent of s and of the forcing.*

Lemma 4.9 (State linearisation of the prediction flow). *Fix $s \in [0, \boldsymbol{T}_h)$. For every $\kappa \in [0, \boldsymbol{T}_h + 1 - s]$ the map $\boldsymbol{y} \mapsto Z(\kappa; s, \boldsymbol{y})$ is C^1 on $\mathbb{R}^2$, with derivative $Y(\kappa; s, \boldsymbol{y}) \stackrel{\text{def}}{=} \partial_{\boldsymbol{y}} Z(\kappa; s, \boldsymbol{y}) \in \mathbb{R}^{2\times 2}$ the solution of the linear Carathéodory system*

$$\partial_\kappa Y = \partial_{\boldsymbol{y}} \mathsf{G}(Z(\kappa; s, \boldsymbol{y}))\, Y \quad \text{a.e. in } \kappa, \qquad Y(0; s, \boldsymbol{y}) = I. \tag{4.13}$$

The map $(\kappa, s, \boldsymbol{y}) \mapsto Y(\kappa; s, \boldsymbol{y})$ is continuous, and

$$e^{-G_1 \kappa}|v| \ \le\ |Y(\kappa; s, \boldsymbol{y})v| \ \le\ e^{G_1 \kappa}|v| \qquad (v \in \mathbb{R}^2); \tag{4.14}$$

in particular Y is invertible with $\|Y^{-1}\| \le e^{G_1 \kappa}$.

Proof. Differentiability. Fix $\boldsymbol{y}_0$, a direction v, and $h \neq 0$; write $Z = Z(\cdot; s, \boldsymbol{y}_0)$, $Z^h = Z(\cdot; s, \boldsymbol{y}_0 + hv)$ and $\xi_h \overset{\text{def}}{=} (Z^h - Z)/h$. By (3.10), $|Z^h - Z| \leq L_B|h||v|$ on a bounded set B containing both. By the fundamental theorem of calculus along the segment between $Z(r)$ and $Z^h(r)$,

$$\begin{aligned}\mathsf{G}(r, Z^h(r)) - \mathsf{G}(r, Z(r)) &= \Big(\int_0^1 \partial_{\boldsymbol{y}}\mathsf{G}(Z(r) + \lambda(Z^h(r) - Z(r)))d\lambda\Big)(Z^h(r) - Z(r)) \\ &= \partial_{\boldsymbol{y}}\mathsf{G}(Z(r))\,(Z^h(r) - Z(r)) + R(r, h),\end{aligned}$$

with $|R(r,h)| \leq \omega(L_B|h||v|)\, L_B|h||v|$, ω the modulus of continuity of $\partial_{\boldsymbol{y}}\mathsf{G}$ on a compact set (here $\omega(\rho) \leq C\rho$). Hence

$$\xi_h(\kappa) = v + \int_0^\kappa \partial_{\boldsymbol{y}}\mathsf{G}(Z(r))\,\xi_h(r)\,dr + E_h(\kappa), \qquad |E_h(\kappa)| \leq \kappa\,\omega(L_B|h||v|)\,L_B|v| \to 0.$$

Let ξ solve $\xi(\kappa) = v + \int_0^\kappa \partial_{\boldsymbol{y}}\mathsf{G}(Z)\xi$, a linear equation with bounded measurable coefficient. Grönwall applied to $\xi_h - \xi$ gives $\sup_{[0,\kappa_1]}|\xi_h - \xi| \leq \sup|E_h|\, e^{G_1\kappa_1} \to 0$. Thus $\partial_v Z(\kappa; s, \boldsymbol{y}_0)$ exists and equals $\xi(\kappa) = Y(\kappa; s, \boldsymbol{y}_0)v$, where Y solves (4.13). Continuity of Y in $(\kappa, s, \boldsymbol{y})$ follows from continuous dependence of the linear system on its coefficient $A(\kappa; s, \boldsymbol{y}) = \partial_{\boldsymbol{y}}\mathsf{G}(Z(\kappa; s, \boldsymbol{y}))$, which is jointly continuous because Z is (lemma 3.3(b)) and $\partial_{\boldsymbol{y}}\mathsf{G}$ is continuous in the state. Continuity of $\boldsymbol{y} \mapsto Y(\kappa; s, \boldsymbol{y})$ gives the C^1 claim.

Two-sided bound. With $w = Yv$ and $\sigma = |w|^2$ (absolutely continuous), $|\dot\sigma| = 2|\langle w, Aw\rangle| \leq 2G_1\sigma$ a.e., so $\frac{d}{d\kappa}(\sigma e^{\pm 2G_1\kappa}) \gtrless 0$ a.e.; integrating (legitimate for absolutely continuous functions) gives $e^{-2G_1\kappa}|v|^2 \leq \sigma(\kappa) \leq e^{2G_1\kappa}|v|^2$. ∎

Lemma 4.10 (Regularity of $\boldsymbol{V}$ on compacts). *Let $\boldsymbol{D} \Subset \mathbb{D}$ be compact, $K \Subset \mathbb{D}$ a compact neighbourhood of $\boldsymbol{D}$, and γ_K, $\tau_{\max}(K)$, $\widehat{L}_\tau(K)$ as in subsections 4.2 and 4.3. Write $\boldsymbol{V} \overset{\text{def}}{=} \tau$.*

(i) ***(C^1 in the state.)*** *For every $(s, \boldsymbol{y}) \in \boldsymbol{D}$ the map $\boldsymbol{y}' \mapsto \boldsymbol{V}(s, \boldsymbol{y}')$ is continuously differentiable on a neighbourhood of $\boldsymbol{y}$, with*

$$\nabla_{\boldsymbol{y}}\boldsymbol{V}(s, \boldsymbol{y}) = -\frac{\boldsymbol{e}_1^\top Y(\tau(z); s, \boldsymbol{y})}{\Lambda(\tau(z); z)}, \qquad |\nabla_{\boldsymbol{y}}\boldsymbol{V}(s, \boldsymbol{y})| \leq \frac{e^{G_1\tau_{\max}(K)}}{\gamma_K}, \tag{4.15}$$

and $(s, \boldsymbol{y}) \mapsto \nabla_{\boldsymbol{y}}\boldsymbol{V}(s, \boldsymbol{y})$ is continuous on $\boldsymbol{D}$.

(ii) ***(Lipschitz in time; a.e. classical HJB with a common exceptional time set.)*** *$\boldsymbol{V}$ is Lipschitz on $\boldsymbol{D}$. Let $N \subset [0, \boldsymbol{T}_h)$ be the null set of times that are not Lebesgue points of a_ℓ. At every $(s, \boldsymbol{y}) \in \boldsymbol{D}$ with $s \notin N$ the time derivative $\partial_s\boldsymbol{V}(s, \boldsymbol{y})$ exists, $\boldsymbol{V}$ is Fréchet-differentiable, and, with $\mathsf{a}_\ell(s)$ the Lebesgue value,*

$$\partial_s\boldsymbol{V}(s, \boldsymbol{y}) + \mathsf{G}(s, \boldsymbol{y}) \cdot \nabla_{\boldsymbol{y}}\boldsymbol{V}(s, \boldsymbol{y}) = -1. \tag{4.16}$$

Consequently (4.16) holds for a.e. s and every *$\boldsymbol{y}$ with $(s, \boldsymbol{y}) \in \boldsymbol{D}$ (the exceptional set of times N does not depend on $\boldsymbol{y}$), and for a.e. $(s, \boldsymbol{y}) \in \boldsymbol{D}$. Moreover, at every $(s, \boldsymbol{y}) \in \boldsymbol{D}$ at which $\partial_s\boldsymbol{V}(s, \boldsymbol{y})$ exists (for fixed $\boldsymbol{y}$, a.e. s), $\boldsymbol{V}$ is Fréchet-differentiable.*

(iii) ***(Joint C^1 where the forcing is continuous.)*** *If a_ℓ is continuous on an open interval $J \subset (0, \boldsymbol{T}_h)$, then $\boldsymbol{V} \in C^1$ on $\mathbb{D} \cap (J \times \mathbb{R}^2)$, with $\partial_s\boldsymbol{V} = -1 - \mathsf{G} \cdot \nabla_{\boldsymbol{y}}\boldsymbol{V}$ there. (No condition is imposed on the forcing* after *s: the prediction from $(s, \boldsymbol{y})$ may cross jumps of a_ℓ.)*

Proof. *(i).* Fix $z^0 = (s^0, \boldsymbol{y}^0) \in \boldsymbol{D}$, $\nu^0 \overset{\text{def}}{=} \tau(z^0)$, $\gamma \overset{\text{def}}{=} -\Lambda(\nu^0; z^0) \geq \gamma_K$. For $\boldsymbol{y}$ near $\boldsymbol{y}^0$ (so that $z \overset{\text{def}}{=} (s^0, \boldsymbol{y}) \in K$), let $\nu \overset{\text{def}}{=} \tau(z)$.

Step 1 (identity at the common elapsed time ν^0). Since the flow is global, $\Xi(\cdot; z)$ is absolutely continuous on $[0, \boldsymbol{T}_h + 1 - s^0]$ with derivative Λ, and $\Xi(\nu; z) = 0 = \Xi(\nu^0; z^0)$. Hence, as a signed identity with no case split,

$$\Xi(\nu^0; z) - \Xi(\nu^0; z^0) = \Xi(\nu^0; z) - \Xi(\nu; z) = -\int_{\nu^0}^{\nu} \Lambda(\theta; z)\, d\theta. \tag{4.17}$$

Step 2 (left side). By lemma 4.9 at the fixed elapsed time ν^0 and start time s^0, $\boldsymbol{y} \mapsto \Xi(\nu^0; s^0, \boldsymbol{y})$ is C^1 on $\mathbb{R}^2$, so

$$\Xi(\nu^0; s^0, \boldsymbol{y}) - \Xi(\nu^0; s^0, \boldsymbol{y}^0) = \boldsymbol{e}_1^\top Y(\nu^0; s^0, \boldsymbol{y}^0)(\boldsymbol{y} - \boldsymbol{y}^0) + o(|\boldsymbol{y} - \boldsymbol{y}^0|). \tag{4.18}$$

Step 3 (right side). For θ between ν^0 and ν,

$$|\Lambda(\theta; z) - \Lambda(\nu^0; z^0)| \leq |\Lambda(\theta; z) - \Lambda(\nu^0; z)| + |\Lambda(\nu^0; z) - \Lambda(\nu^0; z^0)| \leq m_B|\nu - \nu^0| + L_B|\boldsymbol{y} - \boldsymbol{y}^0| =: \varepsilon(\boldsymbol{y}),$$

by elapsed-time Lipschitz continuity along the single trajectory z and (3.10) at the common absolute time $s^0 + \nu^0$ (same start time). So

$$-\int_{\nu^0}^{\nu} \Lambda(\theta; z)\, d\theta = \gamma\,(\nu - \nu^0) + R, \qquad |R| \leq \varepsilon(\boldsymbol{y})|\nu - \nu^0|. \tag{4.19}$$

Step 4 (remainder). By theorem 4.8 on K, $|\nu - \nu^0| \leq \widehat{L}_\tau(K)|\boldsymbol{y} - \boldsymbol{y}^0|$, so $|R| \leq (m_B\widehat{L}_\tau(K) + L_B)\widehat{L}_\tau(K)|\boldsymbol{y} - \boldsymbol{y}^0|^2 = o(|\boldsymbol{y} - \boldsymbol{y}^0|)$.

Step 5 (conclusion). Combining, $\boldsymbol{e}_1^\top Y(\nu^0; s^0, \boldsymbol{y}^0)(\boldsymbol{y} - \boldsymbol{y}^0) = \gamma(\tau(z) - \tau(z^0)) + o(|\boldsymbol{y} - \boldsymbol{y}^0|)$, so $\boldsymbol{y} \mapsto \tau(s^0, \boldsymbol{y})$ is differentiable at $\boldsymbol{y}^0$ with gradient (4.15). The bound uses $\|Y(\nu^0)\| \leq e^{G_1\nu^0} \leq e^{G_1\tau_{\max}(K)}$ from (4.14) and $\gamma \geq \gamma_K$.

Joint continuity of the gradient. $z \mapsto \Lambda(\tau(z); z)$ is continuous on K and $\leq -\gamma_K$ (lemma 4.5); $z \mapsto Y(\tau(z); z)$ is continuous as the composition of the continuous τ with the jointly continuous Y (lemma 4.9). Hence (4.15) is continuous on K, and in particular $\boldsymbol{y} \mapsto \boldsymbol{V}(s^0, \boldsymbol{y})$ is C^1 near $\boldsymbol{y}^0$.

(ii). Lipschitz continuity is theorem 4.8.

Fréchet differentiability. At a point $(s^0, \boldsymbol{y}^0) \in \boldsymbol{D}$ where $\partial_s \boldsymbol{V}$ exists, all three partial derivatives exist, and the two state partials exist on a neighbourhood (in K) and are continuous there by (i). A function whose partial derivatives all exist at a point, all but one of them being continuous on a neighbourhood, is Fréchet-differentiable at that point [3, Thm. 12.11]. (For fixed $\boldsymbol{y}$, $s \mapsto \boldsymbol{V}(s, \boldsymbol{y})$ is Lipschitz, so this applies for a.e. s.)

Existence of $\partial_s \boldsymbol{V}$ and the HJB identity at Lebesgue points. Fix $z^0 = (s^0, \boldsymbol{y}^0) \in \boldsymbol{D}$ with $s^0 \notin N$, and let $h \neq 0$ be small (two-sided if $s^0 > 0$, $h > 0$ if $s^0 = 0$). Set $\tilde{\boldsymbol{y}}_h \overset{\text{def}}{=} Z^{\text{abs}}(s^0; (s^0 + h, \boldsymbol{y}^0))$, the state at absolute time s^0 of the prediction through $(s^0 + h, \boldsymbol{y}^0)$ (defined by lemma 3.3(a), backward as well as forward). By the cocycle property, $(s^0 + h, \boldsymbol{y}^0)$ and $(s^0, \tilde{\boldsymbol{y}}_h)$ lie on one characteristic, at elapsed distance $|h|$. Since $\tilde{\boldsymbol{y}}_h \to \boldsymbol{y}^0$ as $h \to 0$, $\mathbb{D}$ is open, and $\tau \geq \tau_{\min}(K) > 0$

on K, for small $|h|$ both points lie in K and $|h| < \tau$ at the earlier of the two, so (4.2) gives, for either sign of h,

$$\tau(s^0 + h, \boldsymbol{y}^0) = \tau(s^0, \tilde{\boldsymbol{y}}_h) - h. \tag{4.20}$$

(For $h > 0$ the point $(s^0+h, \boldsymbol{y}^0)$ is at elapsed time h after $(s^0, \tilde{\boldsymbol{y}}_h)$; for $h < 0$ the point $(s^0, \tilde{\boldsymbol{y}}_h)$ is at elapsed time $-h$ after $(s^0+h, \boldsymbol{y}^0)$; both readings give (4.20).) Next, $|Z^{\mathrm{abs}}(r; (s^0+h, \boldsymbol{y}^0)) - \boldsymbol{y}^0| \le m_B|h|$ for r between s^0 and $s^0 + h$, and G is Lipschitz in the state, so

$$\begin{aligned}\tilde{\boldsymbol{y}}_h = \boldsymbol{y}^0 - \int_{s^0}^{s^0+h} \mathsf{G}(r, Z^{\mathrm{abs}}(r; (s^0 + h, \boldsymbol{y}^0)))dr &= \boldsymbol{y}^0 - \int_{s^0}^{s^0+h} \mathsf{G}(r, \boldsymbol{y}^0)\, dr + O(h^2) \\ &= \boldsymbol{y}^0 - h\,\mathsf{G}(s^0, \boldsymbol{y}^0) + o(h),\end{aligned}$$

because s^0 is a Lebesgue point of $r \mapsto \mathsf{G}(r, \boldsymbol{y}^0)$ (only a_ℓ is discontinuous, and a Lebesgue point is two-sided). By (i), $\boldsymbol{y} \mapsto \tau(s^0, \boldsymbol{y})$ is differentiable at $\boldsymbol{y}^0$, so $\tau(s^0, \tilde{\boldsymbol{y}}_h) = \tau(z^0) - h\,\nabla_{\boldsymbol{y}}\tau(z^0) \cdot \mathsf{G}(s^0, \boldsymbol{y}^0) + o(h)$. Inserting this in (4.20),

$$\tau(s^0 + h, \boldsymbol{y}^0) - \tau(z^0) = -h(1 + \mathsf{G}(s^0, \boldsymbol{y}^0) \cdot \nabla_{\boldsymbol{y}}\tau(z^0)) + o(h),$$

so $\partial_s \boldsymbol{V}(z^0)$ exists and equals $-1 - \mathsf{G}(s^0, \boldsymbol{y}^0) \cdot \nabla_{\boldsymbol{y}} \boldsymbol{V}(z^0)$, which is (4.16); Fréchet differentiability at z^0 then follows from the previous paragraph. No uniformity in $\boldsymbol{y}^0$ was needed: the exceptional set is N, determined by a_ℓ alone.

Measure statements. N is null, and Fubini gives the a.e. statement on $\boldsymbol{D}$.

(iii). Let $F(\kappa, s, \boldsymbol{y}) \overset{\mathbf{def}}{=} \Xi(\kappa; s, \boldsymbol{y})$ on $\{(\kappa, s, \boldsymbol{y}) : s \in J,\ \kappa \in [0, \boldsymbol{T}_h - s)\}$. We show $F \in C^1$ there. $\partial_\kappa F = \Lambda(\kappa; s, \boldsymbol{y})$ (for every κ, since Ξ is C^1 in elapsed time) and $\partial_{\boldsymbol{y}} F = \boldsymbol{e}_1^\top Y(\kappa; s, \boldsymbol{y})$ (lemma 4.9); both are continuous in $(\kappa, s, \boldsymbol{y})$.

For $\partial_s F$, write $F(\kappa, s, \boldsymbol{y}) = \Xi^{\mathrm{abs}}(s+\kappa; (s, \boldsymbol{y}))$ and vary s at fixed absolute time $t = s+\kappa$ and at fixed κ separately. At fixed t: by the cocycle property, $Z^{\mathrm{abs}}(t; (s + h, \boldsymbol{y})) = Z^{\mathrm{abs}}(t; (s, \tilde{\boldsymbol{y}}_h))$ with $\tilde{\boldsymbol{y}}_h \overset{\mathbf{def}}{=} Z^{\mathrm{abs}}(s; (s + h, \boldsymbol{y}))$ the backward state at time s of the prediction from $(s + h, \boldsymbol{y})$. Since s is a continuity point of a_ℓ (as $s \in J$), $\tilde{\boldsymbol{y}}_h = \boldsymbol{y} - h\,\mathsf{G}(s, \boldsymbol{y}) + o(h)$. By lemma 4.9 for the start time s and absolute end time t (elapsed time $t - s$), $Z^{\mathrm{abs}}(t; (s, \tilde{\boldsymbol{y}}_h)) = Z^{\mathrm{abs}}(t; (s, \boldsymbol{y})) - h\,Y(t - s; s, \boldsymbol{y})\mathsf{G}(s, \boldsymbol{y}) + o(h)$. Hence

$$\partial_s|_t Z^{\mathrm{abs}}(t; (s, \boldsymbol{y})) = -Y(t - s; s, \boldsymbol{y})\, \mathsf{G}(s, \boldsymbol{y}),$$

whatever happens to the forcing between s and t: the only ingredients are the continuity of a_ℓ at s and the forcing-free variational equation. At fixed κ, $\partial_s F = \partial_t \Xi^{\mathrm{abs}} + \partial_s|_t \Xi^{\mathrm{abs}} = \Lambda(\kappa; s, \boldsymbol{y}) - \boldsymbol{e}_1^\top Y(\kappa; s, \boldsymbol{y})\mathsf{G}(s, \boldsymbol{y})$, which is continuous in $(\kappa, s, \boldsymbol{y})$ for $s \in J$ (continuity of Λ, Y, and of $\mathsf{G}(\cdot, \boldsymbol{y})$ on J). So $F \in C^1$.

At $\kappa = \tau(s, \boldsymbol{y})$ we have $F = 0$ and $\partial_\kappa F = \Lambda(\tau; z) \le -\gamma_K < 0$. The implicit function theorem gives a C^1 function $\hat{\kappa}(s, \boldsymbol{y})$ near z with $F(\hat{\kappa}, s, \boldsymbol{y}) = 0$ and $\hat{\kappa}(z) = \tau(z)$; since τ is continuous and $\hat{\kappa}$ is the unique zero of $F(\cdot, s', \boldsymbol{y}')$ near $\tau(z)$, $\hat{\kappa} = \tau$ on a neighbourhood. Hence $\tau \in C^1$ there, with $\nabla_{\boldsymbol{y}}\tau$ as in (4.15) and

$$\partial_s \tau = -\frac{\partial_s F}{\partial_\kappa F} = -\frac{\Lambda(\tau; z) - \boldsymbol{e}_1^\top Y(\tau; s, \boldsymbol{y})\mathsf{G}(s, \boldsymbol{y})}{\Lambda(\tau; z)} = -1 - \mathsf{G}(s, \boldsymbol{y}) \cdot \nabla_{\boldsymbol{y}}\tau.$$

■

Remark 4.11 (**$\boldsymbol{V}$ is not C^1 in time under rough forcing**). *Suppose* a_ℓ *is continuous on* $(t_1-\eta,t_1)$ *and on* $(t_1,t_1+\eta)$ *with distinct one-sided limits* $\mathsf{a}_\ell(t_1^\pm)$. *By* ((iii)) *on each side,* $\partial_s\boldsymbol{V}$ *has one-sided limits at* $s=t_1$ *for each* $\boldsymbol{y}$, *and by the formula in* ((iii)), *using that* a_ℓ *enters* $\mathsf{G}(s,\boldsymbol{y})$ *only through its second component and that* $\nabla_{\boldsymbol{y}}\boldsymbol{V}$ *is continuous across* t_1 *(part (i)),*

$$\partial_s\boldsymbol{V}(t_1^+,\boldsymbol{y})-\partial_s\boldsymbol{V}(t_1^-,\boldsymbol{y})=-\big[\mathsf{a}_\ell(t_1^+)-\mathsf{a}_\ell(t_1^-)\big]\,\partial_\Lambda\boldsymbol{V}(t_1,\boldsymbol{y}).$$

Moreover $\partial_\Lambda\boldsymbol{V}(t_1,\boldsymbol{y})>0$ *whenever* $\tau(t_1,\boldsymbol{y})<\pi/\omega_\alpha$, $\omega_\alpha \overset{\textit{def}}{=} \sqrt{\alpha(1-\alpha/4)}$ *(and at every state if* $\alpha\ge 4$*): by* (4.15), $\partial_\Lambda\boldsymbol{V}=-Y_{12}(\tau;z)/\Lambda(\tau;z)=Y_{12}(\tau;z)/|\Lambda(\tau;z)|$, *and* $w \overset{\textit{def}}{=} Y_{12}$ *(second column of* (4.13)*) solves* $w''+\alpha w'+\alpha\mathcal{V}'(\Xi)w=0$, $w(0)=0$, $w'(0)=1$; *the substitution* $u=e^{\alpha\kappa/2}w$ *gives* $u''+(\alpha\mathcal{V}'(\Xi)-\alpha^2/4)u=0$ *with coefficient* $\le\alpha-\alpha^2/4=\omega_\alpha^2$ *(as* $0<\mathcal{V}'\le 1$*), so by Sturm comparison* u, *hence* w, *has no zero in* $(0,\pi/\omega_\alpha)$, *and none at all if* $\alpha\ge 4$ *(this is the argument of lemma* 3.28*(b),(c) with* $\tilde c=\mathcal{V}'(\Xi)$*). At such points* $\partial_s\boldsymbol{V}$ *jumps and* $\boldsymbol{V}\notin C^1$. *For* $\tau\ge\pi/\omega_\alpha$ *the sensitivity may vanish, and where it does the jump of* $\partial_s\boldsymbol{V}$ *degenerates; the conclusion "*$\boldsymbol{V}\notin C^1$ *near* $(t_1,\boldsymbol{y})$*" requires* $\partial_\Lambda\boldsymbol{V}(t_1,\boldsymbol{y})\neq 0$. *(In the numerical setting* $\alpha=2$, $\boldsymbol{T}_h=1$ *one has* $\pi/\omega_\alpha=\pi>1\ge\tau$, *so the sensitivity is positive everywhere on* $\mathbb{D}$*.) A numerical check, in a separate test with a horizon* $\boldsymbol{T}_h>t_1$ *large enough for the contact to occur before it, with a jump of* a_ℓ *from* -0.5 *to* 0.3 *at* $t_1=1$ *and the state* $(\Xi,\Lambda)=(1.2,-3.0)$, *gives a measured jump of* $\partial_s\tau$ *of* -0.25456 *against the predicted* -0.25458. *The Lipschitz-in-time regularity of* ((ii)) *is therefore sharp.*

5. Hamilton–Jacobi characterisation and learning with guarantees.

Evaluating τ online means simulating the prediction flow at every control instant, and the resulting latency breaks the feedback loop. We therefore approximate τ by a network $\boldsymbol{V}_\theta$ trained on the Hamilton–Jacobi equation that τ satisfies as a value function, and we prove that the L^2 error of any C^1 candidate is bounded by the square root of its training loss. The constant in this estimate depends on the compact training domain only through the maximal predicted collision time and the Lipschitz constants of the prediction flow; it does not involve the transversality margin and does not degenerate at the boundary of $\mathbb{D}$. We keep the notation of section 4.

5.1. Value function, HJB equation, and the loss.

The incoming collision surface.. Let

$$\Sigma^- \overset{\textbf{def}}{=} \{(t,0,\Lambda):\ 0\le t<\boldsymbol{T}_h,\ \Lambda<0\} \tag{5.1}$$

be the *incoming* part of the collision plane $\partial\mathcal{Q}=\{\Xi=0\}$.

Value function.. Set $\boldsymbol{V}\overset{\textbf{def}}{=}\tau$ on $\mathbb{D}$ and $\boldsymbol{V}\overset{\textbf{def}}{=}0$ on Σ^-. By (4.2) (optimality principle), for $z=(s,\boldsymbol{y})\in\mathbb{D}$ and $0\le\kappa\le\boldsymbol{V}(s,\boldsymbol{y})$,

$$\boldsymbol{V}\big(s+\kappa,Z(\kappa;z)\big)=\boldsymbol{V}(s,\boldsymbol{y})-\kappa, \tag{5.2}$$

where at $\kappa=\boldsymbol{V}(s,\boldsymbol{y})$ the left side is evaluated at the collision point $(s+\tau(z),0,\Lambda(\tau(z);z))$, which lies in Σ^- because $\Lambda(\tau(z);z)<0$ and $s+\tau(z)<\boldsymbol{T}_h$ by definition of $\mathbb{D}$, and equals 0.

HJB equation.. Differentiating (5.2) at $\kappa = 0$, formally, gives the transport equation

$$\partial_s \boldsymbol{V} + \mathsf{G}(s, \boldsymbol{y}) \cdot \nabla_{\boldsymbol{y}} \boldsymbol{V} = -1 \ \text{ on } \mathbb{D}, \qquad \boldsymbol{V} = 0 \ \text{ on } \Sigma^-. \tag{5.3}$$

The senses in which $\boldsymbol{V}$ solves (5.3) are made precise in proposition 5.15: it satisfies the integrated form (5.2) exactly along every characteristic; it satisfies (5.3) classically at every state for all times outside a fixed null set determined by a_ℓ, hence a.e. on $\mathbb{D}$ (lemma 4.10((ii))); and it is a classical C^1 solution on every time interval on which a_ℓ is continuous (lemma 4.10((iii))). Writing (5.3) as $\partial_s \boldsymbol{V} + H(s, \boldsymbol{y}, \nabla_{\boldsymbol{y}} \boldsymbol{V}) = 0$, the Hamiltonian $H(s, \boldsymbol{y}, p) = p \cdot \mathsf{G}(s, \boldsymbol{y}) + 1$ is affine in p, so the characteristics are the predicted OVM trajectories, independent of p, and they do not cross. This is what makes the equation integrable along characteristics from the collision set, which is all the proofs below use: a solution that is absolutely continuous along characteristics—for instance a locally Lipschitz a.e. solution—is thereby determined by its values on Σ^- (proposition 5.15(d),(e)), whereas a merely continuous a.e. solution is not (remark 5.16). Whether $\boldsymbol{V}$ is also the unique viscosity solution of (5.3) is discussed in remark 5.16; since a_ℓ is only measurable in time this requires the L^1-in-time theory, and it is not used anywhere in this document.

Topological conventions.. All sets of states lie in the half-space $\mathcal{H} \overset{\text{def}}{=} [0, \boldsymbol{T}_h) \times \mathbb{R}^2$; "open", "neighbourhood" and "interior" refer to the relative topology of $\mathcal{H}$ unless $\mathrm{int}_{\mathbb{R}^3}$ is written. The face $\{s = 0\}$ is $\mathcal{L}^3$-null, so almost-everywhere statements are unaffected by it; Rademacher's theorem and the area and co-area formulas are applied on the $\mathbb{R}^3$-open set $\mathrm{int}_{\mathbb{R}^3} K = \mathrm{int}\, K \setminus \{s = 0\}$ of a compact neighbourhood K, which contains $\boldsymbol{D}$ up to a null set. At states with $s = 0$, derivatives in s are one-sided. (The prediction flow is defined for absolute times in $[0, \boldsymbol{T}_h + 1]$ only, lemma 3.3; there is no negative-time extension, and none is used.)

Forward tube, endpoint map, boundary set.. Fix a compact $\boldsymbol{D} \Subset \mathbb{D}$ and set

$$\boldsymbol{D}^+ \overset{\text{def}}{=} \{(s + \kappa, Z(\kappa; s, \boldsymbol{y})) : \ (s, \boldsymbol{y}) \in \boldsymbol{D}, \ 0 \le \kappa < \boldsymbol{V}(s, \boldsymbol{y})\} \subset \mathbb{D} \quad \text{(by lemma 4.2(b))}, \tag{5.4}$$

$$\Psi(z) \overset{\text{def}}{=} (s + \tau(z), \, Z(\tau(z); z)) \in \Sigma^-, \quad \tilde{\Psi}(z) \overset{\text{def}}{=} (s + \tau(z), \, \Lambda(\tau(z); z)) \in \mathbb{R}^2, \quad z = (s, \boldsymbol{y}) \in \mathbb{D}, \tag{5.5}$$

$$\mathcal{Q}_{\boldsymbol{D}} \overset{\text{def}}{=} \Psi(\boldsymbol{D}^+) = \Psi(\boldsymbol{D}) \subset \Sigma^- \subset \partial\mathcal{Q}. \tag{5.6}$$

$\partial\mathcal{Q} = \{\Xi = 0\}$ is a plane in (t, Ξ, Λ)-space, and $\tilde{\Psi}$ is Ψ written in the chart (t, Λ) of that plane; the chart is an isometry, so $|\Psi(z_1) - \Psi(z_2)| = |\tilde{\Psi}(z_1) - \tilde{\Psi}(z_2)|$ and $\mathcal{H}^2 \llcorner \partial\mathcal{Q}$ corresponds to Lebesgue measure on $\mathbb{R}^2$. Note $\Psi(\boldsymbol{D}^+) = \Psi(\boldsymbol{D})$ because points on one characteristic share their endpoint.

Lemma 5.1 (Continuity of the zero extension on the incoming surface). *$\Sigma^- \subset \boldsymbol{cl}[\mathbb{D}] \setminus \mathbb{D}$, and $\boldsymbol{V}$ is continuous on $\mathbb{D} \cup \Sigma^-$. For every compact $\boldsymbol{D} \Subset \mathbb{D}$ one has $\mathcal{Q}_{\boldsymbol{D}} \subset \Sigma^-$ (indeed $\Lambda \le -\gamma_{\boldsymbol{D}}$ on $\mathcal{Q}_{\boldsymbol{D}}$) and $\boldsymbol{cl}[\boldsymbol{D}^+] = \boldsymbol{D}^+ \cup \mathcal{Q}_{\boldsymbol{D}}$.*

Proof. Let $p = (t^*, 0, \Lambda^*) \in \Sigma^-$ and $\gamma \overset{\text{def}}{=} -\Lambda^*/2 > 0$. The prediction from p exists on $[0, \boldsymbol{T}_h + 1]$ (lemma 3.3); choose $h \in (0, 1)$ with $t^* + h < \boldsymbol{T}_h$ and $\Lambda^{\mathrm{abs}}(r; p) \le -\frac{3}{2}\gamma$ for

$r \in I \stackrel{\text{def}}{=} [\max\{0, t^* - h\},\, t^* + h]$ (the prediction is defined for absolute times in $[0, \boldsymbol{T}_h + 1]$ only, so the interval is clipped at 0; $t^* = 0$ is allowed). Let B be the closed unit ball around p intersected with $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$, and L_B as in lemma 3.3(b); pick $\rho \in (0, h/2)$ with $L_B \rho \le \gamma/2$ and $\rho \le \gamma h/2$. For $z = (s, X, \Lambda) \in \Omega$ with $|z - p| < \rho$ (so $s \ge 0$ and $s \in I$), (3.10) gives $\Lambda^{\text{abs}}(r; z) \le -\gamma$ on $I \ni s$, hence $\Xi^{\text{abs}}(r; z) = X + \int_s^r \Lambda^{\text{abs}} \le X - \gamma(r - s)$ for $r \in [s, t^* + h]$. So the first zero of the predicted gap lies in $[s, s + X/\gamma] \subset [s, t^* + h]$, where $\Lambda \le -\gamma < 0$: it is transversal and occurs before $\boldsymbol{T}_h$. Thus $z \in \mathbb{D}$, which shows $p \in \mathbf{cl}[\mathbb{D}]$ (and $p \notin \mathbb{D} \subset \Omega$), and $\tau(z) \le X/\gamma \to 0$ as $z \to p$. Since $\boldsymbol{V} = 0$ on Σ^-, $\boldsymbol{V}$ is continuous at p; continuity on $\mathbb{D}$ is lemma 4.5(a). The inclusion $\mathcal{Q}_{\boldsymbol{D}} \subset \Sigma^-$ is lemma 4.5(b) together with $s + \tau(z) = \sigma(z) < \boldsymbol{T}_h$. Finally $\boldsymbol{D}^+ \cup \mathcal{Q}_{\boldsymbol{D}}$ is the image of the compact set $\{(\kappa, z) : z \in \boldsymbol{D},\ 0 \le \kappa \le \tau(z)\}$ under the continuous map $(\kappa, z) \mapsto (s + \kappa, Z(\kappa; z))$, hence compact, and every point of $\mathcal{Q}_{\boldsymbol{D}}$ is the limit of $(s + \kappa, Z(\kappa; z))$ as $\kappa \uparrow \tau(z)$. ∎

Remark 5.2 (No global continuous extension). *$\boldsymbol{V}$ has in general no continuous extension to the opening part $\{\Lambda > 0\}$ of the collision plane, nor to the grazing part of $\partial\mathbb{D}$ (remark 3.12). For instance, with $\alpha = 0.1$, $\boldsymbol{T}_h = 3$ and the smooth admissible leader data $v_\ell = u + q'$, $\mathsf{a}_\ell = u' + q''$, where $q(t) = 0.01t(2 - t)$ and $u' = \alpha(\mathcal{V}(q) - u)$, $u(0) = 0.5$ (so $0.33 \le v_\ell \le 0.52$ and $-0.07 \le \mathsf{a}_\ell \le -0.057$), the prediction from $z_0 = (0, 0, 0.02)$ is $(\Xi, \Lambda) = (q, q')$, while $z_\delta = (0, \delta, 0.02) \in \mathbb{D}$ with $\tau(z_\delta) \to 2$ as $\delta \downarrow 0$ (numerically $\tau(z_\delta) = 2.048,\ 2.005,\ 2.0005$ for $\delta = 10^{-3}, 10^{-4}, 10^{-5}$). Extending $\boldsymbol{V}$ by 0 on the whole plane $\{\Xi = 0\}$ would therefore produce a function that is not continuous on $\mathbf{cl}[\mathbb{D}]$: at this opening point the* zero *extension fails (the limiting TTC is 2, so a continuous extension with the value 2 exists there, but it is not the boundary datum). At grazing points of $\partial\mathbb{D}$ inside Ω, by contrast, no continuous extension exists at all (remark 3.12), and $\boldsymbol{V}$ is not defined there. This is why the datum in (5.3) is imposed on Σ^- only, and why all function classes below require continuity on $\mathbb{D} \cup \Sigma^-$ and nothing more. (A neural network is continuous on $\mathbb{R}^3$, so this is no restriction in practice; but the boundary collocation points must be drawn on Σ^-, i.e. on $\{\Xi = 0,\ \Lambda < 0\}$, not on the whole plane: at an opening point $(t, 0, \Lambda > 0)$ the target value is not 0.)*

Loss.. For $\boldsymbol{V}_\theta \in C^1(\mathbb{D}) \cap C(\mathbb{D} \cup \Sigma^-)$, the HJB residual and the two loss components are

$$r_\theta \stackrel{\text{def}}{=} \partial_s \boldsymbol{V}_\theta + \mathsf{G} \cdot \nabla_{\boldsymbol{y}} \boldsymbol{V}_\theta + 1, \qquad J_{\text{HJB}} \stackrel{\text{def}}{=} \int_{\boldsymbol{D}^+} |r_\theta|^2 \, d\mathcal{L}^3, \qquad J_{\partial\mathcal{Q}} \stackrel{\text{def}}{=} \int_{\mathcal{Q}_{\boldsymbol{D}}} |\boldsymbol{V}_\theta|^2 \, d\mathcal{H}^2, \tag{5.7}$$

and the training loss is

$$J_{\text{total}}(\theta) \stackrel{\text{def}}{=} \lambda_{\text{HJB}} J_{\text{HJB}} + \lambda_{\partial\mathcal{Q}} J_{\partial\mathcal{Q}}, \qquad \lambda_{\text{HJB}}, \lambda_{\partial\mathcal{Q}} > 0. \tag{5.8}$$

The residual r_θ is measurable, but under L^∞ forcing it is in general not continuous, since it contains $\mathsf{a}_\ell(s)\, \partial_\Lambda \boldsymbol{V}_\theta$; nothing below requires more than measurability. In implementation the integrals are approximated by Monte-Carlo quadrature over collocation points drawn from measures equivalent to $\mathcal{L}^3 \llcorner \boldsymbol{D}^+$ and $\mathcal{H}^2 \llcorner \mathcal{Q}_{\boldsymbol{D}}$ (the boundary points lying on the incoming surface Σ^-); the analysis is carried out at the continuum level, and the constants below refer to the unweighted Lebesgue and surface measures.

Bounded reference set and constants.. Fix a compact neighbourhood $K \Subset \mathbb{D}$ of $\boldsymbol{D}$ (it exists since $\mathbb{D}$ is open), with forward tube K^+ defined as in (5.4). Let B be the closed 1-neighbourhood of the compact set $\mathbf{cl}[K^+]$, intersected with $[0, \boldsymbol{T}_h) \times \mathbb{R}^2$, and let m_B, L_B be the

constants of lemma 3.3(b) for B. Recall from lemma 3.3 that the prediction flow is defined, forward and backward, on all of $[0, \boldsymbol{T}_h + 1]$, and that (4.3) holds for all absolute times, in particular for *backward* evaluation. Let $\tau_{\max} = \tau_{\max}(\boldsymbol{D})$ and $\widehat{L}_\tau(\boldsymbol{D})$ be as in theorem 4.8.

5.2. The error identity along characteristics.

Lemma 5.3 (Error along a characteristic). *Let $\boldsymbol{V}_\theta \in C^1(\mathbb{D}) \cap C(\mathbb{D} \cup \Sigma^-)$, $e \overset{\text{def}}{=} \boldsymbol{V}_\theta - \boldsymbol{V}$, $z = (s, \boldsymbol{y}) \in \boldsymbol{D}$, $\nu \overset{\text{def}}{=} \boldsymbol{V}(z)$, and $c(\kappa) \overset{\text{def}}{=} (s + \kappa, Z(\kappa; z))$ for $\kappa \in [0, \nu]$.*

(a) *The improper integral $\int_0^{\nu^-} r_\theta(c(\kappa))\, d\kappa \overset{\text{def}}{=} \lim_{\nu' \uparrow \nu} \int_0^{\nu'} r_\theta(c(\kappa))\, d\kappa$ exists, and*

$$e(s, \boldsymbol{y}) = \boldsymbol{V}_\theta(\Psi(z)) - \int_0^{\nu^-} r_\theta\big(s + \kappa, Z(\kappa; z)\big)\, d\kappa. \tag{5.9}$$

(b) *For every $z \in \boldsymbol{D}$, as an inequality in $[0, \infty]$,*

$$|e(s, \boldsymbol{y})|^2 \le 2\,|\boldsymbol{V}_\theta(\Psi(z))|^2 + 2\nu \int_0^{\nu} |r_\theta(s + \kappa, Z(\kappa; z))|^2\, d\kappa. \tag{5.10}$$

(c) *If the integral on the right of (5.10) is finite, then $r_\theta \circ c \in L^1(0, \nu)$ and the integral in (5.9) is an ordinary Lebesgue integral. This is the case for $\mathcal{L}^3$-a.e. $z \in \boldsymbol{D}$ whenever $J_{\text{HJB}} < \infty$ (by Tonelli and proposition 5.12, whose proof does not use this lemma), and for every $z \in \boldsymbol{D}$ if $\boldsymbol{V}_\theta$ is C^1 on a neighbourhood of $\boldsymbol{cl}[\boldsymbol{D}^+]$ (then $r_\theta \circ c \in L^\infty(0, \nu)$).*

Proof. The curve c is Lipschitz, lies in $\boldsymbol{D}^+ \subset \mathbb{D}$ for $\kappa < \nu$, and ends at $\Psi(z) \in \mathcal{Q}_{\boldsymbol{D}} \subset \Sigma^-$ (lemma 5.1). Since $\boldsymbol{V}_\theta \in C^1(\mathbb{D})$, $\boldsymbol{V}_\theta \circ c$ is absolutely continuous on $[0, \nu']$ for every $\nu' < \nu$, with $(\boldsymbol{V}_\theta \circ c)'(\kappa) = \partial_s \boldsymbol{V}_\theta + \nabla_{\boldsymbol{y}} \boldsymbol{V}_\theta \cdot \partial_\kappa Z = \partial_s \boldsymbol{V}_\theta + \nabla_{\boldsymbol{y}} \boldsymbol{V}_\theta \cdot \mathsf{G} = r_\theta(c(\kappa)) - 1$ for a.e. κ (the chain rule for a C^1 function along a Lipschitz curve, and $\partial_\kappa Z = \mathsf{G}(s + \kappa, Z)$ a.e.). Also $\boldsymbol{V}(c(\kappa)) = \nu - \kappa$ by (5.2). Hence $(e \circ c)' = r_\theta \circ c$ a.e. on $[0, \nu']$ and

$$e(c(\nu')) - e(z) = \int_0^{\nu'} r_\theta(c(\kappa))\, d\kappa \qquad (0 \le \nu' < \nu). \tag{5.11}$$

As $\nu' \uparrow \nu$, $c(\nu') \to \Psi(z)$, so $\boldsymbol{V}_\theta(c(\nu')) \to \boldsymbol{V}_\theta(\Psi(z))$ by continuity of $\boldsymbol{V}_\theta$ on $\mathbb{D} \cup \Sigma^-$ at $\Psi(z)$, and $\boldsymbol{V}(c(\nu')) = \nu - \nu' \to 0$. Thus the left side of (5.11) converges, the improper integral exists, and (a) follows. Note that interior C^1 regularity and continuity at the endpoint do *not* by themselves make $r_\theta \circ c$ Lebesgue integrable on $[0, \nu]$: the candidate $\boldsymbol{V}_\theta = \Xi \sin(\Xi^{-2})$ (extended by 0 at $\Xi = 0$) is in $C^1(\mathbb{D}) \cap C(\mathbb{D} \cup \Sigma^-)$, and along a characteristic its exact residual is $r_\theta(c(\kappa)) = 1 + \Lambda(\kappa)\big[\sin(\Xi(\kappa)^{-2}) - 2\Xi(\kappa)^{-2}\cos(\Xi(\kappa)^{-2})\big]$. Near a transversal endpoint Λ is bounded and bounded away from 0, and Ξ is strictly decreasing with $\Xi' = \Lambda$; substituting $x = \Xi(\kappa)$, the leading term contributes $2\int_0^a x^{-2}|\cos(x^{-2})|\, dx = \int_{a^{-2}}^\infty u^{-1/2}|\cos u|\, du = \infty$ (with $u = x^{-2}$), while the remaining terms have finite integral; so $r_\theta \circ c \notin L^1(0, \nu)$, although the truncated integrals converge.

(b) By Cauchy–Schwarz on $[0, \nu']$ and monotonicity of the integral, $|\int_0^{\nu'} r_\theta \circ c|^2 \le \nu' \int_0^{\nu'} |r_\theta \circ c|^2 \le \nu \int_0^{\nu} |r_\theta \circ c|^2 \in [0, \infty]$ for every $\nu' < \nu$; let $\nu' \uparrow \nu$ in (5.11) and use $(a + b)^2 \le 2a^2 + 2b^2$.

(c) If $\int_0^\nu |r_\theta \circ c|^2 < \infty$ then $r_\theta \circ c \in L^2(0, \nu) \subset L^1(0, \nu)$ and the improper integral equals the Lebesgue integral (dominated convergence). If $J_{\text{HJB}} < \infty$, proposition 5.12 gives

$$\int_{\boldsymbol{D}} \int_0^{\boldsymbol{V}(z)} |r_\theta(c_z(\kappa))|^2 d\kappa\, dz < \infty,$$

so the inner integral is finite for a.e. $z \in \boldsymbol{D}$ (the integrand is Borel on $[0,\tau_{\max}]\times\boldsymbol{D}$ once a Borel representative of a_ℓ is fixed; r_θ is measurable but, under L^∞ forcing, not continuous, since it contains $\mathsf{a}_\ell(s)\partial_\Lambda \boldsymbol{V}_\theta$). If $\boldsymbol{V}_\theta$ is C^1 on a neighbourhood of the compact set $\mathbf{cl}[\boldsymbol{D}^+]$, then $\nabla \boldsymbol{V}_\theta$ is bounded there and, since $\mathsf{a}_\ell \in L^\infty$, r_θ is essentially bounded on $\boldsymbol{D}^+$; so $r_\theta \circ c \in L^\infty(0,\nu)$. ∎

Integrating (5.10) (valid for every $z \in \boldsymbol{D}$ as an inequality in $[0,\infty]$) over $\boldsymbol{D}$ and using $\nu \le \tau_{\max}$,

$$\begin{aligned} &\|e\|^2_{L^2(\boldsymbol{D})} \;\le\; 2I_1 + 2\tau_{\max} I_2, \\ &I_1 \overset{\mathbf{def}}{=} \int_{\boldsymbol{D}} |\boldsymbol{V}_\theta(\Psi(z))|^2 dz, \qquad I_2 \overset{\mathbf{def}}{=} \int_{\boldsymbol{D}} \int_0^{\boldsymbol{V}(z)} |r_\theta(s+\kappa, Z(\kappa;z))|^2 d\kappa\, dz. \end{aligned} \tag{5.12}$$

5.3. The endpoint map: Lipschitz bound, slice estimate, Jacobian.

Lemma 5.4 ($\tilde\Psi$ is Lipschitz on compacts). *For $z_1, z_2 \in K$,*

$$|\tilde\Psi(z_1) - \tilde\Psi(z_2)| \le C_\Psi(K)|z_1 - z_2|, \qquad C_\Psi(K) \overset{\boldsymbol{def}}{=} (1 + \widehat{L}_\tau(K)) + (m_B \widehat{L}_\tau(K) + L_B + m_B).$$

Proof. Write $z_i = (s_i, \boldsymbol{y}_i)$, $\tau_i \overset{\mathbf{def}}{=} \tau(z_i)$ and $t_i \overset{\mathbf{def}}{=} s_i + \tau_i \in [0, \boldsymbol{T}_h)$ (the predicted contact time; $t_i < \boldsymbol{T}_h$ because $z_i \in \mathbb{D}$). First component: $|t_1 - t_2| \le |s_1 - s_2| + |\tau_1 - \tau_2| \le (1 + \widehat{L}_\tau(K))|z_1 - z_2|$ by theorem 4.8 on K. Second component: $\Lambda(\tau_i; z_i) = \Lambda^{\mathrm{abs}}(t_i; z_i)$ is the predicted relative velocity of z_i at the *absolute* time t_i. We compare in absolute time, inserting $\Lambda^{\mathrm{abs}}(t_2; z_1)$, which is defined since $t_2 \in [0, \boldsymbol{T}_h) \subset [0, \boldsymbol{T}_h + 1]$ (lemma 3.3(a)):

$$\begin{aligned} |\Lambda^{\mathrm{abs}}(t_1; z_1) - \Lambda^{\mathrm{abs}}(t_2; z_2)| &\le |\Lambda^{\mathrm{abs}}(t_1; z_1) - \Lambda^{\mathrm{abs}}(t_2; z_1)| + |\Lambda^{\mathrm{abs}}(t_2; z_1) - \Lambda^{\mathrm{abs}}(t_2; z_2)| \\ &\le m_B |t_1 - t_2| + L_B |z_1 - z_2| \end{aligned}$$

by (4.3) ($z_1, z_2 \in K \subset B$; all absolute times in $[0, \boldsymbol{T}_h)$). With the first-component bound this is $\le (m_B \widehat{L}_\tau(K) + L_B + m_B)|z_1 - z_2|$. Adding the two components gives the claim. (Comparing at a common *elapsed* time τ_2 instead would evaluate the prediction from z_1 at the absolute time $s_1 + \tau_2 = t_2 + (s_1 - s_2)$, which may exceed $\boldsymbol{T}_h + 1$ and leave the domain of the flow; this is why lemma 4.4 carries its domain condition.) ∎

Remark 5.5. *$C_\Psi(K)$ is the only constant in this part that is not controlled uniformly in K: through $\widehat{L}_\tau(K)$ it depends on the transversality margin γ_K (the local term $2(L_B + m_B)/\gamma_K$ of (4.10)) and on the secant radius $\varrho_0(K)$ of proposition 4.7, and our bound for it is not uniform as K approaches $\partial\mathbb{D}$. It is used only to guarantee that $\tilde\Psi$ is Lipschitz, hence differentiable a.e. (Rademacher) and admissible in the co-area formula; it does not enter the final constant.*

Lemma 5.6 (The flow map at fixed start time is bi-Lipschitz). *Fix $s \in [0, \boldsymbol{T}_h)$ and let $\mathcal{F}_s(\kappa, \boldsymbol{y}) \overset{\boldsymbol{def}}{=} (s + \kappa, Z(\kappa; s, \boldsymbol{y}))$. For $(\kappa_i, \boldsymbol{y}_i)$ with $(s, \boldsymbol{y}_i) \in B$, $0 \le \kappa_i$, $s + \kappa_i < \boldsymbol{T}_h$, and $\mathcal{F}_s(\kappa_i, \boldsymbol{y}_i) \in B$:*

$$|\mathcal{F}_s(\kappa_1, \boldsymbol{y}_1) - \mathcal{F}_s(\kappa_2, \boldsymbol{y}_2)| \;\ge\; \frac{1}{1 + L_B} |(\kappa_1, \boldsymbol{y}_1) - (\kappa_2, \boldsymbol{y}_2)|. \tag{5.13}$$

Proof. Write $p_i = (t_i, \boldsymbol{x}_i) \overset{\text{def}}{=} \mathcal{F}_s(\kappa_i, \boldsymbol{y}_i)$. By uniqueness of the prediction flow, $\boldsymbol{y}_i = Z^{\text{abs}}(s; p_i)$, the state at absolute time s of the trajectory through p_i (backward evaluation), and $\kappa_i = t_i - s$. Hence, by (4.3) with absolute time s,

$$|(\kappa_1, \boldsymbol{y}_1) - (\kappa_2, \boldsymbol{y}_2)| \le |t_1 - t_2| + |Z^{\text{abs}}(s; p_1) - Z^{\text{abs}}(s; p_2)| \le (1 + L_B)|p_1 - p_2|. \qquad \blacksquare$$

Lemma 5.7 (Slice estimate). *For $z_i = (s, \boldsymbol{y}_i) \in K$ with the* same *start time s,*

$$|\tilde{\Psi}(s, \boldsymbol{y}_1) - \tilde{\Psi}(s, \boldsymbol{y}_2)| \;\ge\; c_B\, |\boldsymbol{y}_1 - \boldsymbol{y}_2|, \qquad c_B \overset{\textit{def}}{=} \frac{1}{1 + L_B}. \tag{5.14}$$

Proof. $\Psi(s, \boldsymbol{y}_i) = \mathcal{F}_s(\tau_i, \boldsymbol{y}_i)$ with $\tau_i = \tau(s, \boldsymbol{y}_i)$, and all points lie in $\mathbf{cl}[K^+] \subset B$. By (5.13) and the chart isometry, $|\tilde{\Psi}(s, \boldsymbol{y}_1) - \tilde{\Psi}(s, \boldsymbol{y}_2)| = |\Psi(s, \boldsymbol{y}_1) - \Psi(s, \boldsymbol{y}_2)| \ge c_B|(\tau_1, \boldsymbol{y}_1) - (\tau_2, \boldsymbol{y}_2)| \ge c_B|\boldsymbol{y}_1 - \boldsymbol{y}_2|$. $\blacksquare$

Remark 5.8 (Transversality is used, but not quantitatively). *(5.14) says that two states at the same time with different relative states have endpoints at least $c_B|\boldsymbol{y}_1 - \boldsymbol{y}_2|$ apart on the collision plane: the two endpoints cannot coincide because the characteristics through them are distinct and do not cross. The margin $\gamma_{\boldsymbol{D}}$ does not appear; it enters only through the Lipschitz regularity of τ, which is needed to make sense of $D\tilde{\Psi}$ at all.*

Proposition 5.9 (Jacobian lower bound). *$\tilde{\Psi}$ is differentiable at $\mathcal{L}^3$-a.e. point of $\operatorname{int}_{\mathbb{R}^3} K \supset \boldsymbol{D} \setminus \{s = 0\}$, and at every such point*

$$J_2\tilde{\Psi}(z) \overset{\textit{def}}{=} \sqrt{\det(D\tilde{\Psi}(z)\, D\tilde{\Psi}(z)^\top)} \;\ge\; c_* \overset{\textit{def}}{=} c_B^2 = \frac{1}{(1 + L_B)^2}. \tag{5.15}$$

In particular $D\tilde{\Psi}(z)$ has rank 2 there.

Proof. Differentiability a.e. on the open set $\operatorname{int}_{\mathbb{R}^3} K$ is Rademacher's theorem (lemma 5.4). Let $z = (s, \boldsymbol{y}) \in \operatorname{int}_{\mathbb{R}^3} K$ be a point of differentiability and $A \overset{\text{def}}{=} D\tilde{\Psi}(z) \in \mathbb{R}^{2\times 3}$; write $A = [a_t \;\; A_{\boldsymbol{y}}]$ with $A_{\boldsymbol{y}} \in \mathbb{R}^{2\times 2}$ the derivative in the state directions. For $v \in \mathbb{R}^2$ and small h, $(s, \boldsymbol{y} + hv) \in K$, so (5.14) gives $|\tilde{\Psi}(s, \boldsymbol{y} + hv) - \tilde{\Psi}(s, \boldsymbol{y})| \ge c_B h|v|$; dividing by h and letting $h \to 0$, $|A_{\boldsymbol{y}} v| \ge c_B|v|$. Hence both singular values of $A_{\boldsymbol{y}}$ are $\ge c_B$ and $|\det A_{\boldsymbol{y}}| \ge c_B^2$. By the Cauchy–Binet formula, $\det(AA^\top) = \sum_S \det(A_S)^2$, the sum over the three 2×2 minors A_S of A; keeping only $S = \{\text{state columns}\}$, $\det(AA^\top) \ge (\det A_{\boldsymbol{y}})^2 \ge c_B^4$. $\blacksquare$

Lemma 5.10 (Fibres are arcs of characteristics). *Let $p = (t^*, \Lambda^*) \in \mathbb{R}^2$ and $E_p \overset{\textit{def}}{=} \tilde{\Psi}^{-1}(p) \cap \boldsymbol{D}$. Set $r_p \overset{\textit{def}}{=} \min\{t^*, \tau_{\max}(\boldsymbol{D})\}$ if $E_p \neq \varnothing$ and $r_p \overset{\textit{def}}{=} 0$ otherwise. If $E_p \neq \varnothing$, then $t^* \in (0, \boldsymbol{T}_h)$, and with*

$$\gamma_p : [0, r_p] \to [0, \boldsymbol{T}_h) \times \mathbb{R}^2, \qquad \gamma_p(r) \overset{\textit{def}}{=} (t^* - r,\; Z^{\text{abs}}(t^* - r; (t^*, 0, \Lambda^*)))$$

(the backward prediction from the collision point) we have $E_p \subset \gamma_p([0, r_p])$, and γ_p is Lipschitz with constant $\sqrt{1 + m_B^2}$. Consequently, for every $p \in \mathbb{R}^2$,

$$\mathcal{H}^1(\tilde{\Psi}^{-1}(p) \cap \boldsymbol{D}) \;\le\; \sqrt{1 + m_B^2}\, r_p \;\le\; \ell_{\max} \overset{\textit{def}}{=} \sqrt{1 + m_B^2}\, \tau_{\max}(\boldsymbol{D}). \tag{5.16}$$

Proof. Let $z = (s, \boldsymbol{y}) \in E_p$. Then $t^* = s + \tau(z)$ with $0 < \tau(z) \le \tau_{\max}(\boldsymbol{D})$, and $t^* = \sigma(z) < \boldsymbol{T}_h$ since $z \in \mathbb{D}$; so $t^* \in (0, \boldsymbol{T}_h)$ and $\tau(z) = t^* - s \le \min\{t^*, \tau_{\max}(\boldsymbol{D})\} = r_p$. The prediction from z is at the state $(0, \Lambda^*)$ at absolute time t^*; by uniqueness of the global prediction flow, forward and backward (lemma 3.3(a)), it coincides on $[0, \boldsymbol{T}_h + 1]$ with the prediction from the datum $q_p \overset{\text{def}}{=} (t^*, 0, \Lambda^*)$. Hence $\boldsymbol{y} = Z^{\text{abs}}(s; q_p)$, i.e. $z = \gamma_p(\tau(z))$ with $\tau(z) \in [0, r_p]$. This proves $E_p \subset \gamma_p([0, r_p])$.

The datum $q_p = \Psi(z)$ lies in $\mathbf{cl}[\boldsymbol{D}^+] \subset B$, and for $r \in [0, r_p]$ the absolute time $t^* - r$ lies in $[0, t^*] \subset [0, \boldsymbol{T}_h + 1]$. Since (4.3) holds for data in B at all absolute times in $[0, \boldsymbol{T}_h + 1]$, whether or not the trajectory remains in B (it is proved from the a priori bound $|Z| \le R_B$ along the whole trajectory, lemma 3.3(b)),

$$|\gamma_p(r_1) - \gamma_p(r_2)|^2 = |r_1 - r_2|^2 + |Z^{\text{abs}}(t^* - r_1; q_p) - Z^{\text{abs}}(t^* - r_2; q_p)|^2 \le (1 + m_B^2)|r_1 - r_2|^2.$$

A Lipschitz image of an interval of length r_p has $\mathcal{H}^1$-measure at most $\text{Lip}(\gamma_p)\, r_p$, which gives (5.16); if $E_p = \varnothing$ the bound is trivial. (The parametrisation must stop at r_p: for $r > t^*$ the absolute time $t^* - r$ is negative and the prediction is not defined.) ∎

5.4. The two bounds and the theorem.

Proposition 5.11 (Boundary term). $I_1 \le \dfrac{\ell_{\max}}{c_*}\, J_{\partial\mathcal{Q}} = (1 + L_B)^2 \sqrt{1 + m_B^2}\; \tau_{\max}(\boldsymbol{D})\; J_{\partial\mathcal{Q}}.$

Proof. Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be a Lipschitz extension of $\tilde{\Psi}|_K$ (McShane, componentwise); $f = \tilde{\Psi}$ on $\text{int}_{\mathbb{R}^3} K \supset \boldsymbol{D} \setminus \{s = 0\}$, so $Df = D\tilde{\Psi}$ and $J_2 f = J_2 \tilde{\Psi}$ at every differentiability point of $\text{int}_{\mathbb{R}^3} K$. Let $E \subset \boldsymbol{D}$ be the null set where $\tilde{\Psi}$ is not differentiable, and set $g \overset{\text{def}}{=} \mathbf{1}_{\boldsymbol{D}\setminus E}\, |\boldsymbol{V}_\theta \circ \Psi|^2 / J_2\tilde{\Psi} \ge 0$, measurable and well defined by (5.15). The co-area formula [20, Thm. 3.2.12], [19, Thm. 3.11] for Lipschitz $f : \mathbb{R}^3 \to \mathbb{R}^2$ gives

$$\begin{aligned} I_1 = \int_{\boldsymbol{D}\setminus E} |\boldsymbol{V}_\theta(\Psi(z))|^2 dz = \int_{\mathbb{R}^3} g\, J_2 f\, dz &= \int_{\mathbb{R}^2} \int_{f^{-1}(p)} g\, d\mathcal{H}^1\, dp \\ &= \int_{\mathbb{R}^2} |\boldsymbol{V}_\theta(p)|^2 \int_{\tilde{\Psi}^{-1}(p) \cap (\boldsymbol{D}\setminus E)} \frac{d\mathcal{H}^1}{J_2\tilde{\Psi}}\, dp, \end{aligned}$$

where $\boldsymbol{V}_\theta(p)$ denotes the value at the point of $\partial\mathcal{Q}$ with chart coordinates p (constant on the fibre). By (5.15) and (5.16) the inner integral is $\le \ell_{\max}/c_*$, and the outer integral runs over $\tilde{\Psi}(\boldsymbol{D})$, which in the chart is $\Psi(\boldsymbol{D}) \subset \mathcal{Q}_{\boldsymbol{D}}$ with $dp = d\mathcal{H}^2$. Hence $I_1 \le (\ell_{\max}/c_*) \int_{\mathcal{Q}_{\boldsymbol{D}}} |\boldsymbol{V}_\theta|^2 d\mathcal{H}^2$. ∎

Proposition 5.12 (Interior term). $I_2 \le \dfrac{e^{\alpha\tau_{\max}(\boldsymbol{D})} - 1}{\alpha}\, J_{\text{HJB}} \le \tau_{\max}(\boldsymbol{D})\, e^{\alpha\tau_{\max}(\boldsymbol{D})}\, J_{\text{HJB}}.$

Proof. By Tonelli, $I_2 = \int_0^{\tau_{\max}} \int_{\boldsymbol{D}_\kappa} |r_\theta(\mathcal{F}_\kappa(z))|^2 dz\, d\kappa$ with $\boldsymbol{D}_\kappa \overset{\text{def}}{=} \{z \in \boldsymbol{D} : \tau(z) > \kappa\}$ and $\mathcal{F}_\kappa(s, \boldsymbol{y}) \overset{\text{def}}{=} (s + \kappa, Z(\kappa; s, \boldsymbol{y}))$, which is defined on $\boldsymbol{D}_\kappa$ since there $s + \kappa < s + \tau(z) < \boldsymbol{T}_h$. For fixed $\kappa \in [0, \tau_{\max}]$, $\mathcal{F}_\kappa$ is injective (uniqueness of the flow) and Lipschitz on $\boldsymbol{D}_\kappa$ with constant $\le 1 + L_B + m_B$ (lemma 4.4, whose domain condition holds); its inverse $w = (t, \boldsymbol{x}) \mapsto (t - \kappa, Z^{\text{abs}}(t - \kappa; w))$ is Lipschitz with constant $\le 1 + L_B + m_B$ on $\mathcal{F}_\kappa(\boldsymbol{D}_\kappa) \subset \boldsymbol{D}^+ \subset B$: for w_1, w_2 there, the start times are $s_i = t_i - \kappa$ and $|Z^{\text{abs}}(s_1; w_1) - Z^{\text{abs}}(s_2; w_2)| \le m_B|s_1 - s_2| + L_B|w_1 - w_2|$ by (4.3) (first move the evaluation time along trajectory 1, then compare the two trajectories at the common absolute time s_2).

The Jacobian of $\mathcal{F}_\kappa$ is explicit. At every point of differentiability of $\mathcal{F}_\kappa$ (a.e. on $\boldsymbol{D}_\kappa$, by Rademacher), its Jacobian matrix is block lower-triangular,

$$D\mathcal{F}_\kappa(s,\boldsymbol{y}) = \begin{pmatrix} 1 & 0 \\ \partial_s Z(\kappa;s,\boldsymbol{y}) & Y(\kappa;s,\boldsymbol{y}) \end{pmatrix},$$

with $Y = \partial_{\boldsymbol{y}} Z$ the state linearisation of lemma 4.9, which exists everywhere. Since $\kappa \mapsto Y(\kappa;s,\boldsymbol{y})$ is absolutely continuous and solves (4.13) with $\operatorname{tr}\partial_{\boldsymbol{y}}\mathsf{G} \equiv -\alpha$ (see (4.12): the trace is independent of the state and of the forcing), Jacobi's formula $(\det Y)' = \operatorname{tr}(\partial_{\boldsymbol{y}}\mathsf{G})\det Y$ a.e. gives Liouville's identity

$$\det Y(\kappa;s,\boldsymbol{y}) = e^{-\alpha\kappa} \qquad \text{for all } (s,\boldsymbol{y}) \text{ and } \kappa. \tag{5.17}$$

Hence $J\mathcal{F}_\kappa = |\det D\mathcal{F}_\kappa| = e^{-\alpha\kappa}$ a.e. on $\boldsymbol{D}_\kappa$, and the area formula for the injective Lipschitz map $\mathcal{F}_\kappa$ [19, Thm. 3.9] gives

$$\int_{\boldsymbol{D}_\kappa} |r_\theta \circ \mathcal{F}_\kappa|^2 dz = e^{\alpha\kappa} \int_{\boldsymbol{D}_\kappa} |r_\theta \circ \mathcal{F}_\kappa|^2 \, J\mathcal{F}_\kappa \, dz = e^{\alpha\kappa} \int_{\mathcal{F}_\kappa(\boldsymbol{D}_\kappa)} |r_\theta(w)|^2 \, dw \le e^{\alpha\kappa} J_{\mathrm{HJB}}.$$

Integrating in κ over $[0,\tau_{\max}]$ gives the claim. (Bounding instead the Jacobian of the Lipschitz inverse by $(1+L_B+m_B)^3$ gives the cruder $I_2 \le \tau_{\max}(1+L_B+m_B)^3 J_{\mathrm{HJB}}$, which is also valid; (5.17) removes the flow constants from the interior term altogether.) ∎

Theorem 5.13 (L^2 error is controlled by the training loss). *Let $\boldsymbol{D} \Subset \mathbb{D}$ be compact and $\boldsymbol{V}_\theta \in C^1(\mathbb{D}) \cap C(\mathbb{D} \cup \Sigma^-)$. Then*

$$\|\boldsymbol{V}_\theta - \boldsymbol{V}\|^2_{L^2(\boldsymbol{D})} \;\le\; C^2_{\boldsymbol{D}} (J_{\mathrm{HJB}} + J_{\partial\mathcal{Q}}), \tag{5.18}$$

with

$$C^2_{\boldsymbol{D}} \overset{\text{def}}{=} 2\,\tau_{\max}(\boldsymbol{D}) \max\Big\{(1+L_B)^2\sqrt{1+m_B^2},\; \frac{e^{\alpha\tau_{\max}(\boldsymbol{D})}-1}{\alpha}\Big\}.$$

The interior coefficient depends only on α and $\tau_{\max}(\boldsymbol{D}) \le \boldsymbol{T}_h$; the boundary coefficient depends on $\boldsymbol{D}$ only through the bounded set $B \supset \boldsymbol{cl}[K^+]$ (through m_B, L_B). In particular the constant does not *depend on the transversality margin $\gamma_{\boldsymbol{D}}$ and does not degenerate as $\boldsymbol{D}$ approaches $\partial\mathbb{D}$.*

Proof. (5.12) with propositions 5.11 and 5.12; if $J_{\mathrm{HJB}} = \infty$ there is nothing to prove. The estimate is one-directional: it bounds the error by the loss. The reverse inequality is false in general (perturb $\boldsymbol{V}$ on a smooth interior patch by $n^{-1/2}\chi(z)\sin(ns)$ with χ smooth and compactly supported: the L^2 error tends to zero while the residual grows like $n^{1/2}$), which is why we do not call the estimate "two-sided". ∎

Remark 5.14 (Where $\boldsymbol{D} \Subset \mathbb{D}$ is used). *The positive distance of $\boldsymbol{D}$ to $\partial\mathbb{D}$ (the margin γ_K and the secant radius $\varrho_0(K)$ of a compact neighbourhood K) is used only in lemma 5.4: $\tilde{\Psi}$ must be Lipschitz on a neighbourhood of $\boldsymbol{D}$ for Rademacher and the co-area formula. The Lipschitz constant $C_\Psi(K)$ there is bounded only in terms of γ_K and $\varrho_0(K)$ and may be large, but it never enters (5.18). Boundedness of $\boldsymbol{D}$ enters only through m_B, L_B, and since (4.3) for a bounded set implies it for every subset, m_B, L_B may be replaced by the constants of any bounded*

$B' \supset B$. Hence, if $\boldsymbol{D}_\infty \subset \mathbb{D}$ is bounded and open (or bounded and σ-compact, so that it admits a compact exhaustion $\boldsymbol{D}_n \uparrow \boldsymbol{D}_\infty$; an arbitrary bounded subset need not even be measurable) and B_∞ is a bounded set containing the closed 1-neighbourhood of the set reached on $[0, \boldsymbol{T}_h)$ by predictions from the closed 1-neighbourhood of $\boldsymbol{D}_\infty$ (bounded by the Grönwall bound in lemma 3.3(b)), then (5.18) holds for every compact $\boldsymbol{D} \subset \boldsymbol{D}_\infty$ with the single constant obtained from $m_{B_\infty}, L_{B_\infty}$ and $\tau_{\max} \le \boldsymbol{T}_h$; since $\boldsymbol{D}_\infty^+ = \bigcup_n \boldsymbol{D}_n^+$ and $\Psi(\boldsymbol{D}_\infty) = \bigcup_n \Psi(\boldsymbol{D}_n)$, monotone convergence along the exhaustion gives the estimate on $\boldsymbol{D}_\infty$ itself, with the losses taken over $\boldsymbol{D}_\infty^+$ and $\Psi(\boldsymbol{D}_\infty)$. Thus the transversality margin governs the regularity of τ (theorem 4.8) and the constant C_Ψ, but not the stability estimate.

5.5. The HJB equation: what is proved and what is cited.

Proposition 5.15 (Senses in which $\boldsymbol{V}$ solves (5.3)).

(a) ***(Characteristic form.)*** *$\boldsymbol{V}$ satisfies (5.2) on $\mathbb{D}$; this is the integrated form of (5.3) along characteristics and holds exactly.*

(b) ***(Almost everywhere classical.)*** *$\boldsymbol{V}$ is Lipschitz on compacts of $\mathbb{D}$, Fréchet-differentiable at every point where $\partial_s \boldsymbol{V}$ exists, and (5.3) holds at every $\boldsymbol{y}$ for every s outside the null set N of non-Lebesgue points of a_ℓ, hence a.e. on $\mathbb{D}$ (lemma 4.10((ii))); the exceptional set of times is N and does not depend on $\boldsymbol{y}$.*

(c) ***(Classical where the forcing is continuous.)*** *On $\mathbb{D} \cap (J \times \mathbb{R}^2)$, J an interval of continuity of a_ℓ, $\boldsymbol{V} \in C^1$ and (5.3) holds everywhere (lemma 4.10((iii))).*

(d) ***(Uniqueness among a.e. solutions with the same characteristic structure.)*** *If $W \in C(\mathbb{D} \cup \Sigma^-)$ is absolutely continuous on compact subintervals of every predicted characteristic in $\mathbb{D}$ with $\frac{d}{d\kappa} W(s+\kappa, Z(\kappa; z)) = -1$ a.e. in κ, and $W = 0$ on Σ^-, then $W = \boldsymbol{V}$. (Integrate from the collision point; the derivative is bounded, so no endpoint-integrability issue arises.)*

(e) ***(Uniqueness among locally Lipschitz a.e. solutions.)*** *If $W \in C(\mathbb{D} \cup \Sigma^-)$ is locally Lipschitz on $\mathbb{D}$, satisfies $\partial_s W + \mathsf{G} \cdot \nabla_{\boldsymbol{y}} W = -1$ a.e. on $\mathbb{D}$ and $W = 0$ on Σ^-, then $W = \boldsymbol{V}$. (The proof of proposition 5.18 applies verbatim with C^1 replaced by locally Lipschitz: add the null set of non-differentiability points of W to the set N in its Step 1, and exhaust $\mathbb{D}$ by compact sets $\boldsymbol{D} = \boldsymbol{cl}[\operatorname{int} \boldsymbol{D}]$.)*

Remark 5.16 (On the viscosity characterisation). *One expects $\boldsymbol{V}$ to be the unique viscosity solution of (5.3) on $\mathbb{D}$, by the standard argument from (5.2); we do not prove this, and nothing in this paper depends on it. Two points deserve care. The Hamiltonian $H(s, \boldsymbol{y}, p) = p \cdot \mathsf{G}(s, \boldsymbol{y}) + 1$ (convention $\partial_s \boldsymbol{V} + H = 0$) is only measurable in s (through a_ℓ), so the classical theory [13, 7] does not apply directly and the relevant framework is the L^1-in-time theory of Ishii [28] and Nunziante [50]; moreover a pointwise definition would depend on the representative of a_ℓ, whereas the flow, τ and the losses do not. And the datum is prescribed only on the incoming surface Σ^-, so a comparison principle requires that characteristics exit $\mathbb{D}$ only through Σ^-, which is lemma 4.2(b) with lemma 5.1. When a_ℓ is continuous, $\boldsymbol{V} \in C^1(\mathbb{D})$ by lemma 4.10((iii)) and is then a classical, hence viscosity, solution; we make no claim in the measurable case. The stability estimate uses only (5.2), and proposition 5.18 proves $\boldsymbol{V}_\theta = \boldsymbol{V}$ directly.*

The structural reason is (d),(e) above: H is affine in p, the characteristics are the predicted trajectories and do not cross, so a solution that is absolutely continuous along characteristics

(for instance a locally Lipschitz a.e. solution) is already determined by its values on Σ^-. *Absolute continuity cannot be dropped: if* C *is a nonconstant Cantor function with* $C(0)=0$ *and* $C'=0$ *off a null set* N_C, *then* $W \overset{\text{def}}{=} \tau + C(\tau)$ *on* $\mathbb{D}$, $W \overset{\text{def}}{=} 0$ *on* Σ^-, *is continuous on* $\mathbb{D}\cup\Sigma^-$, *vanishes on* Σ^- *and satisfies* (5.3) *a.e. on* $\mathbb{D}$ *(where* τ *is differentiable and* $\tau(z)\notin N_C$, $DW=(1+C'(\tau))D\tau=D\tau$; *and* $\tau^{-1}(N_C)\cap \boldsymbol{D}$ *is* $\mathcal{L}^3$*-null for every compact* $\boldsymbol{D}$ *by the co-area formula, because* $|\nabla_{\boldsymbol{y}}\tau| \geq e^{-G_1\tau_{\max}}/m_B > 0$ *there by* (4.15) *and* (4.14)*), but* $W \neq \boldsymbol{V}$.

Definition 5.17 (Strong fixed point). $W \in C(\mathbb{D}\cup\Sigma^-)$ *is a* strong fixed point *on* $\mathbb{D}$ *if for every* $(s,\boldsymbol{y})\in\mathbb{D}$ *and every* $\kappa\in[0,\boldsymbol{V}(s,\boldsymbol{y})]$,

$$W(s+\kappa, Z(\kappa;s,\boldsymbol{y})) = W(s,\boldsymbol{y}) - \kappa. \tag{5.19}$$

By (5.2) and lemma 5.1, $\boldsymbol{V}$ is a strong fixed point; by proposition 5.15(d) it is the only one vanishing on Σ^-. Continuity is required on $\mathbb{D}\cup\Sigma^-$ only: $\boldsymbol{V}$ is not continuous up to the opening part of the collision plane (remark 5.2).

5.6. Zero residual implies $V_\theta = V$.

Proposition 5.18 (Exact fixed point in the zero-residual limit). *Let* $\boldsymbol{D}\Subset\mathbb{D}$ *be compact with* $\boldsymbol{D}=\boldsymbol{cl}[\operatorname{int}\boldsymbol{D}]$, *and let* $\boldsymbol{V}_\theta\in C^1(\mathbb{D})\cap C(\mathbb{D}\cup\Sigma^-)$ *satisfy* $J_{\mathrm{HJB}}=0$ *and* $J_{\partial\mathcal{Q}}=0$ *(losses over* $\boldsymbol{D}^+$ *and* $\mathcal{Q}_{\boldsymbol{D}}$ *as in* (5.7)*). Then*

$$\boldsymbol{V}_\theta(\mathcal{F}_\kappa(z)) = \tau(z)-\kappa \qquad \text{for every } z\in\boldsymbol{D} \text{ and every } \kappa\in[0,\tau(z)], \tag{5.20}$$

where $\mathcal{F}_\kappa(s,\boldsymbol{y}) \overset{\text{def}}{=} (s+\kappa, Z(\kappa;s,\boldsymbol{y}))$. *In particular* $\boldsymbol{V}_\theta=\boldsymbol{V}$ *on* $\boldsymbol{D}^+\cup\mathcal{Q}_{\boldsymbol{D}}\supset\boldsymbol{D}$, *and* $\boldsymbol{V}_\theta$ *satisfies* (5.19) *for every* $(s,\boldsymbol{y})\in\boldsymbol{D}$.

Proof. Throughout, B, m_B, L_B are the bounded reference set and the flow constants fixed in subsection 5.1 ("Bounded reference set and constants"), so that $\mathbf{cl}[\boldsymbol{D}^+]\subset\mathbf{cl}[K^+]\subset B$; $\boldsymbol{D}_\kappa \overset{\text{def}}{=} \{z\in\boldsymbol{D}:\tau(z)>\kappa\}$, and $\Psi(z)=\mathcal{F}_{\tau(z)}(z)\in\mathcal{Q}_{\boldsymbol{D}}\subset\Sigma^-$ (lemma 5.1).

Step 0: the space–time flow map is bi-Lipschitz. On $A \overset{\text{def}}{=} \{(\kappa,z)=(\kappa,s,\boldsymbol{y}):\ z\in B,\ 0\leq\kappa\leq\tau_{\max},\ s+\kappa<\boldsymbol{T}_h\}$ let $\Phi(\kappa,z)\overset{\text{def}}{=}(\kappa,\mathcal{F}_\kappa(z))\in\mathbb{R}^4$. By (4.3), Φ is Lipschitz on A; it is injective by uniqueness of the prediction flow, and its inverse $\Phi^{-1}(\kappa,t,\boldsymbol{x})=(\kappa,\ t-\kappa,\ Z^{\mathrm{abs}}(t-\kappa;(t,\boldsymbol{x})))$ is Lipschitz on $\Phi(A)\cap(\mathbb{R}\times B)$ with constant $\leq 1+2m_B+L_B$ (as in the proof of proposition 5.12: move the evaluation time along one trajectory, then compare trajectories at a common absolute time). Hence, on the target region $\Phi(A)\cap(\mathbb{R}\times B)$, Φ^{-1} maps $\mathcal{L}^4$-null sets to $\mathcal{L}^4$-null sets, and preimages under Φ of Lebesgue measurable subsets of that region are Lebesgue measurable (a measurable set is a Borel set up to a null set, and Φ is continuous). This is all that Step 1 needs, since the null set used there lies in $[0,\tau_{\max}]\times\boldsymbol{D}^+\subset\mathbb{R}\times B$.

Step 1: the residual vanishes along a.e. characteristic. The residual r_θ is Lebesgue measurable on $\mathbb{D}$ ($\partial_s\boldsymbol{V}_\theta$, $\nabla_{\boldsymbol{y}}\boldsymbol{V}_\theta$, Λ, $\mathcal{V}$, v_ℓ are continuous and a_ℓ is measurable); it is in general *not* continuous under L^∞ forcing, since it contains $\mathsf{a}_\ell(s)\,\partial_\Lambda\boldsymbol{V}_\theta$, and continuity is not needed. The set $\boldsymbol{D}^+$ is Borel (the image of the σ-compact set $\{(\kappa,z):z\in\boldsymbol{D},\ 0\leq\kappa<\tau(z)\}$ under the continuous map $(\kappa,z)\mapsto\mathcal{F}_\kappa(z)$). Let $N\overset{\text{def}}{=}\{w\in\boldsymbol{D}^+:r_\theta(w)\neq 0\}$; $\mathcal{L}^3(N)=0$ because

$J_{\mathrm{HJB}} = \int_{\boldsymbol{D}^+} |r_\theta|^2 = 0$. Set

$$\begin{aligned}\tilde{N} &\overset{\mathbf{def}}{=} \{(\kappa, z) \in [0, \tau_{\max}] \times \boldsymbol{D} : \ \kappa < \tau(z), \ \mathcal{F}_\kappa(z) \in N\} \\ &= \Phi^{-1}([0, \tau_{\max}] \times N) \cap \{(\kappa, z) : z \in \boldsymbol{D}, \ \kappa < \tau(z)\} \subset A.\end{aligned}$$

Since $[0, \tau_{\max}] \times N$ is $\mathcal{L}^4$-null, Step 0 shows that $\tilde{N}$ is measurable and $\mathcal{L}^4(\tilde{N}) = 0$. By Fubini, for $\mathcal{L}^3$-a.e. $z \in \boldsymbol{D}$ the slice $\tilde{N}_z = \{\kappa \in [0, \tau(z)) : \mathcal{F}_\kappa(z) \in N\}$ is $\mathcal{L}^1$-null. Let $E_1 \subset \boldsymbol{D}$ be the set of such z; $\mathcal{L}^3(\boldsymbol{D} \setminus E_1) = 0$.

Step 2: integration along such a characteristic. Let $z = (s, \boldsymbol{y}) \in E_1$ and $c(\kappa) \overset{\mathbf{def}}{=} \mathcal{F}_\kappa(z)$, $\kappa \in [0, \tau(z)]$. The curve c is Lipschitz, lies in $\boldsymbol{D}^+ \subset \mathbb{D}$ for $\kappa < \tau(z)$, and $c(\tau(z)) = \Psi(z) \in \Sigma^-$. Since $\boldsymbol{V}_\theta \in C^1(\mathbb{D})$, $g \overset{\mathbf{def}}{=} \boldsymbol{V}_\theta \circ c$ is locally Lipschitz on $[0, \tau(z))$ with $g'(\kappa) = \partial_s \boldsymbol{V}_\theta(c(\kappa)) + \nabla_{\boldsymbol{y}} \boldsymbol{V}_\theta(c(\kappa)) \cdot \mathsf{G}(c(\kappa)) = r_\theta(c(\kappa)) - 1$ for a.e. κ (chain rule for $C^1 \circ$Lipschitz, and $\partial_\kappa Z = \mathsf{G}(s+\kappa, Z)$ a.e.). By Step 1, $g' = -1$ a.e. on $[0, \tau(z))$, so $g(\kappa) = g(0) - \kappa$ for $0 \le \kappa < \tau(z)$. Letting $\kappa \uparrow \tau(z)$ and using the continuity of $\boldsymbol{V}_\theta$ on $\mathbb{D} \cup \Sigma^-$ at $\Psi(z)$,

$$\boldsymbol{V}_\theta(\mathcal{F}_\kappa(z)) = \boldsymbol{V}_\theta(\Psi(z)) + \tau(z) - \kappa \qquad \text{for all } \kappa \in [0, \tau(z)], \ z \in E_1. \tag{5.21}$$

Step 3: the boundary value vanishes at a.e. endpoint. Let $N_\partial \overset{\mathbf{def}}{=} \{p \in \mathcal{Q}_{\boldsymbol{D}} : \boldsymbol{V}_\theta(p) \neq 0\}$, relatively open in the compact set $\mathcal{Q}_{\boldsymbol{D}}$ (continuity of $\boldsymbol{V}_\theta$ on Σ^-), hence Borel; $\mathcal{H}^2(N_\partial) = 0$ because $J_{\partial\mathcal{Q}} = 0$. The set $\Psi^{-1}(N_\partial) \cap \boldsymbol{D}$ is Borel (Ψ is continuous on $\mathbb{D}$). Apply the co-area formula as in the proof of proposition 5.11, with $g \overset{\mathbf{def}}{=} \mathbf{1}_{(\Psi^{-1}(N_\partial) \cap \boldsymbol{D}) \setminus E}$, E the null set of non-differentiability of $\tilde{\Psi}$:

$$\int_{\Psi^{-1}(N_\partial) \cap \boldsymbol{D}} J_2 \tilde{\Psi} \, dz = \int_{\tilde{\Psi}(\Psi^{-1}(N_\partial))} \mathcal{H}^1(\tilde{\Psi}^{-1}(p) \cap \boldsymbol{D}) \, dp = 0,$$

since the outer domain of integration is the chart image of N_∂, which is $\mathcal{L}^2$-null. As $J_2 \tilde{\Psi} \ge c_* > 0$ a.e. on $\boldsymbol{D}$ (proposition 5.9), $\mathcal{L}^3(\Psi^{-1}(N_\partial) \cap \boldsymbol{D}) = 0$. Put $E \overset{\mathbf{def}}{=} E_1 \setminus \Psi^{-1}(N_\partial)$. Then $\mathcal{L}^3(\boldsymbol{D} \setminus E) = 0$ and, by (5.21),

$$\boldsymbol{V}_\theta(\mathcal{F}_\kappa(z)) = \tau(z) - \kappa \qquad \text{for all } \kappa \in [0, \tau(z)], \ z \in E. \tag{5.22}$$

Step 4: from almost every to every point of $\boldsymbol{D}$, by density. E is dense in $\boldsymbol{D}$: if $U \subset \mathbb{R}^3$ is open with $U \cap \boldsymbol{D} \neq \varnothing$, then $U \cap \operatorname{int} \boldsymbol{D} \neq \varnothing$ because $\boldsymbol{D} = \mathbf{cl}[\operatorname{int} \boldsymbol{D}]$, and this nonempty (relatively) open set has positive measure, so it meets E. Fix $z \in \boldsymbol{D}$ and choose $z_n \in E$ with $z_n \to z$.

Let $0 \le \kappa < \tau(z)$. Since τ is continuous on $\mathbb{D}$ (lemma 4.5(a)), $\kappa < \tau(z_n)$ for all large n, so $\boldsymbol{V}_\theta(\mathcal{F}_\kappa(z_n)) = \tau(z_n) - \kappa$ by (5.22). As $n \to \infty$, $\mathcal{F}_\kappa(z_n) \to \mathcal{F}_\kappa(z) \in \boldsymbol{D}^+ \subset \mathbb{D}$ (lemma 3.3(b)) and $\boldsymbol{V}_\theta$ is continuous on $\mathbb{D}$; hence $\boldsymbol{V}_\theta(\mathcal{F}_\kappa(z)) = \tau(z) - \kappa$.

Let $\kappa = \tau(z)$. By joint continuity of $(\kappa, z) \mapsto \mathcal{F}_\kappa(z)$ and of τ, $\Psi(z_n) = \mathcal{F}_{\tau(z_n)}(z_n) \to \mathcal{F}_{\tau(z)}(z) = \Psi(z)$; all these points lie in $\mathcal{Q}_{\boldsymbol{D}} \subset \Sigma^-$, $\boldsymbol{V}_\theta(\Psi(z_n)) = 0$ by (5.22) at $\kappa = \tau(z_n)$, and $\boldsymbol{V}_\theta$ is continuous on $\mathbb{D} \cup \Sigma^-$; hence $\boldsymbol{V}_\theta(\Psi(z)) = 0 = \tau(z) - \tau(z)$. This proves (5.20).

Step 5: conclusions. Taking $\kappa = 0$ in (5.20) gives $\boldsymbol{V}_\theta = \tau = \boldsymbol{V}$ on $\boldsymbol{D}$. Every point of $\boldsymbol{D}^+$ is $\mathcal{F}_\kappa(z)$ with $z \in \boldsymbol{D}$, $\kappa < \tau(z)$, and $\tau(z) - \kappa = \tau(\mathcal{F}_\kappa(z)) = \boldsymbol{V}(\mathcal{F}_\kappa(z))$ by (4.2); every point of $\mathcal{Q}_{\boldsymbol{D}}$ is some $\Psi(z)$, where $\boldsymbol{V} = 0$. Thus $\boldsymbol{V}_\theta = \boldsymbol{V}$ on $\boldsymbol{D}^+ \cup \mathcal{Q}_{\boldsymbol{D}}$. Finally, for $(s, \boldsymbol{y}) \in \boldsymbol{D}$ and $\kappa \in [0, \boldsymbol{V}(s, \boldsymbol{y})]$, (5.20) reads $\boldsymbol{V}_\theta(s + \kappa, Z(\kappa; s, \boldsymbol{y})) = \boldsymbol{V}_\theta(s, \boldsymbol{y}) - \kappa$, which is (5.19). ∎

Remark 5.19 (What the proof does not use). *The proof uses neither continuity of r_θ (which fails under L^∞ forcing), nor continuity of $\boldsymbol{V}_\theta$ beyond $\mathbb{D} \cup \Sigma^-$, nor any claim that the characteristics through almost every point of $\boldsymbol{D}$ fill almost all of $\boldsymbol{D}^+$. The latter claim is false in general: a null transversal subset of $\boldsymbol{D}$ can sweep out positive volume under the flow (in flow-box coordinates $(p, r) \in \mathbb{R}^2 \times \mathbb{R}$, take $\boldsymbol{D} = \{|r| \le \operatorname{dist}(p, C)\}$ with C compact of positive area and empty interior; then $\boldsymbol{D} = \boldsymbol{cl}[\operatorname{int} \boldsymbol{D}]$, $C \times \{0\}$ is $\mathcal{L}^3$-null, but its forward saturation contains $C \times (0, a)$, of positive measure). This is why Step 4 argues by density. The hypothesis $\boldsymbol{D} = \boldsymbol{cl}[\operatorname{int} \boldsymbol{D}]$ enters only in Step 4.*

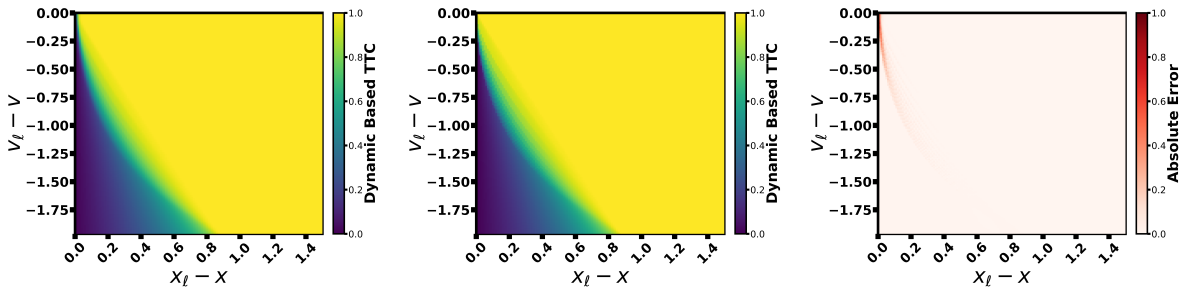
Remark 5.20 (Scope: the zero-residual limit under rough forcing). *proposition 5.18 concludes $\boldsymbol{V}_\theta = \boldsymbol{V}$ with $\boldsymbol{V}_\theta \in C^1$. By remark 4.11, if a_ℓ has a jump at a time t_1 such that some $(t_1, \boldsymbol{y})$ lies in $\operatorname{int}(\boldsymbol{D}^+)$ with $\partial_\Lambda \boldsymbol{V}(t_1, \boldsymbol{y}) \neq 0$ (automatic when $\tau(t_1, \boldsymbol{y}) < \pi/\omega_\alpha$, $\omega_\alpha = \sqrt{\alpha(1 - \alpha/4)}$, in particular whenever $\boldsymbol{T}_h \le \pi/\omega_\alpha$ or $\alpha \ge 4$; see remark 4.11), then $\boldsymbol{V}$ is not C^1 near that point, so* no *C^1 function satisfies the hypotheses: the zero-loss set is empty. The proposition is therefore a statement about the continuous-forcing case (or about compact sets whose forward tube avoids the jumps of a_ℓ). Under rough forcing the operative result is the quantitative theorem 5.13: small loss forces $\boldsymbol{V}_\theta$ close to $\boldsymbol{V}$ in L^2, with no claim of attainment. Whether the infimum of the loss over C^1 functions is zero under rough forcing is plausible (mollify $\boldsymbol{V}$ in time; the residual of the mollification is small in L^2 where $\boldsymbol{V}$ is Lipschitz) but is not proved here.*

6. Numerical Validation.

The analysis above is constructive in two respects that admit direct numerical assessment. Thm. 3.38 excludes impacts by contradiction, and therefore does not exhibit the avoidance behavior it guarantees; and Thm. 5.13 bounds the error of a learned value function by the training loss, without indicating how small that loss becomes in practice. This section reports simulations addressing both. The definition of different types of simulation is detailed in Section SM2. Implementation details — the architectures of $\boldsymbol{V}_\theta$, $\mathcal{N}_V$ and $\mathcal{N}_X$, the biased sampling strategy, the training procedure and its performance metrics — are deferred to Sections SM3, SM6 and SM8, and the full set of trajectory comparisons to Section SM10.

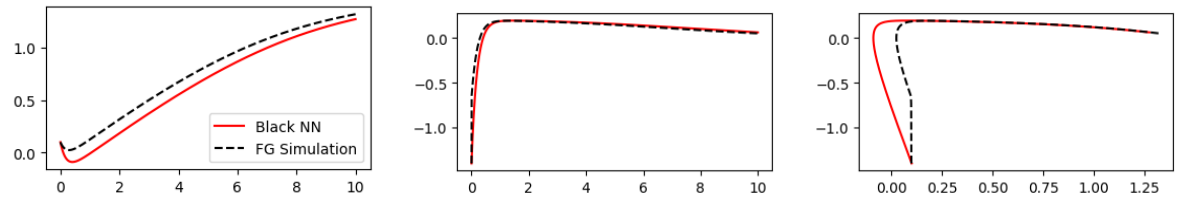
Learning the TTC map and the full trajectory. The value network $\boldsymbol{V}_\theta$ is a feedforward multilayer perceptron trained on the total loss (5.8), with sampling concentrated near the collision boundary where the transversality margin $\gamma_{\boldsymbol{D}}$ — and hence the Lipschitz constant of τ in Section 4 — is small. Compared against direct numerical integration of the relative OVM dynamics (3.8), the learned map reproduces the level-set structure of τ with small absolute error away from the boundary and growing error as it is approached, consistent with the degeneration of the Lipschitz constant of τ as $\gamma_{\boldsymbol{D}} \to 0$ (Figure 3, Table SM1 in Section SM3). A purely black-box network trained without the analytical structure fails to capture the dynamics in the near-collision regime at all (Figure 4), which is the practical counterpart of the observation that the strong-form residual is only meaningful where the solution is regular (Lem. 4.10). Composing $\boldsymbol{V}_\theta$ with the trajectory network $\mathcal{N}_X$ yields the two-stage map $\mathcal{N}_{\text{sol}}$, whose predictions of τ, spacing and relative velocity track the delay-free reference simulation over the full horizon across the initial conditions tested (Section SM10).

Platoon behavior. To examine the closed loop rather than the approximation in isolation, we simulate a six-vehicle platoon in which the leader follows a non-constant acceleration pro-



(a) Neural Approximation ($\boldsymbol{V}_\theta$) (b) Numerical Solution (c) Absolute Error

Figure 3: Comparison between the dynamic-based TTC values approximated by V_θ and dynamic-based TTC values computed through the direct numerical simulation of the relative OVM dynamics (3.8) with $\alpha = 2$ and $\boldsymbol{T}_h = 1$.



(a) Spacing vs Time (b) Relative Velocity vs Time (c) Relative Velocity vs Spacing

Figure 4: Black Box Neural Network Performance compared with the delay-free trajectory of the proposed model (FG-simulation; see Section SM2 for the definition), illustrating the limitation of non-physically informed training to capture the proposed dynamics in the near-collision regime. Same parameters, simulation time, and x profile as in Figure 2, with $\mathsf{a}_\ell = 0$. Initial Conditions: $x_\ell = 1.21$, $v = 1.52$, $v_\ell = 0.13$.

file and the pair $(0, 1)$ is driven by $\mathcal{N}_{\text{sol}}$, the remaining pairs being integrated numerically. The initial spacing of the first pair is set to 0.1, deliberately close to the collision boundary, with the remaining vehicles started at their OVM equilibrium spacing. Figure SM8 and Figure SM9 show the resulting spacing and relative velocity along the platoon. Two features are worth noting. Spacing remains bounded away from zero throughout, despite the tight initial configuration; this is consistent with Thm. 3.38, which excludes any contact with nonzero closing speed for the exact controller (it does not by itself exclude a zero-speed touch, remark 3.41, and it is a statement about the exact TTC, not about the learned surrogate; see below). And the disturbance introduced by the leader attenuates as it propagates downstream, with later pairs exhibiting smaller excursions than earlier ones, an attenuation consistent with the local stability analysis of the underlying model (cf. Remark SM3.1); no string-stability theorem for the learned controller is claimed.

Scope of these results. These simulations illustrate the theory rather than test it: a finite set of scenarios does not prove the theorems above, and a numerical discrepancy would have to be checked analytically and numerically before being treated as a counterexample to them; the reported behavior is consistent with, but not evidence for, the analytical guarantees. Three limitations should be stated. The impact-prevention theorem concerns the exact coefficient τ, and Thm. 5.13 bounds the L^2 error of the learned value; neither gives a near-contact bound on the reciprocal gain $1/\boldsymbol{V}_\theta^2$, and a uniformly small absolute TTC error does not by itself preserve impact-freeness: with a stationary leader and $\boldsymbol{V}_\theta = \tau + \varepsilon$ the feedback gain is bounded by $\alpha+\beta/\varepsilon^2$ near contact, and a follower started close enough to the leader reaches it with negative relative velocity. A safety statement for the surrogate controller would need a relative near-contact estimate (of the type $0 < \boldsymbol{V}_\theta \le CX/|V|$ on the approach region) or a separate safety filter; neither is established here. The evaluation uses the same leader acceleration profile as training, so the results speak to in-distribution accuracy only; assessing out-of-distribution leader behavior is left to future work. And the analysis of Subsection 5.1 is at the continuum level, bounding the error by the *population* loss, whereas training minimizes an empirical loss over finitely many collocation points. Closing that gap requires a generalization bound; we regard this as the principal open problem on the theoretical side.

7. Conclusions. A dynamically informed TTC-based feedback control framework for nonlinear microscopic traffic dynamics is proposed in this paper. Unlike conventional TTC formulations based on kinematic prediction, the proposed controller computes dynamic-based TTC using the OVM dynamics in this work, producing a predictive feedback law that anticipates vehicle interactions.

We established several analytical properties of the resulting closed-loop system. The closed-loop field is Lipschitz away from the grazing set and discontinuous across it; we proved existence of Filippov solutions for all physical data, uniqueness in the regular set, transversal crossing of the grazing set at nondegenerate states, and that every contact has zero closing speed, so that no solution ever suffers an impact (a contact with negative relative velocity). We then established the regularity of the dynamic-based TTC map on compact subsets of the transversal domain — Lipschitz, C^1 in the state with an explicit gradient, and Lipschitz in time with a time derivative that jumps where the leader acceleration does — providing the basis for neural network approximation.

Building on these analytical results, we developed a Hamilton–Jacobi based learning framework for approximating the dynamic-based TTC map for a delay-free control law, with an L^2 error bound in terms of the training loss whose constant is uniform up to the boundary of the transversal domain. In addition, we introduced a two-stage semi-supervised physics-informed neural network architecture for approximating the full system trajectories, considering the near-collision singularities and oscillatory dynamics.

Future work will focus on extending the present framework to more complex dynamical systems involving interactions among multiple control agents. In particular, it is important to generalize the proposed reachability-based analysis to systems in which agents interactions are not limited to one-dimensional car-following behavior, but may include lane changing, different vehicle types, network-level interactions, and more general multi-agent control structures. Such extensions would broaden the applicability of the framework to safety-critical

transportation systems and other interacting-agent systems where control decisions, dynamic predictions, and collision-avoidance constraints are strongly coupled.

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SUPPLEMENTARY MATERIALS: Regularity of the Time-to-Collision Map and Convergence of Learned Hamilton–Jacobi Solutions

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SM1. Leading acceleration. In this research, we assume a_ℓ can be approximated/learned. Such assumption is justified by the following remarks.

- **Sequential Learning in vehicle platoons.** First, in a platoon setting, the trajectory of each vehicle becomes the input for the subsequent follower. Specifically, once the trajectory (x_ℓ, v_ℓ) of a leading vehicle is learned or observed, it can be treated as a known signal for the following vehicle. In this sense, the learning framework could be applied recursively along the platoon: learning the dynamics of one vehicle would provide the input (i.e., a_ℓ) for the next. This recursive use is a conceptual possibility; the networks reported in this paper are trained under the single profile (SM4.1), and the platoon experiment of Subsection SM6.4 states separately how the predecessor's motion enters.
- **Interpretation as control input.** Second, the leading acceleration a_ℓ can be interpreted as a control input. From a control-theoretic standpoint, the leader's behavior is not arbitrary, but rather reflects a control strategy—either human or automated.
  Under this interpretation, assuming a_ℓ is known corresponds to analyzing the follower dynamics under a fixed control policy. This is standard in reachability and optimal control, where the system is studied under a given control signal or as part of the predictive trajectory in feedback control.
  Moreover, this viewpoint allows for future extensions in which acceleration is designed or optimized to achieve specific objectives such as safety, efficiency, or stability.
- **Parametric approximation.** The leading acceleration can be modeled parametrically. For example,
$$\mathsf{a}_\ell(t) \approx \sum_{k=0}^{n} c_k t^k,$$
or even more generally, representing the a_ℓ using proper basis (polynomials, NN, functions, and so on). The coefficients can then be estimated from observed trajectory data in a reliable way.

SM2. Simulation: definitions.

Definition SM2.1 (**Simulation Term**).

Throughout this work, we use several notions of simulation to support our results and discussions. For clarity, we define an distinguish the following terms:

(i) ***Freeze-and-Go Simulation (FG-Simulation)*** *refers to calculating the solution step by step without taking the delay of TTC-approximation into account. In particular, at*

*École Polytechnique
†University of California Berkeley
‡University of California Berkeley

each time step, we freeze the dynamical model, compute the TTC from the current state by integrating the prediction over a window of length $\boldsymbol{T}_h$ (with the leader's prescribed acceleration profile over the window, and $\tau = \boldsymbol{T}_h$ if no contact is predicted or the gap is opening), and update the numerical solution for the next time step. See Figure 2 for trajectories.

(ii) ***Neural-Numerical Simulation (NeuNum-Simulation)*** *refers to a hybrid computational framework that integrates the conventional numerical simulation of vehicle dynamics with the trained NN for dynamic-based TTC approximation (queried at each instant with the prediction clock reset and the current leader speed, Subsection SM8.1.2), thereby avoiding delay in TTC computation. See Figure SM8.*

(iii) ***NN-based Approximation*** *is designed to provide the solution learned by a DNN. More precisely, in this approach, the TTC, spacing, and relative solution are approximated through DNNs. See Figure SM7 for the corresponding behavior of trajectories.*

Remark SM2.2 (Length of Vehicles). *The length of vehicles is neglected for the analytical part of this paper—the vehicles are modeled* (as particles)*—since considering a constant length and safety headway only shifts the analytical results by a constant without altering their fundamental structures [SM3].*

SM3. Architecture and performance of V_θ. Validated by the analytical results, the total loss Equation (5.8), the biased sampling strategy, and the well-established stability properties of the OVM (see e.g., [SM2, SM1]), we employ the neural network $\boldsymbol{V}_\theta(s, \boldsymbol{y})$ as a standard feedforward MLP and trained according to Algorithm SM7.1 to approximate the TTC mapping τ. The architecture and training procedure are described in this section and detailed in Subsection SM8.1 in Section SM8.

Remark SM3.1 (Local stability and the choice of the training region). *Local stability of the non-delayed system motivates where the training effort is concentrated: near a locally stable equilibrium, trajectories starting in a neighborhood of each other remain close, so a finite set of training trajectories is representative of its neighbors, whereas near the collision boundary small differences in the initial state produce large differences in the trajectories, which is why the sampling is biased toward that region. This is a heuristic for the design of the training set; it is not a generalization theorem for the learned map, and the accuracy reported below is the measured accuracy on the finite test sets described in Section SM8. A local stability analysis of the class of car-following dynamics encompassing the proposed model is presented in [SM7, SM6]; the platoon experiment of Subsection SM6.4 reports the observed behavior of the implementation in one scenario.*

The architecture of $\boldsymbol{V}_\theta$ is informed by the analytical properties established earlier. Training performance metrics summarized in Table SM1 and the comparison between V_θ and numerical simulation in Figure 3 show high predictive reliability and robustness of the trained model.

SM4. Leading Vehicle Acceleration. To evaluate the general performance of the neural network framework, it is important to consider the cases of non-zero acceleration of the leading vehicle, causing the leading vehicle to move at non-constant velocity. Although we can train the proposed neural network framework with various a_ℓ profiles, it requires a lot of training efforts and time to do so. Therefore, in this paper, we present the performance of the proposed

Table SM1: Training performance summary for V_θ, $\mathcal{N}_V$, and $\mathcal{N}_X$. All models trained on Apple M2 Max with 96GB RAM. Full training configurations are detailed in Section SM8.

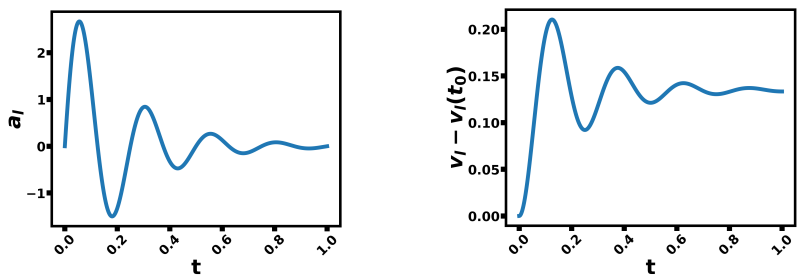
Model	Epochs	Final Loss	R^2 (Train)	R^2 (Val)	R^2 (Test)	Primary Variable
V_θ	500,000	7.3273E-5	–	0.9847	0.9887	TTC τ
$\mathcal{N}_V$	3,000	0.2389	–	0.9994	–	Relative velocity V
$\mathcal{N}_V$	3,000	0.2389	–	0.9997	–	Spacing X
$\mathcal{N}_X$	3,000	2321.265	–	0.9999	–	Spacing X

neural network frameworks with a particular leading vehicle's acceleration profile.

For the general evaluation, we choose the acceleration profile such that the profile consists of a series of accelerations and decelerations. The chosen acceleration profile is as follows:

$$\mathsf{a}_\ell(t) = \begin{cases} 3.5\exp\left(-\ln\left(100\right)t\right)\sin\left(8\pi t\right), & \forall t \in [0,1], \\ 0, \quad \text{otherwise} \end{cases} \tag{SM4.1}$$

where the visualization of the acceleration profile and the resulting profile of the $v_\ell(t) - v_\ell(t_\circ)$ is illustrated in Figure SM1. This acceleration profile a_ℓ is the attenuating oscillation function, making the leading vehicle accelerate and decelerate while its magnitude decreases within the time horizon. With the setup, the neural network needs to learn the effect of the leading vehicle's dynamics on the following vehicle's positions and velocity, especially the oscillating dynamics.



(a) a_ℓ profile considered (b) $v_\ell(t) - v_\ell(t_\circ)$

Figure SM1: Visualization of a_ℓ profile and $v_\ell(t) - v_\ell(t_\circ)$ profile.

SM5. Flow Charts.

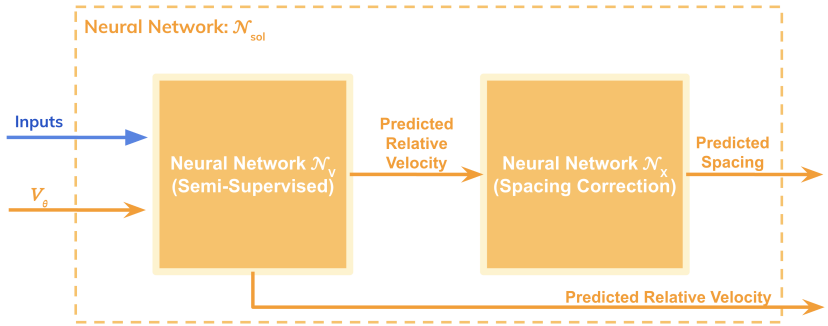


Figure SM2: Overall Framework for Non-Delayed Solution with Neural Network

SM6. Deep Neural Network Approximation of the solution. Building on $\boldsymbol{V}_\theta$, we now extend the framework to learn the entire system dynamics. While $\boldsymbol{V}_\theta$ provides instantaneous dynamic-based TTC estimates τ, a well-coordinated vehicle platoon requires non-delayed, long-horizon prediction of spacing and relative velocity, as each follower depends on its leader's acceleration, velocity, and position. Learning the full trajectory removes the latency of online simulation from the prediction of the leader–follower states; the platoon behavior obtained with the learned predictor is reported, as an observation for the tested scenario, in Subsection SM6.4.

However, a non-physically informed learning approach, without incorporating analytical structure, proves ineffective in this context. Due to strong irregularities and nonlinearities (see Section 3), a black-box neural network fails to capture the proposed dynamics or to maintain stability across varying initial conditions, as shown in Figure 4. This limitation highlights the need for an analytically guided, biased-sampling framework that aligns the learning process with the system's underlying physical and stability properties.

To address the limitation of non-physically informed black-box learning in capturing the proposed irregular and nonlinear dynamics, we proposed a semi-supervised Physics-Informed Neural Network (semi-supervised PINN) framework that applies data-driven supervision with biased sampling in the safety-critical region and a PINN formulation in the remaining region. The overall architecture of the proposed framework is shown in Figure SM3.

As illustrated in Figure SM3, the proposed neural network framework $\mathcal{N}_{\text{sol}}$ takes the following inputs: (i) the initial positions and velocities, (ii) the acceleration profile of the leading vehicle, and (iii) the timestep of interest for prediction. By leveraging these inputs along with the dynamic-based TTC approximation from the pre-trained $\boldsymbol{V}_\theta$, $\mathcal{N}_{\text{sol}}$ predicts the full state evolution of spacing X and relative velocity V, enabling delay-free, physically consistent trajectory prediction across any forecast horizon. The detailed framework of $\mathcal{N}_{\text{sol}}$ is shown in Figure SM2 in Section SM5.

One motivation for the semi-supervised PINN framework is to address loss-imbalance in purely physics-informed training, as illustrated in Figure SM4 for the single-stage unsupervised PINN with joint spacing–relative velocity prediction. In the near-collision regime, the proposed model generates collision-free yet aggressive maneuvers with abrupt changes in relative-velocity, producing larger residuals in relative velocity than in spacing and causing

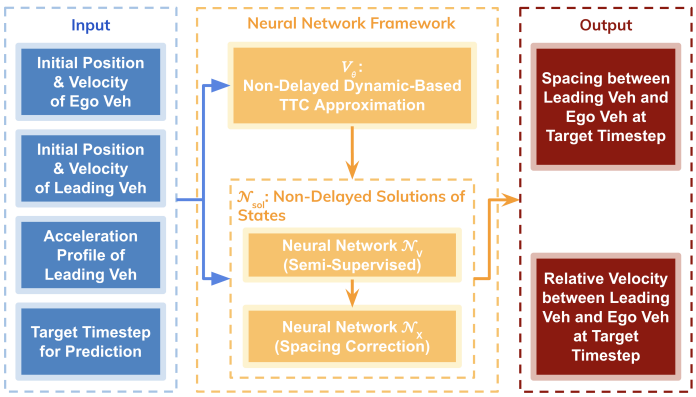
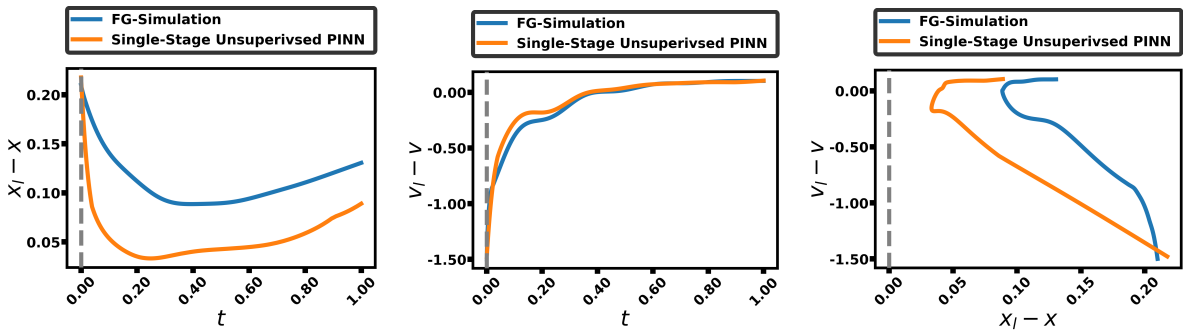


Figure SM3: Overall Framework using Neural Network

training to prioritize the relative-velocity profile over spacing [SM4, SM5].

To capture the coupled spacing and relative velocity dynamics, $\mathcal{N}_{\text{sol}}$ adopts a two-stage architecture with $\mathcal{N}_V$ and $\mathcal{N}_X$. The first-stage model $\mathcal{N}_V$ predicts the relative velocity profiles through a semi-supervised strategy that integrates ODE-based physical constraint with data-driven supervision, conditioned on τ. Then, the second-stage model $\mathcal{N}_X$ refines the corresponding spacing profile using the output of $\mathcal{N}_V$. This architecture enforces the kinematic relation $\dot{X} = V$ between the two predicted states and removes the online-simulation delay; the measured errors in the safety-critical region, for the tested initial conditions and the single leader profile used in training, are reported in Table SM1 and Subsection SM6.3.

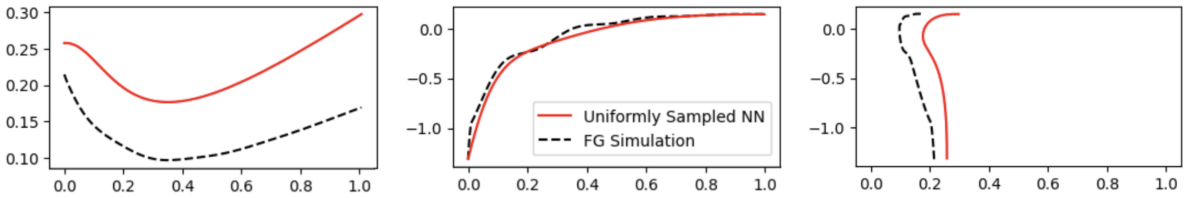


(a) Spacing vs Time (b) Relative Velocity vs Time (c) Relative Velocity vs Spacing

Figure SM4: Predictions from trained single-stage unsupervised PINN, showing degraded spacing prediction in the near-collision regime due to loss-imbalance issue in purely physics-informed training. Same parameters, simulation time, a_ℓ, and x profile as in Figure 2. Initial Conditions: $x_\ell = 0.21$, $v = 1.65$, $v_\ell = 0.15$.

While the proposed two-stage learning framework mitigates imbalance-related errors, uniform random sampling remains insufficient as it ignores the underlying system behavior, which

is particularly characterized by sharp decelerations, abrupt transitions, and irregular non-uniform behavior near the collision boundary. Consequently, the safety-critical region with small τ is not emphasized enough during training, leading to degraded prediction accuracy (see Figure SM5 for an example).



(a) Spacing vs Time (b) Relative Velocity vs Time (c) Relative Velocity vs Spacing

Figure SM5: Uniform Random Sampling and Performance of NN , illustrating degraded prediction in the near-collision regime due to insufficient emphasis on safety-critical cases. Same parameters, simulation time, a_ℓ, and x profile as in Figure 2. Initial Conditions: $x_\ell = 0.26$, $v = 1.53$, $v_\ell = 0.21$.

To improve the prediction reliability and robustness in the safety-critical region, we utilize a biased sampling strategy analogous to that for $\boldsymbol{V}_\theta$ in Section 5, increasing the representation of near-collision cases according to their safety importance.

Having established the analytical foundations, safety-critical region, local stability analysis, the concept of semi-supervised PINN, biased sampling strategy, and two-stage learning, we now detail each stage of the proposed framework, $\mathcal{N}_V$ and $\mathcal{N}_X$.

SM6.1. Neural Network $\mathcal{N}_V$. We begin with the first-stage network, $\mathcal{N}_V$, implemented using a semi-supervised PINN formulation to capture the nonlinear coupling between X and V. Within this framework, $\mathcal{N}_V$ acts as a neural operator parameterized by θ, mapping inputs including $X_\circ$, $V_\circ$, $v_{\ell,\circ}$, a_ℓ, τ from $\boldsymbol{V}_\theta$, and a query time t to the corresponding system states. This setup enables $\mathcal{N}_V$ to model the full solution map and predict the state evolution for any $t \in [0, T]$.

To train this operator, θ of $\mathcal{N}_V$ is learned by minimizing the following composite loss:

$$J_{\mathcal{N}_V} = \lambda_r J_{r,\mathcal{N}_V} + \lambda_{\text{ic}} J_{\text{ic},\mathcal{N}_V} + \lambda_{\text{data}} J_{\text{data},\mathcal{N}_V}, \tag{SM6.1}$$

where $J_{r,\mathcal{N}_V}$, $J_{\text{ic},\mathcal{N}_V}$, and $J_{\text{data},\mathcal{N}_V}$ denote the residual, initial condition, and data-driven, respectively. Their weights are set to $\lambda_r = 6$, $\lambda_{\text{ic}} = 75$, and $\lambda_{\text{data}} = 10$, as determined by grid search. Each loss component is described below.

ODE Loss $J_{r,\mathcal{N}_V}$. The ODE loss is formulated as the residual between predicted and true derivatives, promoting physically consistent trajectories that adhere to the governing dynamical equations. For each sample i and randomly selected time step t_j, the residuals are

computed as follows:

(SM6.2)
$$\begin{cases} r_X^{(i,j)} = \frac{d\hat{X}^{(i)}(t_j)}{dt} - \hat{V}^{(i)}(t_j) \\ r_V^{(i,j)} = \frac{d\hat{V}^{(i)}(t_j)}{dt} - \left\{ \mathsf{a}_\ell - \left(\alpha \left\{ \mathcal{V}(\hat{X}^{(i)}(t_j)) + \hat{V}^{(i)}(t_j) - v_\ell^{(i)}(t_j) \right\} + \beta \frac{\hat{V}^{(i)}(t_j)}{\left(\hat{\tau}^{(i)}(t_j)\right)^2} \right) \right\}. \end{cases}$$

Here, $\hat{V}^{(i)}(t_j)$ and $\hat{X}^{(i)}(t_j)$ are the predictions from $\mathcal{N}_V$, while $\hat{\tau}^{(i)}(t_j)$ is the prediction from $\boldsymbol{V}_\theta$, evaluated at the current state with the prediction clock reset (see Subsection SM8.1.2). All quantities are in physical time; the training horizon is $T = 1$, so this coincides with the normalized form $\hat{\tau}/T$ used in the implementation, and the TTC term is the one of the analytical model (2.1).

In the semi-supervised PINN framework, these residuals are enforced selectively, with the ODE loss computed only when the dynamic-based TTC exceeds a specified threshold $\underline{\tau}$:

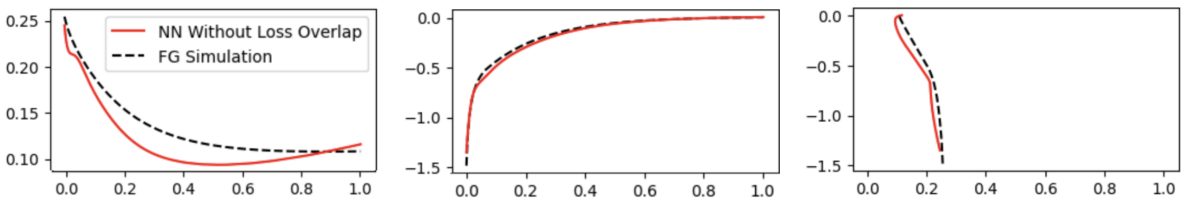
(SM6.3)
$$J_{r,\mathcal{N}_V} = \frac{1}{|\mathcal{I}_{\text{ode}}|} \sum_{(i,j)\in\mathcal{I}_{\text{ode}}} \left(r_X^{(i,j)}\right)^2 + \left(r_V^{(i,j)}\right)^2,$$

where $\mathcal{I}_{\text{ode}} = \left\{(i,j) : \tau^{(i)}(t_j) \geq \underline{\tau}\right\}$ and $\underline{\tau}$ is set to be 0.5 in this article.

Data-Driven Loss $J_{\text{data},\mathcal{N}_V}$. The data-driven loss acts as a stabilizing regularizer, aligning predictions with data and enhancing robustness by providing supervision in a safety-critical region with small dynamic-based TTC, where sharp decelerations, stiffness, or singularities occur. This loss is computed for timesteps where the dynamic-based TTC falls below a threshold $\bar{\tau}$:

(SM6.4)
$$J_{\text{data},\mathcal{N}_V} = \frac{1}{|\mathcal{I}_{\text{data}}|} \sum_{(i,j)\in\mathcal{I}_{\text{data}}} \left(\hat{X}^{(i)}(t_j) - X^{(i)}(t_j)\right)^2 + \left(\hat{V}^{(i)}(t_j) - V^{(i)}(t_j)\right)^2,$$

where $\mathcal{I}_{\text{data}} = \left\{(i,j) : \tau^{(i)}(t_j) \leq \bar{\tau}\right\}$ and $\bar{\tau} = 0.7$ in this article.



(a) Spacing vs Time (b) Relative Velocity vs Time (c) Relative Velocity vs Spacing

Figure SM6: Example trajectory trained without overlap between the ODE and data-driven losses, resulting in a jagged transition in the middle of the trajectories. Same parameters, simulation time, and x profile as in Figure 2, with $\mathsf{a}_\ell = 0$. Initial Conditions: $x_\ell = 0.25$, $v = 1.63$, $v_\ell = 0.14$.

From (SM6.3) and (SM6.4), some timesteps can be included in both the ODE and data-driven loss domains. This overlap improves sampling stability while preserving physical consistency. Without it, the transition between the two loss regimes can produce inaccurate and jagged behavior in the middle of the trajectory, as illustrated in Figure SM6.

Initial Condition Loss, $J_{\text{ic},\mathcal{N}_V}$. The solution map of the proposed model is determined by the initial state, so $\mathcal{N}_V$ must reproduce the initial condition at time $t = 0$. We enforce this through a band-penalty initial-condition loss, which penalizes deviations only when they exceed a tolerance $\delta_{\text{tol}} > 0$. This accounts for finite neural-network precision and avoids hindering convergence due to negligible deviations; it also means that the initial condition is matched only within the tolerance, see below.

We set $\delta_{\text{tol}} = 10^{-4}$, and define the loss as:

$$J_{\text{ic},\mathcal{N}_V} = \frac{1}{N} \sum_{i\in[1,N]} \left(\rho_{\delta_{\text{tol}}}\left(\hat{X}^{(i)}(0) - X^{(i)}(0)\right)\right)^2 + \left(\rho_{\delta_{\text{tol}}}\left(\hat{V}^{(i)}(0) - V^{(i)}(0)\right)\right)^2, \tag{SM6.5}$$

where N denotes the number of samples, and $\rho_{\delta_{\text{tol}}}(\cdot)$ denotes the band penalty function defined by:

$$\rho_{\delta_{\text{tol}}}(e) = |e - \delta_{\text{tol}}| + |e + \delta_{\text{tol}}| - 2\delta_{\text{tol}} = 2\big(|e| - \delta_{\text{tol}}\big)_+. \tag{SM6.6}$$

Here, the band-penalty function is zero for $|e| \le \delta_{\text{tol}}$ and grows linearly beyond this tolerance band; its square is C^1 and convex in the scalar error e. The weight λ_{ic} balances initial-condition enforcement against the residual and data terms. Two limitations of this construction should be stated. First, convexity in the scalar error does not make the training objective convex in the network parameters: a sum of squared residuals of a nonlinear network is in general nonconvex in θ, and no optimization guarantee is claimed. Second, a zero initial-condition loss only constrains the initial error to the band $|e| \le \delta_{\text{tol}}$; it does not single out one solution. For instance, for a fixed predicted relative velocity $\hat{V}_{\mathcal{N}_V}$, every spacing $\hat{X}_c(t) = X_0 + c + \int_0^t \hat{V}_{\mathcal{N}_V}(r)\,dr$ with $|c| \le \delta_{\text{tol}}$ has zero residual r_X and zero initial-condition loss. The uniqueness of the exact solution map (Picard–Lindelöf for the model with exact initial data) is a property of the model, not a consequence of this loss; if exact initial data are required they can be imposed by construction, e.g. $\hat{X}(t) = X_0 + t\,\mathcal{N}_\theta(t, \dots)$.

With all loss components defined, the neural network $\mathcal{N}_V$ is trained as a standard sequential feedforward MLP; the architecture is summarized in Algorithm SM7.2 in Section SM7 and the implementation details are provided in Subsection SM8.2. Its performance is reported in Table SM1 in Section SM3. The results show that the neural operator formulation effectively learns the nonlinear characteristics of the proposed system dynamics, even in the near-collision cases.

SM6.2. Neural Network $\mathcal{N}_X$. Having fully trained $\mathcal{N}_V$, we now introduce the second-stage network $\mathcal{N}_X$ to address the spacing prediction degradation of $\mathcal{N}_V$ caused by residual loss imbalance. Specifically, $\mathcal{N}_X$ refines spacing predictions from $\mathcal{N}_V$ by enforcing the dynamical consistency between spacing and relative velocity.

The loss function for $\mathcal{N}_X$ consists of two components: 1) a physics-based loss using V from $\mathcal{N}_V$, and 2) an initial condition loss ensuring well-posedness:

$$J_{\mathcal{N}_X} = \lambda_r J_{r,\mathcal{N}_X} + \lambda_{\text{ic}} J_{\text{ic},\mathcal{N}_X} \tag{SM6.7}$$

where $\lambda_r = 2$ and $\lambda_{\text{ic}} = 50{,}000$ are selected through grid search to balance physical fidelity and numerical stability. Each loss component is described below.

ODE Loss Based on $\hat{V}$ from $\mathcal{N}_V$, $J_{r,\mathcal{N}_X}$. Similar to $\mathcal{N}_V$, the ODE loss enforces a physical consistency of $\mathcal{N}_X$ with the dynamical relationship $\dot{X} = V$ by minimizing the residual between the derivative of the predicted spacing and the predicted relative velocity. For each sample i and randomly selected time step t_j, the residual is computed as: :

$$r_X^{(i,j)} = \frac{d\hat{X}^{(i)}(t_j)}{dt} - \hat{V}_{\mathcal{N}_V}^{(i)}(t_j). \tag{SM6.8}$$

The total residual loss is then given by

$$J_{r,\mathcal{N}_X} = \sum_{i,j} \left(r_X^{(i,j)}\right)^2. \tag{SM6.9}$$

Initial Condition Loss, $J_{\text{ic},\mathcal{N}_X}$. The initial condition loss for $\mathcal{N}_X$ is enforced through the band-penalty formulation analogous to that imposed for $\mathcal{N}_V$, as defined in (SM6.5) and (SM6.6), with $\delta_{\text{tol}} = 10^{-4}$:

$$J_{\text{ic},\mathcal{N}_X} = \sum_{i=1}^{N} \left(\rho_{\delta_{\text{tol}}}\left(\hat{X}^{(i)}(0) - X^{(i)}(0)\right)\right)^2. \tag{SM6.10}$$

The loss function of $\mathcal{N}_X$ therefore has the same structure as that of $\mathcal{N}_V$: squared dynamical residuals plus a tolerance-band penalty on the initial condition. It is differentiable in the parameters but, as for $\mathcal{N}_V$, generally nonconvex in them, and the initial condition is matched within the tolerance δ_{tol} rather than exactly. With the loss function (SM6.7), $\mathcal{N}_X$ employs a sequential feedforward MLP architecture detailed in Algorithm SM7.3. Then, $\mathcal{N}_X$ is trained using the strategy described in Subsection SM8.3. Table SM1 in Section SM3 summarizes the performance of $\mathcal{N}_X$. The results show that $\mathcal{N}_X$ exhibits a clear improvement in spacing accuracy, validating the effectiveness of the two-stage framework in refining the spacing prediction of $\mathcal{N}_V$ by enforcing the physical relationship $\dot{X} = V$.

SM6.3. Performance of $\mathcal{N}_{\text{sol}}$. With the two-stage framework validated by the improved spacing prediction from $\mathcal{N}_X$, we now evaluate the overall performance of the complete neural network framework $\mathcal{N}_{\text{sol}}$, composed of the trained $\mathcal{N}_V$ and $\mathcal{N}_X$. The evaluation compares trajectories predicted by the *NN-based approximation* ($\mathcal{N}_{\text{sol}}$) with the reference trajectories from the *FG-Simulation*. Three key quantities are examined: the dynamic-based TTC τ, spacing X, and relative velocity V, under the same non-zero leading vehicle acceleration profile a_ℓ defined in (SM4.1) in Section SM4.

As shown in Figure SM7, the trajectories predicted by $\mathcal{N}_{\text{sol}}$ closely match those from the *FG-Simulation* across all scenarios, exhibiting strong long-horizon consistency and convergence to the reference solution, confirming its ability to capture the delay-free nonlinear dynamics. Moreover, Figure SM13 in Section SM10 highlights that $\boldsymbol{V}_\theta$ maintains accurate TTC prediction throughout each scenario, reinforcing the reliability of the proposed neural network framework for safety-critical applications.

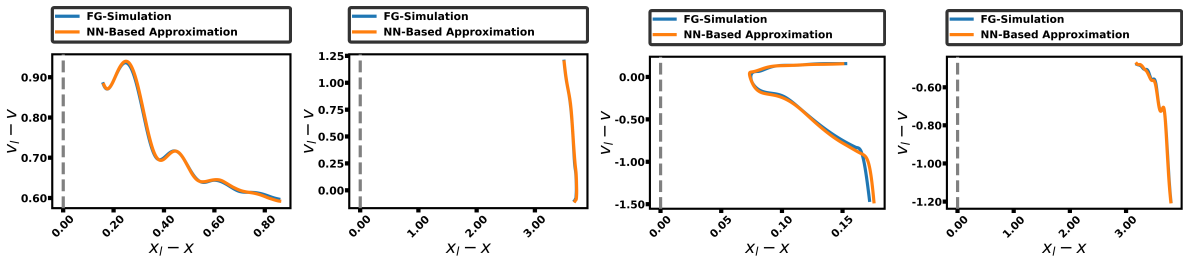


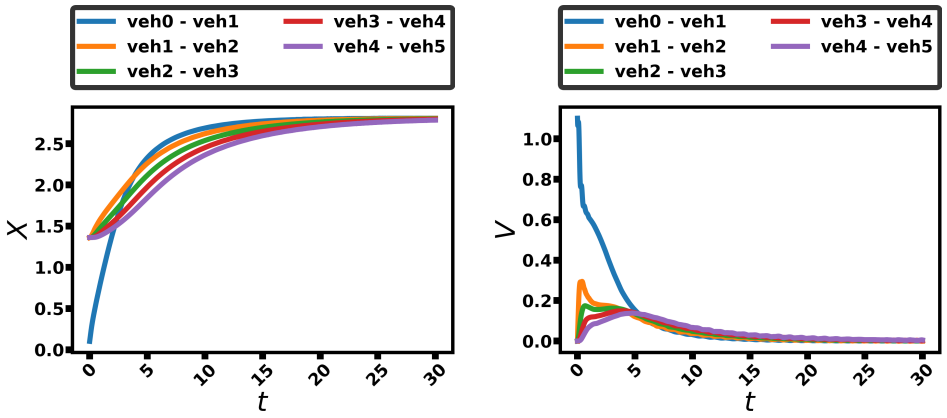
Figure SM7: *NN-based Approximation* vs *FG-Simulation* Framework, Parameters: α =2, $\beta = 3$, $x = 0$, Simulation Time = 1, a_ℓ as (SM4.1) in Section SM4. Initial Conditions: (a) $x_\ell = 0.16$, $v = 0.59$, $v_\ell = 1.47$; (b) $x_\ell = 3.5$, $v = 0.3$, $v_\ell = 1{,}5$; (c) $x_\ell = 0.17$, v =1.74, $v_\ell = 0.29$; (d) $x_\ell = 3.8$, $v = 1.7$, $v_\ell = 0.5$.

SM6.4. Numerical Stability Analysis of the Model with Dynamic-Based Time-to-Collision: Platoon Simulation. To observe the behavior of the proposed model Equation (2.1) under $\mathcal{N}_{\text{sol}}$ in a platoon, we conduct a numerical platoon simulation involving six vehicles as a representative example. The model itself is defined for arbitrary platoon sizes; the analytical results of the main text concern one pair with a prescribed leader. The leading vehicle follows the same non-constant acceleration a_ℓ defined in Equation (SM4.1) in Section SM4, consistent with the training setup of $\boldsymbol{V}_\theta$, $\mathcal{N}_V$, and $\mathcal{N}_X$, while the following vehicles $n \in [1, 5]$ evolve according to the proposed dynamics. The initial conditions are specified as:

$$\begin{cases} v_\ell(t_\circ) = 1.5, \\ v_n(t_\circ) = 0.4, \quad n \in [1, 5], \\ x_\ell(t_\circ) - \boldsymbol{x}_1(t_\circ) = 0.1, \\ x_{n-1} - x_n = \mathcal{V}^{-1}(v_n(t_\circ)), \quad n \in [2, 5]. \end{cases} \tag{SM6.11}$$

To examine disturbance propagation, we track spacing and relative velocity along the platoon over $t \in [0, 30]$. In the simulation, the leader-follower pair ($n = 0, 1$) employs the *NN-based approximation* ($\mathcal{N}_{\text{sol}}$) while remaining pairs use the *NeuNum-Simulation*. Two features of this setup go beyond the training and analysis setting and should be stated. First, the 30-second run is organized as 30 consecutive one-second segments; the leader follows (SM4.1) during the first segment and then keeps the speed reached at $t = 1$ (its acceleration is zero for $t > 1$). The two-stage networks $\mathcal{N}_V, \mathcal{N}_X$, trained on $[0, 1]$, drive the first pair during the first segment only, with the inputs $(X_0, V_0, v_\ell, \hat{\tau}(0), t)$ of Algorithms SM7.2 and SM7.3; from the second segment on, the first pair is integrated numerically like the others, each segment being restarted from the end state of the previous one. Second, with the initial data (SM6.11) the first pair starts with $V(t_\circ) = v_\ell - v_1 = 1.1 > 0$ (opening), outside the value-network training support $\Delta\zeta \leq 0$; following the rule of Subsection SM8.1.2, the value network is then evaluated with the relative velocity clipped to $\min\{V, 0\} = 0$ for $\hat{\tau}(0)$, and inside the numerically integrated segments an opening pair uses $\hat{\tau} = \boldsymbol{T}_h$. Third, for the pairs

$n \geq 2$ the TTC at each instant is $\boldsymbol{V}_\theta(t', X_n, \min\{V_n, 0\}, v_{n-1}(t'))$ with t' the time within the current segment and $v_{n-1}(t')$ the predecessor's realized speed at that instant (the predecessors are integrated first); the predecessor is thus predicted at its current speed over the window, and its actual future acceleration is not used. These pairs are controlled followers, not the prescribed leader of the analysis.



(a) Spacing between vehicles.

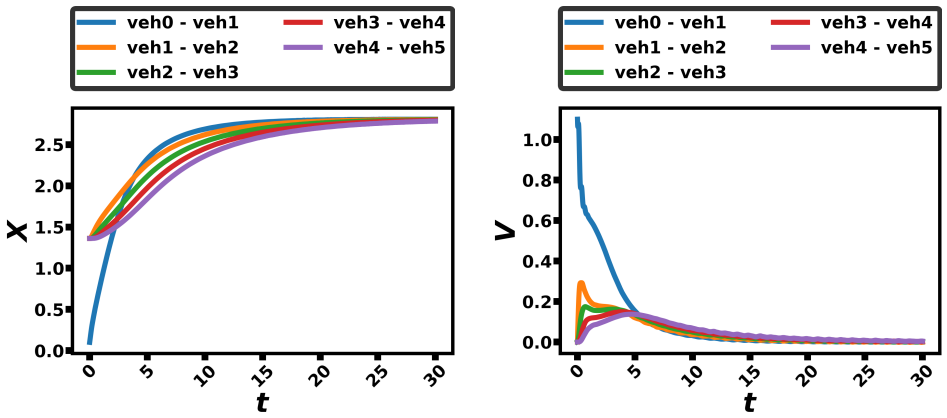
(b) Relative velocity between vehicles.

Figure SM8: Simulated nonlinear stability of the proposed model through *NN-based Approximation* and *NeuNum-Simulation* Approach: (a) spacing $X(t)$ and (b) relative velocity $V(t)$ between vehicles. Parameters $\alpha = 2$, $\beta = 3$ as in Figure 2; simulation time 30; a_ℓ as in (SM4.1) in Section SM4. Initial Conditions as stated in (SM6.11) with $x_5 = 0$.

As shown in Figure SM8, in this scenario the fluctuations in spacing and relative velocity decay along the platoon and the states approach the equilibrium spacing. This is an observation for the tested configuration; it is consistent with the local stability analysis of car-following dynamics in [SM7, SM6], but no string-stability or stability-preservation theorem for the learned implementation is claimed.

To further evaluate the neural network framework, we perform the same simulation using the reference *FG-Simulation* for comparison. As shown in Figure SM8 and Figure SM9, trajectories from the *NN-based Approximation* and the reference *FG-Simulation* are nearly indistinguishable in this scenario, which indicates that the learned predictor reproduces the reference dynamics for these initial data while removing the online-simulation delay.

SM7. Pseudo-codes for architectures of $\boldsymbol{V}_\theta$, $\mathcal{N}_{\mathsf{sol}}$, $\mathcal{N}_V$, and $\mathcal{N}_X$. The details of NN-architecture for the HJB solution $\boldsymbol{V}_\theta$ is detailed in the following algorithm:



(a) Spacing between vehicles.

(b) Relative velocity between vehicles.

Figure SM9: Simulated nonlinear stability of the proposed model through *FG-Simulation* Offline Approach: (a) spacing $X(t)$ and (b) relative velocity $V(t)$ between vehicles. Parameters $\alpha = 2$, $\beta = 3$ as in Figure 2; simulation time 30; $\mathtt{a}_\ell$ as in (SM4.1) in Section SM4. Initial Conditions as stated in (SM6.11) with $x_5 = 0$.

Algorithm SM7.1 Neural Network Framework $\boldsymbol{V}_\theta$ for Value Function Approximation

Input: Prediction-clock time $s \in [0, \boldsymbol{T}_h)$, current spacing $\Delta\xi = X(s)$, current relative velocity $\Delta\zeta = V(s) = v_\ell(s) - v(s)$, and current leader speed $v_\ell = v_\ell(s)$ (see Subsection SM8.1.2 for the meaning of the fourth input)

Output: Approximated value function $\boldsymbol{V}_\theta(s, \Delta\xi, \Delta\zeta, v_\ell)$

Initialize a sequential neural network model
Normalize s, $\Delta\xi$, $\Delta\zeta$, and v_ℓ
Add Dense layer with 256 neurons, activation `tanh`
Add Dense layer with 128 neurons, activation `tanh`
Add Dense layer with 256 neurons, activation `swish`
Add Dense layer with 512 neurons, activation `tanh`
Add Dense layer with 256 neurons, activation `tanh`
Add Dense layer with 128 neurons, activation `swish`
Add Dense layer with 64 neurons, activation `tanh`
Apply `tanh` after each residual addition
Add Dense layer with 1 neuron
Apply the sigmoid function $\sigma(\cdot) \in (0, 1)$ to the output and multiply by $\max\{\boldsymbol{T}_h - s, 0\}$
return Value function approximation $\boldsymbol{V}_\theta(s, \Delta\xi, \Delta\zeta, v_\ell) \in [0, \boldsymbol{T}_h - s]$

The NN-architecture for approximating the relative velocity V is elaborated in the following algorithm:

The NN-architecture for approximating the Spacing X is elaborated in the following algorithm:

Algorithm SM7.2 Neural Network Framework $\mathcal{N}_V$ for Proposed Model Solution Approximation

Input: Initial conditions $x_\ell(0) - x(0)$, $v_\ell(0) - v(0)$, $v_\ell(0)$, $\tau(0)$, t.
Output: Spacing X and Relative Velocity V at t
Initialize a sequential neural network model
Add Dense layer with 256 neurons, activation `LeakyReLU` with slope $\alpha = 0.001$
Add Dense layer with 128 neurons, activation `tanh`
Add Dense layer with 256 neurons, activation `sinusoidal` with frequency $\omega_0 = 10$
Add Dense layer with 128 neurons, activation `swish`
Add Dense layer with 128 neurons, activation `tanh`
Add Dense layer with 128 neurons, activation `swish`
Add Dense layer with 2 neurons
Apply `softplus` to the first output (spacing X) to ensure strict positivity
return Spacing X and Relative Velocity V Approximation.

Algorithm SM7.3 Neural Network Framework $\mathcal{N}_X$ for Proposed Model Solution Approximation

Input: Initial conditions $x_\ell(0) - x(0)$, $v_\ell(0) - v(0)$, $v_\ell(0)$, $\tau(0)$, t.
Output: Corrected Spacing X at t
Initialize a sequential neural network model
Add Dense layer with 256 neurons, activation `tanh`
Add Dense layer with 256 neurons, activation `swish`
Add Dense layer with 256 neurons, activation `tanh`
Add Dense layer with 256 neurons, activation `swish`
Add Dense layer with 256 neurons, activation `tanh`
Add Dense layer with 256 neurons, activation `swish`
Add Dense layer with 1 neuron (linear output; no positivity constraint is applied to the corrected spacing)
return Corrected Spacing X

SM8. Neural Network Training Strategies.

Training of the Neural Network is done on an Apple M2 Max ARM Processor with a maximum clock frequency of 3696 MHz, 38 GPU cores, 12 CPU cores (8 Performance and 4 Efficiency), and 96 GB of RAM.

SM8.1. V_θ Training Implementation.

SM8.1.1. V_θ Training Setup.

Training is carried out using the Adam optimizer with gradient clipping. Instead of employing a validation-loss-based learning rate reduction rule, the learning rate follows a warmup–cosine decay schedule: it increases linearly over the first 300 training steps to a peak value 3×10^{-4}, and then decreases according to a cosine decay down to 10^{-6} over the course of training.

In the interior loss, the Bellman residual J_{HJB} is scaled by a constant weight $\lambda_{\mathrm{HJB}} = 10$. In addition, the boundary loss $J_{\partial\mathcal{Q}}$ is multiplied by a separate coefficient $\lambda_{\partial\mathcal{Q}}$, set to 1 in the implementation. Consequently, the training objective places stronger emphasis on satisfying

the HJB equation in the compact interior region while still enforcing the boundary condition.

SM8.1.2. V_θ Training Dataset.

What is learned, and the meaning of the four inputs.. The analysis of Section 5 concerns, for one prescribed leader signal $(v_\ell(\cdot), \mathsf{a}_\ell(\cdot))$, the map $(s, \boldsymbol{y}) \mapsto \tau(s, \boldsymbol{y})$ on $\mathbb{D}$ with the horizon $\boldsymbol{T}_h$ fixed. The value network is trained for the leader signal in which the leader keeps its *current* speed over the whole prediction window, $\mathsf{a}_\ell \equiv 0$ and $v_\ell(r) \equiv w$ for $r \in [s, \boldsymbol{T}_h]$: the fourth input w is the current leader speed, sampled jointly with $\Delta\zeta$ so that the follower speed $v = w - \Delta\zeta$ lies in $[0, v_{\max}]$, and it parametrizes a one-parameter family of problems of the analyzed type (one for each constant w). No leader acceleration enters the training: the prediction used for the collision boundary (the line method) and the characteristics of the residual are those of

$$\dot{\Xi} = \Lambda, \qquad \dot{\Lambda} = -\alpha\big[\mathcal{V}(\Xi) + \Lambda - w\big],$$

and, since w is constant along these characteristics, the residual of (5.3) on the slice $w = \text{const}$ is exactly the three-variable expression

$$r_\theta = \partial_s \boldsymbol{V}_\theta + \Delta\zeta\, \partial_{\Delta\xi} \boldsymbol{V}_\theta - \alpha\big[\mathcal{V}(\Delta\xi) + \Delta\zeta - w\big]\partial_{\Delta\zeta} \boldsymbol{V}_\theta + 1, \tag{SM8.1}$$

which is what the released code computes (no derivative with respect to w is used, and none is needed). This leader signal is admissible for Asm. 3.1 for every $w \in [0, v_{\max}]$, so Sections 4 and 5 apply to the trained object slice by slice, with constants uniform in w; moreover, since the forcing is continuous (indeed zero), the frozen-speed value function is C^1 on $\mathbb{D}$ by Lem. 4.10, so a C^1 candidate with zero residual is not excluded by the time-regularity obstruction of Prop. 5.15. The trained $\boldsymbol{V}_\theta$ is therefore an approximation of the *frozen-leader-speed* dynamic TTC, not of the TTC under the acceleration profile (SM4.1); the two differ by the effect of the leader acceleration over the window, and for (SM4.1) the leader speed deviates from its current value by at most 0.21 (the maximum of $\int_0^t \mathsf{a}_\ell$) over the whole profile.

Horizon convention in the closed-loop simulations.. The value network is trained for the fixed horizon $\boldsymbol{T}_h = 1$ and returns, at prediction-clock time s, values in $[0, \boldsymbol{T}_h - s]$. In the closed-loop simulations the network is queried at every control instant t with the prediction clock reset and the current leader speed,

$$\hat{\tau}(t) = \boldsymbol{V}_\theta\big(0,\, X(t),\, \min\{V(t), 0\},\, v_\ell(t)\big) \in [0, \boldsymbol{T}_h],$$

so that the prediction horizon is a sliding window $[t, t + \boldsymbol{T}_h)$ over which the leader is predicted at its current speed; the inputs are clipped to the training box ($X \in [0, 1.5]$, $V \in [-v_{\max}, 0]$, $v_\ell \in [0, v_{\max} + V]$) and the output to $[0, \boldsymbol{T}_h]$. When the gap is opening, $V(t) \geq 0$, the network is not queried and $\hat{\tau}(t) \overset{\text{def}}{=} \boldsymbol{T}_h$, which is the horizon value $\boldsymbol{T}_h - s$ at $s = 0$ of Def. 3.6. The reference (FG-) simulations recompute the TTC at every instant in the same sliding-window manner by integrating the prediction from the current state over $[t, t + \boldsymbol{T}_h)$, with $\tau = \boldsymbol{T}_h$ if the predicted gap does not reach zero and $\tau = \boldsymbol{T}_h$ when $V(t) \geq 0$; in the trajectory-data generation and in the comparisons of Subsection SM6.3 the prediction uses the leader's actual future acceleration (SM4.1) over the window, whereas in the NeuNum-simulation and in the validation of $\boldsymbol{V}_\theta$ against the numerical TTC (Figure 3, Table SM1) the leader speed is frozen, exactly as in training; the validation grid is $s = 0$, $\Delta\xi \in [0, 1.5]$, $\Delta\zeta \in [-v_{\max}, 0]$, $v_\ell \in [0, v_{\max}+$

$\Delta\zeta]$ (step 0.01), with the first zero of the predicted gap as ground truth. The analytical statements of Section 3 are proved for the fixed-horizon closed loop with one prescribed leader signal; the sliding-window, frozen-leader-speed closed loop used in the simulations is a different construction, to which they are not applied, and the simulations are to be read as observations about that implementation.

Training batches and the sampled region.. With $\boldsymbol{T}_h = 1$, each training step uses two distinct batches: one interior batch and one boundary batch. The interior batch contains 65,536 samples and the boundary batch contains 8,192 samples. This procedure is repeated for 500,000 training steps. Before training, we first identify the dynamic collision boundary using the line method. For each sampled combination of $\Delta\zeta$, v_ℓ, and s, the line method determines the dynamic boundary: the critical spacing value separating states for which a dynamic-based TTC exists from states outside the collision-relevant region.

The interior batch is then sampled only from the compact collision-relevant region. Specifically, values of $\Delta\xi$ are extracted between the collision boundary $\Delta\xi = 0$ and the dynamic boundary found by the line method, $\Delta\zeta \in [-v_{\max}, 0]$, $v_\ell \in [0, v_{\max} + \Delta\zeta]$, and $s \in [0, \boldsymbol{T}_h)$. The boundary batch is sampled on the collision boundary itself $\Delta\xi = 0$, together with the same feasible sampling logic for $\Delta\zeta$, v_ℓ, and s. The boundary batch is introduced to enforce the near-collision zero-value condition, while the interior batch is used to learn the value function over the admissible safe domain.

For both batches, the samples are generated on-the-fly at every training step rather than being drawn from a fixed offline dataset. This design allows the training procedure to continuously expose $\boldsymbol{V}_\theta$ to fresh states while focusing sampling effort on the safety-critical region. For the boundary-batch, candidate points are first sampled from the feasible domain and are then retained only if they lie inside the compact set determined by the line method.

Relation between the empirical loss and the continuum loss.. The interior samples are drawn as follows: $u = (s, \Delta\zeta, v_\ell)$ is sampled according to the Lebesgue measure on $U = [0, \boldsymbol{T}_h) \times \{(\Delta\zeta, v_\ell) : -v_{\max} \le \Delta\zeta \le 0,\ 0 \le v_\ell \le v_{\max} + \Delta\zeta\}$, and then $\Delta\xi$ is sampled uniformly on the admissible interval $(10^{-5}, \Delta\xi_{\text{crit}}(u))$ determined by the dynamic boundary. The joint density of a sample is therefore $p(u, \Delta\xi) = 1/(|U|\,\ell(u))$ with $\ell(u) = \Delta\xi_{\text{crit}}(u) - 10^{-5}$ the length of the admissible gap interval; it is not the uniform density on the sampled region R, so the unweighted sample mean of $|r_\theta|^2$ estimates the weighted population loss $\int_R |r_\theta|^2 p\, dz$, not $\int_R |r_\theta|^2 dz$. Since $\ell(u) \le \ell_{\max} \overset{\text{def}}{=} \sup_U \ell < \infty$, the density is bounded below, $p \ge 1/(|U|\ell_{\max})$, and hence

$$\int_R |r_\theta|^2\, dz \ \le\ |U|\,\ell_{\max} \int_R |r_\theta|^2\, p\, dz,$$

so the continuum interior loss of Equation (5.8) is controlled by the weighted loss that training minimizes, up to the explicit constant $|U|\ell_{\max}$; the same holds for the boundary batch, which is sampled uniformly in u on $\{\Delta\xi = 0\}$ (a constant density on the sampled part of the incoming surface). The sampled region R consists of the states $(s, \Delta\xi, \Delta\zeta, v_\ell)$ whose predicted gap reaches zero before the horizon; it is forward-invariant under the prediction flow until contact (a state on a characteristic has the same contact time as the characteristic's starting state), so R is its own forward tube in the sense of (5.4), and the boundary batch samples the set of contact points on the incoming collision surface $\{\Delta\xi = 0,\ \Delta\zeta \le 0\}$ ($\Delta\zeta = 0$ is a null set), which is the set $\mathcal{Q}_{\boldsymbol{D}}$ of the main text. States of R whose first predicted contact is tangential are

not in the transversal domain $\mathbb{D}$; the constants of Thm. 5.13 apply to compact subsets of $R \cap \mathbb{D}$. Finally, the sampling constraint $0 \le v_\ell \le v_{\max} + \Delta\zeta$ enforces the speed box at the sampled instant only; along a prediction window the leader speed $v_\ell(s) + \int_s^r \mathsf{a}_\ell$ must also remain in $[0, v_{\max}]$ for Asm. 3.1 to apply, which with the profile (SM4.1) restricts the admissible current leader speeds by the extrema of the integrated future acceleration.

In summary, each training step combines one interior batch and one boundary batch, both generated on-the-fly. The interior batch is restricted to the region identified by the line method, and the boundary batch enforces the zero-value condition on the incoming part of the collision boundary. The empirical objective is a weighted version of the continuum objective of Equation (5.8), with the explicit density bound above.

SM8.2. $\mathcal{N}_V$ Training Implementation.

SM8.2.1. $\mathcal{N}_V$ Training Setup. Training is carried out using the Adam optimizer with a learning rate of 10^{-3}. A learning rate scheduler reduces the rate by a factor of 0.75 whenever the loss does not improve for 200 consecutive iterations. Exponential moving averages of gradients and squared gradients are used with decay rates of 0.9 and 0.999. The batch size is set to $N = 256$.

SM8.2.2. $\mathcal{N}_V$ Training Dataset. With the architecture and loss established, we now describe the training procedure for $\mathcal{N}_V$. The training dataset is composed of full-trajectory samples generated under various initial conditions. During each epoch, the data are divided into mini-batches of size N. For every trajectory $i \in \{1, \dots, N\}$ within a batch, the neural network is evaluated at M randomly sampled time points $t_j \in [0, T]$ where $j \in \{1, \dots, M\}$. To ensure temporal boundary accuracy, we always include both $t_1 = 0$ and $t_M = T$. This randomized temporal sampling allows $\mathcal{N}_V$ to learn the full operator mapping from initial conditions to continuous trajectories, rather than overfitting to a fixed time grid. In this study, we set $T = 1$ with a fine timestep size Δt being 1e-5, resulting in a temporal grid of 100,000 discrete steps. At each training iteration, $M = 17,500$ timesteps are randomly selected from this grid to promote temporal robustness and reduce overfitting.

To improve predictive performance in the safety-critical region, the initial condition sampling is intentionally biased toward near-collision scenarios. This strategy mirrors the biased sampling used for training $\boldsymbol{V}_\theta$ and is guided by the safety-critical region, following the local stability analysis for car-following dynamics presented in [SM7, SM6]. Accordingly, initial spacings are sampled uniformly over the 31 intervals of width 0.01 from $[0.00, 0.01]$ up to $[0.30, 0.31]$, with 4,500 samples per interval, and over the 13 intervals of width 0.1 from $[0.2, 0.3]$ up to $[1.4, 1.5]$, where the number of samples per interval is reduced progressively by a factor of 0.8 with a floor of 1,000 samples per interval (163,280 trajectories in total); the initial leader speed v_ℓ and the initial follower speed v_0 are sampled uniformly on $[0, v_{\max}]$.

Full trajectories are then simulated for each selected initial condition over $[0, T]$, $T = 1$, with the closed-loop model (2.1) under the leader acceleration profile (SM4.1) of Section SM4. In the data generation the TTC entering the feedback is recomputed at every integration step by the reference (FG) rule of Subsection SM8.1.2: the prediction is integrated from the current state over a window of length $\boldsymbol{T}_h$ with the leader's actual acceleration profile, $\tau = \boldsymbol{T}_h$ if no contact is predicted or if the gap is opening. The value network is used only

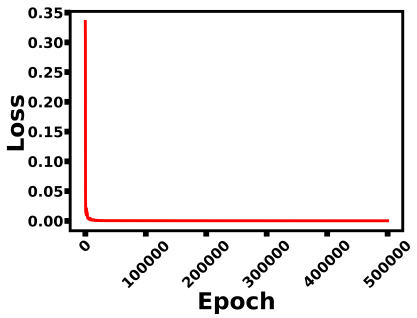
to supply the input $\hat{\tau}(0) = \boldsymbol{V}_\theta(0, X_0, \min\{V_0, 0\}, v_\ell)$ of Algorithms SM7.2 and SM7.3; it does not enter the generated trajectories. Note that $\boldsymbol{V}_\theta$ itself is trained for the frozen-leader-speed prediction (Subsection SM8.1.2), so the trajectory data and the value network refer to two slightly different TTC conventions, which differ by the effect of the leader acceleration over the one-second window.

SM8.3. $\mathcal{N}_X$ Training Implementation.

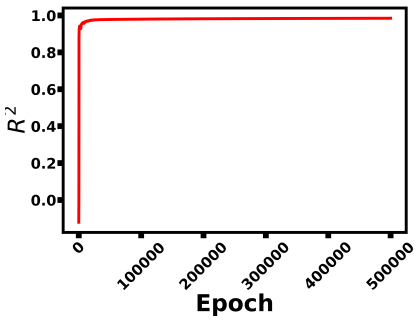
SM8.3.1. $\mathcal{N}_X$ Training Setup. The training setup for $\mathcal{N}_X$ closely follows that of $\mathcal{N}_V$, employing the same Adam optimizer, learning-rate schedule, number of epochs, and mini-batch size ($N = 256$). See Subsection SM8.2.1 for the corresponding $\mathcal{N}_V$ setup.

SM8.3.2. $\mathcal{N}_X$ Training Dataset. Similar to the training setup, the training dataset of $\mathcal{N}_X$ is generated based on the same approach as for $\mathcal{N}_V$ — a randomized temporal sampling strategy, where 17,500 random steps are selected from a temporal grid of 100,000 timesteps over the interval $[0, 1]$ with resolution $\Delta t = 10^{-5}$. Likewise, the same biased sampling approach is applied to initial conditions to ensure reliable performance in the safety-critical region. Throughout training, $\mathcal{N}_V$ provides the predicted relative velocity profiles used in the residual computation. All models are trained under the same non-zero leading vehicle acceleration a_ℓ defined in (SM4.1) in Section SM4, consistent with the setup used for $\boldsymbol{V}_\theta$ in Section 5 and Subsection SM8.1, and for $\mathcal{N}_V$ in Subsection SM8.2.

SM9. Neural Network Training Performance.

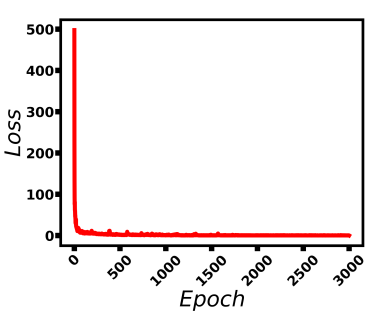


(a) Training loss.

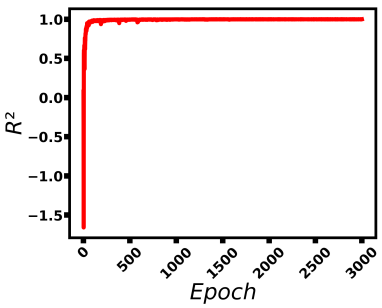


(b) R^2 scores on training sets.

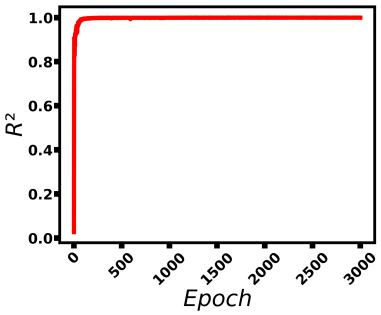
Figure SM10: Training performance of $\boldsymbol{V}_\theta$: (a) loss curves and (b) R^2 scores.



(a) Training loss over epochs.

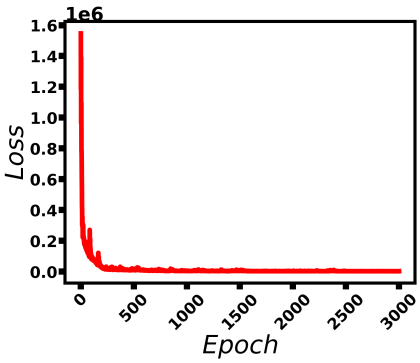


(b) R^2 score for V.

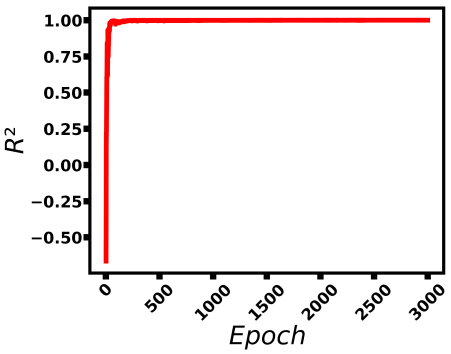


(c) R^2 score for X.

Figure SM11: Training performance of $\mathcal{N}_V$ with a_ℓ defined by (SM4.1) in Section SM4: (a) training loss, (b) R^2 score for relative velocity V, and (c) R^2 score for spacing X.



(a) Training loss over epochs.



(b) R^2 score for X during training.

Figure SM12: Training performance of $\mathcal{N}_X$ with a_ℓ defined by (SM4.1) in Section SM4: (a) training loss and (b) R^2 score for spacing X.

SM10. Dynamic-Based TTC for Each Region.

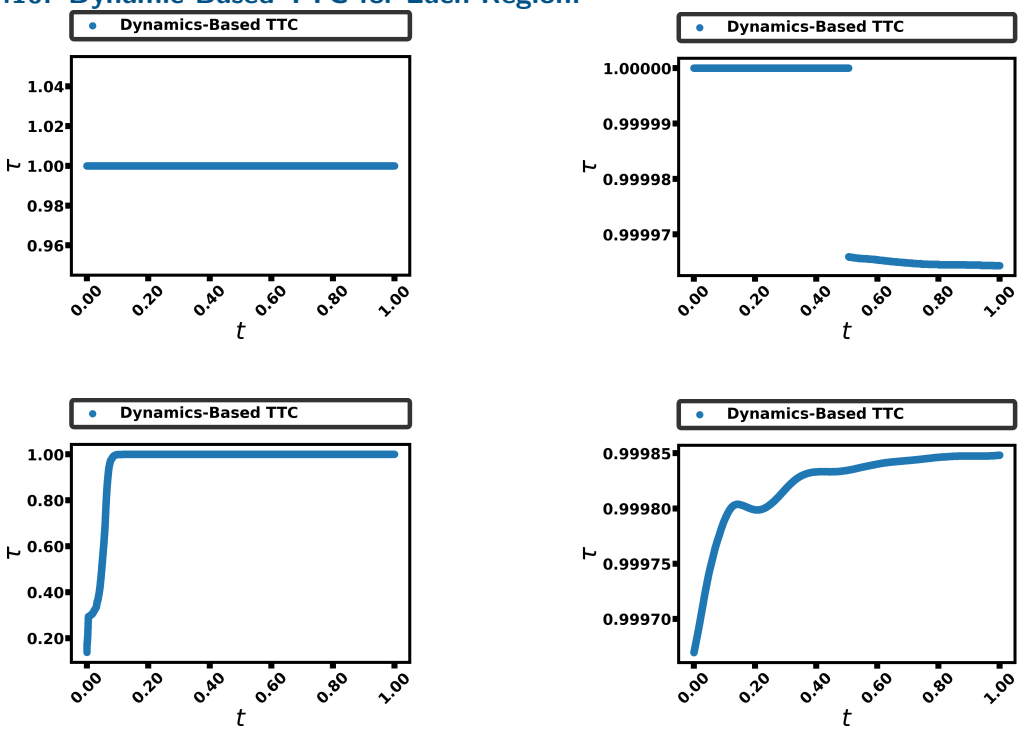


Figure SM13: Dynamic-Based TTC from $\boldsymbol{V}_\theta$ for different initial conditions, Parameters: α =2, β = 3, $x = 0$, Simulation Time = 1, a_ℓ as (SM4.1) in Section SM4. Initial Conditions: (a) $x_\ell = 0.16$, $v = 0.59$, $v_\ell = 1.47$; (b) $x_\ell = 3.5$, $v = 0.3$, $v_\ell = 1{,}5$; (c) $x_\ell = 0.17$, v =1.74, $v_\ell = 0.29$; (d) $x_\ell = 3.8$, $v = 1.7$, $v_\ell = 0.5$

Code Availability. The code used to perform the simulations, train and validate the neural network models, and generate the numerical results and figures in this paper is available from https://github.com/yuneily97/hjb-ttc-neural-workflow.

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